

600-12 Volt DC-DC Converter

DESIGN DOCUMENT

sdmay26-45

Formula SAE Club

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Executive Summary

The Formula SAE team is currently designing an electric racecar, and due to an increase in components, their low-voltage systems (including cooling systems, controls, etc.) demand a substantial increase in power. However, since they have a 600-Volt, 6.8 kWh tractive system battery used to drive the car's powertrain, they are looking to tap into this surplus of available power using a DC-DC converter. The requirements for the DC-DC converter are that it should be able to convert 600 volts – 12 volts relatively efficiently, with the ability to continuously draw 30 amps of current, or in other words, supply 360 watts of power. Since no specific efficiency requirement was listed (due to their power surplus), we set our goal at 90% efficiency to start.

With limited knowledge of the design space, we began researching several DC-DC converter topologies and concluded that the best option was a Half-Bridge LLC Resonant Converter, a state-of-the-art topology that offered substantial design complexity and extensive documentation. With a chosen topology agreed upon, we then searched for resources to support our design. We found an open-source GitHub repository from an FSAE team in Germany that had been developing a 600-24-volt half-bridge LLC topology for several years. However, it was limited to an output current of 20 amps due to the large power dissipation of its rectifier stage. After reading through the team's documentation as well as collaborating with the original design team, we were able to modify the design to fit our design requirements.

The first issue we addressed was to step down to 12 volts rather than 24 volts. We discussed many possibly ways of doing this, but the best way was to simply change the winding ratio from a 28:2 to 28:1 while maintaining similar leakage inductance (forms the resonant inductor). Since we did not have the materials on hand yet to confirm that we could physically achieve this, we asked the previous team to test this theory. They confirmed they could confidently make this change in the winding ratio without altering the leakage inductance.

The next issue we addressed was to increase the output current capacity to meet our requirement of continuously drawing 30 amps, providing 360 watts to the low-voltage systems. Initially, we aimed to replace the synchronous rectification stage with a passive rectification stage; however, we were unable to find diodes that could handle the required current. We were also heavily urged by the previous team to pursue a synchronous rectifier. This was also an issue that the previous design team faced, which is why they designed a synchronous rectifier using MOSFETs instead. Through discussing with the previous design team and reading the MOSFET datasheets, we realized that we could increase our current output from 22 amps to around 80 amps by simply adding heatsinks. We then adjusted our output filtering as well to compensate for the increase in ripple current.

The overall design of the system has been validated through a SPICE simulation containing the switching network, resonant tank, transformer, output filter, passive rectifier (substitute), and control loop. A separate SPICE model contains validation of the precharge circuit. The SPICE simulation results we have obtained give us confidence that our design properly meets the requirements of providing a regulated 12-volt output with a maximum continuous output current of 30 amps at 97.7% efficiency. The converter also responded extremely well to transient response testing with a voltage “droop” of only 18 mV. However, without a proper synchronous rectifier model, some uncertainty remains.

In addition to funding from the department, we were able to secure a sponsorship with Milwaukee Tool through the FSAE club, devising a roughly 50/50 cost split between the department and client. We used JLC PCB to fabricate and assemble many of the SMD components on our boards but did a majority of the soldering ourselves. Since we needed flexibility in tuning the leakage and primary inductances, we designed and fabricated our transformer ourselves.

During our testing process, we first tested the DC supply on the board independently. Furthermore, we tuned the transformer independently and tested it up to 1.8 kV of isolation. After these preliminary tests, we then tested the entire system using a 600-volt power supply, high-voltage differential probes for measurement, and a DC electronic load at the output. After confirming that the converter properly regulated 12 volts at the output, we load-tested the converter at 32 A of output current. We also tuned the switching frequency to 111 kHz to achieve an input voltage range of roughly 350-600 volts, covering the entire SOC range of the high-voltage battery. Finally, we derived the converter's efficiency curve from our load test data, achieving around 95% efficiency at the car's normal operating load.

Finally, we implemented the DC-DC converter into the Formula SAE's electric car. Currently, there is an issue from the client's end that is causing the input voltage to drop to 330 volts. However, despite this, our converter was still able to power the car's low voltage load. Overall, we exceeded the design requirements and delivered a powerful product to the client.

Learning Summary

Development Standards & Practices Used

- IEEE Code of Conduct
- FSAE Design Standards

Summary of Requirements

- Input voltage of 400-600 volts
- 12-volt stable output
- Output current of at least 30 amps
- Efficiency must be greater than 90%
 - Efficiency is defined by the power at the output divided by the power at the input
- Per Formula SAE standards, the converter requires galvanic isolation
- Voltage separation spacing of 4 millimeters

Applicable Courses from Iowa State University Curriculum

- EE 456
- EE 303
- EE 201
- EE 230
- EE 475
- EE 476
- EE 332
- EE 330
- EE 232

New Skills/Knowledge acquired that was not taught in courses

- LLC Converter Design
- Creating a smaller system designed to fit into a larger system
- Physical creation of a transformer
- Physical creation of a large electric system

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1. Introduction

1.1. Problem Statement

The problem our team aimed to solve was to design a 600-12-volt DC-DC converter for Iowa State's Formula SAE team. This converter will be used in the Formula car to power the car's sensors, fans, pumps, and other important low-voltage electronics using a 600V tractive system battery. Without the DC-DC converter, the FSAE team would have to rely solely on a second low-voltage battery in the car, which would lack the capacity necessary to meet their needs.

DC-DC converters are currently used in cars around the world. Our team had found an open-source converter used for another FSAE team car that referenced when we created our own design; however, our design required a lower output voltage than the one we found (12V instead of 24V), and therefore, we created a new transformer for our converter to step down to 12V and modified other portions of the converter to accommodate the new transformer.

1.2. Intended Users

There are three users of our 600V to 12V DC-DC converter:

- The driver:
 - The car driver, David, is only interested in setting fast lap times and needs the car to be reliable and to 'just work' so he can focus on winning races. The car gauges always need to be powered, which requires 12v dc to work. The converter is necessary for the car's gauges to be powered.
- The team members of the Formula SAE club
 - The team members of the FSAE club must work together to maintain and repair components on the race car, and require a converter that is relatively small, lightweight, durable, reliable, and most easy to repair. The design of the converter should take into account these needs. Additionally, the FSAE team members benefit from our project because they will be keeping the converter for use in their race car after we finish creating and testing it.
- Car inspectors.
 - The car inspector, Greg, wants all racecars and their components to adhere to Formula SAE rules and regulations, especially regarding safety, and demands that the DC-DC converter be safe and not short-circuit or exceed EMI limits, potentially causing an accident to the driver or other drivers on the track. Greg needs to make sure that all components of the car (including the converter) are safe and reliable, so our solution should take this into account.

Our DC-DC converter meets the needs of all these groups of people who will use, work on, or inspect it, and ensures they meet their requirements for a reliable, safe, and easy-to-maintain DC-DC converter.

2. Requirements, Constraints, and Standards

2.1. Requirements & Constraints

There are multiple standards that we followed to ensure a successful outcome for our project. In its most basic form, our design must convert 600 volts to 12 volts; however, it needs to do this with at least 90% efficiency. The system must also supply at least 30 Amps of current.

The system must also have detailed documentation and instructions to facilitate troubleshooting any future issues. Additionally, it should provide a way to incorporate it into the client's workflow, allowing for potential future revisions.

Finally, the specification requirements are that the system must have each of its components fully operational. If one part of the system is faulty, the entire system will not function; therefore, we need a properly specified power supply, including primary, secondary, and transformer components, as well as an oscillator with proper calibration and no overheating.

We also had constraints while working on this project, such as needing to have a minimum of 90% efficiency with our converter as stated earlier. We would also had to comply with FSAE design competition rules, such as minimum spacing of 14 mils between high voltage and low voltage traces. We would also need to comply with safety, especially when dealing with high voltage and current. This is why the system must be galvanically isolated.

Although not specified, we had to comply with the client's size requirements. The product must be clean and able to fit where it is needed to, other than that there are no other aesthetic requirements. We worked on a rigorous timeline, as the circuit must function properly by May for the competition, which placed a heavy responsibility on the entire team.

Economically, we were limited to a budget of \$1200 in total. Finally, it was difficult to test our product since we had to test it with a 600-volt power supply, which is not something easily obtainable.

2.2. Engineering Standards

Engineering standards are exactly as they sound: standards. Engineering standards hold certain levels of competency for new solutions. Engineering standards can range from reliability to safety, and as such, when building a relatively high voltage converter, standards will apply – safety more so than reliability. When applying standards to the 600-volt converter, both automotive standards and the standards specified by the FSAE design competition must be followed.

IEC 60664 is likely to apply to the 600-volt converter, as the standard outlines the insulation requirements between high-voltage systems and low-voltage systems, which are relevant to the converter. Likewise, IEEE 1100 outlines standards for grounding and voltage isolation. The purpose of the standard is to reduce voltage disturbances between systems on the same board, therefore improving power and grounding quality.

When it comes to safety, the IEEE 1584 standard is a reputable guideline for handling arc flash situations caused by high voltage. The standard outlines the PPE that should be used to prevent

burns and the components that can be used to mitigate the risk of an arc flash. Alternatively, the IEEE 519 aims to reduce the size of conductors by minimizing the total harmonic distortion caused by switching in DC/DC circuits. IEEE 519 outlines recommended voltage and current limits to reduce harmonics in the converter circuit, thereby reducing potential failure points.

All four of these standards directly apply to the quality and safety of building a 600-volt converter. We also plan on abiding by the FSAE standards to make the board usable in competition.

3. Project Plan

3.1. Project Management/Tracking Procedures

The team used a waterfall and agile management style due to having certain steps that needed to be completed before another, such as doing the circuit design before creating the PCB layout. However, the team also needed to be agile to address issues as they arose as our team wanted to do testing and revisions throughout the two semesters to ensure our design functions as intended. With these requirements in mind, the use of both project management styles was necessary to have a successful project.

The circuit is broken into five sub-schematics that each team member was responsible for completing. These sub-schematics are as follows: Primary Side, Secondary Side, Oscillator, Power Supply, and Pre-charge. There is also the transformer in which a separate team member was responsible for simulating and designing.

The team tracked the progress with weekly meetings by reviewing each of our respective circuit schematics, research updates, or any other idea or question a team member might have during this weekly meeting. This allowed us to see the progress we have made, as well as kept ourselves accountable for the work that we were responsible for completing. The Teams app was also used for general communication and to share our progress, along with any questions or concerns as they arose, so our team was able to continue making progress.

Task Decomposition

- Research LLC Converter topologies
- Source Components
- Get Ki-CAD installed and working
- Download the Git repository and make sure that everything in KiCAD is properly modeled
- Begin building and editing schematics to fit our requirements, each member has their assigned subcircuit
- Build LTSpice models to validate subcircuit functionality and component choice
- Continue to meet and review designs as a team
- Order Transformer parts
- Tune transformer to required specifications
- Layout PCB and coordinate funding
- Begin building the physical 600-12 V converter

- Create detailed test plan
 - Gather proper test equipment (LCR meter, Voltage source, power sink, tractive battery)
 - Preliminary tests of individual subcircuits such as the power supply
 - Isolation test on the transformer
 - Starting by testing with lower voltage, lower wattage and work up to max voltage and load
- Document results
- Iterate until design meets requirements

3.2. Project Milestones, Metrics, and Evaluation Criteria

The first milestone in this project was to ensure that we had a proper understanding of how each of our subcircuits works. With the weekly meetings, we updated the team's progress by either showing any simulations we had done or explaining how certain sections of each person's subcircuit work. We also had a documentation word file in teams, and were able to keep track of progress there as well.

We made sure that the transformer is properly specified since the requirement is to step down 600 Volts to 12 Volts. The transformer is the main part that will do that. We were able to keep track by simulating the transformer and checking if the output matches 12-volts. We also had to test the actual transformer that we designed by first sending in a low voltage to make sure the part works and do different test phases to make sure that it is tuned correctly. Then we followed the FSAE isolation test which states that it should not break down when testing three times the voltage of the battery maximum of 600-volts. This means our transformer must be able to operate at 1800-volts AC RMS for one minute.

We needed to also make sure that the schematic was simulated properly and how we wanted it. We used specific manufacturer parts in the simulation. We needed to make sure that we simulated it before ordering anything. There could always be last-minute changes depending on part availability.

There are many parts that needed to be ordered for the converter; however, by the time we were ready to order them, we predicted that it would be likely that some of them would be out of stock, therefore, we found proper replacements for any part that would be out of stock to mitigate this issue before it occurred. If we didn't do this, any missing part would take us back to the simulation stage; the only difference is that we would need to be quick with simulating the second time around, and ordering. We made a spreadsheet with every single part that we needed, and once a part was ordered, we marked it as ordered or edited in the case we needed to replace it.

Once all the parts were ordered, we assembled the product on the PCB we ordered, along with all the other parts that were ordered. Luckily, we decided we must order a minimum of 5 PCB's so in the case of any error in the circuit or something that doesn't work properly, we have backup PCBs. Once the assembly was completed, we proceeded to test the converter.

Testing a converter at this high voltage was a big milestone, as we didn't have access to a single power supply that reaches the required high voltage. We ended up using cascading power supplies

to reach the required voltage and ensure that the converter worked properly before giving it to FSAE. The ideal test is to have the actual battery that will go into the car and connect our converter to it to make sure it works properly.

Another major milestone was making sure that the PCBs we ordered are tested properly and pass all the tests we have for them. This is to make sure that all the low voltage systems work properly before putting the transformer on the board and start testing at high voltage levels. At the same time as testing the low voltage systems, we were also testing and tuning the transformer to have the proper inductance values.

3.3. Project Timeline/Schedule

To ensure success, we developed an Excel Gantt chart, as illustrated in Figure 1 below. The Gantt chart is essential to maintaining accountability and steady progress towards our goals. Tasks are color-coded to indicate high, medium, or low risk based on their urgency. Additionally, milestones and goals are established to further emphasize urgency. Each task is also annotated and assigned as shown in the zoomed-in image in Figure 2.

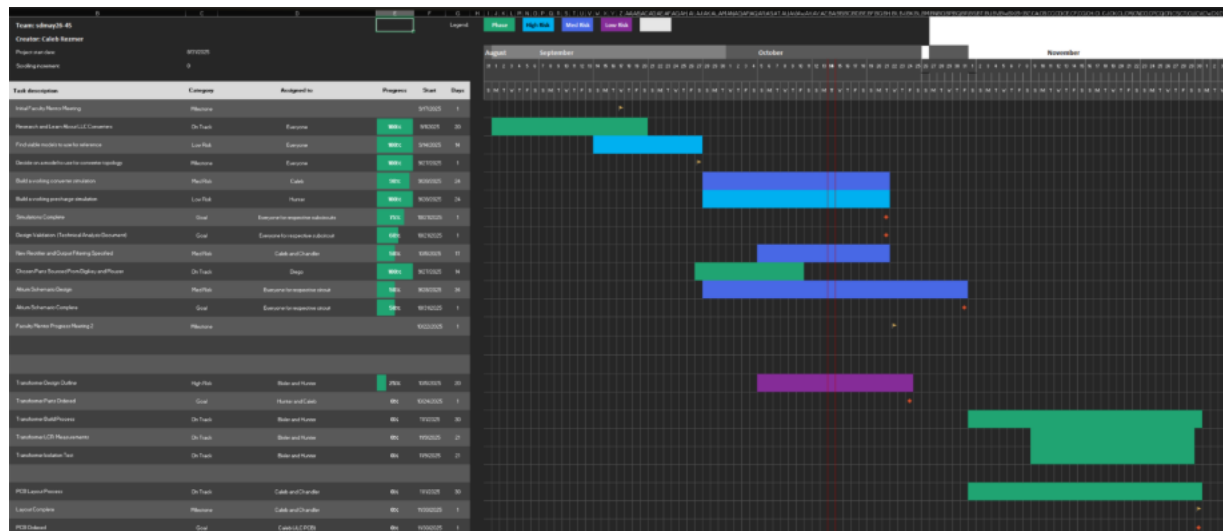


Figure 1: Gantt Chart for Timeline

Task description	Category	Assigned to	Progress	Start	Days
Initial Faculty Mentor Meeting	Milestone			9/17/2025	1
Research and Learn About LLC Converters	On Track	Everyone	100%	9/1/2025	20
Find viable models to use for reference	Low Risk	Everyone	100%	9/14/2025	14
Decide on a model to use for converter topology	Milestone	Everyone	100%	9/27/2025	1
Build a working converter simulation	Med Risk	Caleb	90%	9/28/2025	24
Build a working precharge simulation	Low Risk	Hunter	100%	9/28/2025	24
Simulations Complete	Goal	Everyone for respective subcircuits	75%	10/21/2025	1
Design Validation (Technical Analysis Document)	Goal	Everyone for respective subcircuit	60%	10/21/2025	1
New Rectifier and Output Filtering Specified	Med Risk	Caleb and Chandler	50%	10/5/2025	17
Chosen Parts Sourced From Digkey and Mouser	On Track	Diego	100%	9/27/2025	14
Altium Schematic Design	Med Risk	Everyone for respective circuit	50%	9/28/2025	34
Altium Schematic Complete	Goal	Everyone for respective circuit	50%	10/31/2025	1
Faculty Mentor Progress Meeting 2	Milestone			10/22/2025	1
Transformer Design Outline	High Risk	Bider and Hunter	25%	10/5/2025	20
Transformer Parts Ordered	Goal	Hunter and Caleb	0%	10/24/2025	1
Transformer Build Process	On Track	Bider and Hunter	0%	11/1/2025	30
Transformer LCR Measurements	On Track	Bider and Hunter	0%	11/9/2025	21
Transformer Isolation Test	On Track	Bider and Hunter	0%	11/9/2025	21
PCB Layout Process	On Track	Caleb and Chandler	0%	11/1/2025	30
Layout Complete	Milestone	Caleb and Chandler	0%	11/30/2025	1
PCB Ordered	Goal	Caleb (JLC PCB)	0%	11/30/2025	1

Figure 2: Tasks and Annotations Close Up

We also created a high-level project timeline that encapsulated the entire project, as shown in Figure 3. We approached this project in phases, from research to implementation, which made our progress feel more tangible. In retrospect, we closely followed this timeline, which led to the successful delivery of our converter. The transformer design took slightly longer than we initially thought, which was the only major deviation from the original timeline.

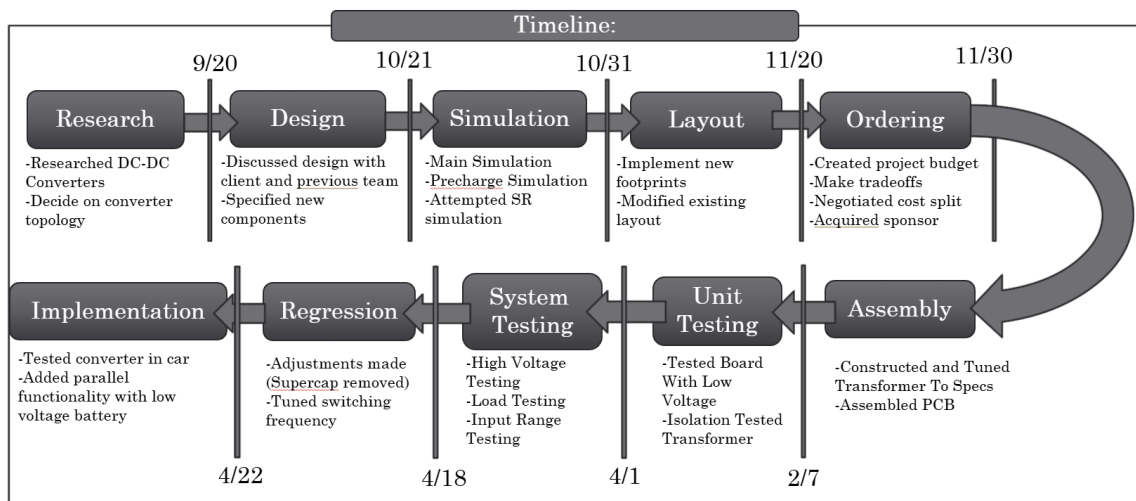


Figure 3: High-Level Project Timeline

3.4. Risks and Risk Management/Mitigation

We used the variable R to estimate the risk associated to each task required for the creation of our 600-12 V DC to DC converter.

- Converter design
 - o Selecting components with proper parameters (R = 30%)
 - o Creation of the full system with everyone's circuit implemented (R = 65%)
 - o Simulation of the design once completed (R = 5%)
- Creation of physical converter
 - o Order components (R = 15%)
 - o Create PCB (R = 10%)
 - o Order PCB (R = 5%)
 - o Assemble components on PCB (R = 70%)

Overall, we do not believe there is a large risk to this project. If the worst-case scenario had come to pass and our project was not completed on time, the Formula SAE team still could have ordered a less efficient and more expensive part, such as a buck converter, to replace the converter in their race car. Furthermore, there are a limited number of ways in which this task introduces safety risks to our team. Regardless, there are a few tasks that present more risk, both in terms of completing the project on time and maintaining personal safety, than other tasks.

The first task that presented more risk was the simulation of the design once completed. The reason that we believed this task presented more risk was that it was the last opportunity to catch issues with the design before ordering parts. If we made any mistakes in the selecting components task or the schematic creation tasks, and they were not found here, we would not know until we had completely ordered the parts and put together the LLC. To remedy this, we all be constructed our own section of the simulation, and then everyone reviewed each portion of the simulation to

ensure that everyone believed each part was correct. By doing this, we would be able to significantly reduce the risk of simulation being incorrect.

The next task that involved more risk was creating the PCB. PCBs are not very expensive, so there isn't much financial risk involved in this process. On the other hand, custom PCBs can sometimes take a fairly long time to ship. This means that if we had messed up the design, it was possible that it could have sidelined us for an extended period. This would have been very bad, as at the stage of the project where we were ordering a PCB, there isn't that much else we could have worked on at the same time. To mitigate the risk of time delay at this portion of the project, we split up creating the PCB and ordering the PCB into two distinct tasks. This gave us an opportunity to review the PCB before we ordered it and ensure it matched the design we had. Furthermore, we aimed to order the PCB before winter break, so if we made an error in the creation stage, we would still have plenty of time to rectify it and finish the project on schedule.

The final task that presents an elevated risk is assembling the components on the PCB. There are a number of ways in which this task can prove harmful to the project. The first of which is if we were to solder a part on in the wrong location or fry a component. This would have forced us to delay working on the project until we had time to order and receive a new PCB or component, depending on the severity of the problem. To mitigate this risk, we ordered multiple PCBs and multiple of each required component, so that if we had messed up at this step, it would not have forced us to delay the project. We had calculated the cost of the project already and ensured that it is within the projects budget to order multiple of each part to reduce the risk in this step. The second risk in this portion is the possibility of someone accidentally getting hurt due to the high voltage required for this project. To reduce the risk of injury, we had a meeting after we received the PCB, where every member of the group would review the IEEE safety standards, as well as general best safety practices when soldering and working with high-voltage components. This gave us a chance to ensure that we all understand both the risks associated with our project and how we can work safely to complete it on time.

The only major risk outside of testing that has affected this project so far was the budget issue. We had miscalculated the project's original cost; this issue was resolved when we secured funding from Milwaukee Tool. During testing, there were many risks and potential problems. Since we were dealing with a board that would eventually handle high voltage, we had components such as capacitors right next to each other, and if we weren't careful, we could have easily created a short circuit and destroyed a chip or another component. We also had a supercapacitor that is dangerous. We had to be careful when handling it and when discharging it. We also tested with high voltage which itself is a risk not only to the components but to ourselves.

3.5. Personnel Effort Requirements

Task	Number of person-hours to perform
Building the schematics	30
Simulation	30
Layout of Schematic	30
Building the 600-12 V converter	40
Documentation of design	30
Meetings	2-3 per week

Table 1: Personnel effort requirements (first semester estimate)

Team Member	Total Hours
Caleb Rezmer	155
Andrew Chandler	138
Drew Bixler	119
Diego Rivera	103
Wesley Mao	68
Hunter Kanne	94
Meetings	2-3 per week

Table 2: Team member hours

Task	Number of person-hours to perform
Building the schematics	20
Simulation	10
Layout of Schematics	20
Building the 600-12 V converter	50
Documentation of design	20
Testing (New section)	60 (Current phase)

Table 3: Personnel effort requirements (second semester revision)

We spent most of our hours testing the board and designing the transformer. We did not spend as much time as we estimated we would for each section listed above.

3.6. Other Resource Requirements

Software: Ki-CAD, LTSpice, Microsoft Excel, Microsoft Word, AutoCAD inventor, Waveforms

Hardware: High Voltage Battery, Cooling Fans, LCR Meter, Multi-meter, Oscilloscope, Waveform generator, Power Supply, Thermal imaging camera, Soldering Iron, PCB oven, & 3D Printer.

Parts & Materials: Capacitors, Inductors, Resistors, HV Fuse, PCB, PLA, Ferrite core, Transformer winding wires, Cooling fans, plastic clamp, Solder, Flux, Comparators, Transistors, FETs/MOSFETs, Optoisolators, & Buck Converters.

4. Design

4.1. Design Context

4.1.1 Broader Context

This design is for the FSAE community. Considering a broader context, this product is used in electric vehicles. However, it is not used in all electric vehicles, even though it is a more efficient alternative than the buck converters they currently use ubiquitously. The societal need that this project addresses is to increase the efficiency of our electric vehicles, resulting in reduced power consumption.

We would reduce overall power consumption with this LLC converter. Not just reducing power consumption but replacing the current solution that the FSAE has for the low voltage systems of hot swapping batteries, which means in the long term, having to purchase multiple batteries and constantly charging them instead of having a system that uses the main battery and there would be no need for a stop to replace the low voltage batteries.

The project would reflect the practices of the FSAE, an engineering design competition that focuses on what engineers do: making products as efficient and effective as possible. Especially in a competition setting, where every small optimization would give an advantage.

The environmental impact that this project has is that it has less power waste compared to buck converters and less physical waste, such as batteries. It is a more efficient alternative to both solutions.

The economic impact of the LLC converter is that although it is a large initial purchase, it is a much more efficient and convenient solution than the other two options. The cost would make an overall normal electric vehicle more expensive however for commercial use it provides high efficiency and reduces power dissipation which would save on power costs. It also provides superior performance in steady state and transient conditions.

Area	Description	Examples
Public health, safety, and welfare	The LLC converter positively affects the stakeholders in the project.	The converter increases power efficiency for the ISU FSAE team, allowing them to use less energy on low voltage electronics such as gauges for the car. The converter also positively affects financial stakeholders like Milwaukee tool as a working converter reflects well on their financial contribution.
Global, cultural, and social	The converter adheres to both IEEE and FSAE guidelines and reflects the standards that our converter is evaluated at.	Such guidelines as transformer isolation, EMI, Low voltage and high voltage signal isolation, High voltage labeling, PCB trace widths and lengths, and PCB dimensions.
Environmental	The LLC converter is more power efficient than other converter types. However, the LLC converter utilizes ICs, PCBs, Copper, and other materials in the manufacturing process.	The manufacturing and refinement of these materials is known to have negative impacts on the environment.
Economic	Large upfront cost especially in lower quantities.	May not be as economically viable for some smaller FSAE clubs with less funding; however, for automotive companies the design is more practical.

Table 4: Areas of relevant consideration related to project

4.1.2 Prior Work/Solutions

Several DC-DC converter products exist on the Digi-Key and Mouser markets. These include products made by TDK-lambda, XP Power, Cincon Electronics, Mean Well, OmniOn, Vicor Corporation, and Artesyn. Such products include the PAF600F48-12 DC-DC Converter (12V, 600W), QHL600 DC-DC converter, LTD CFB600-48S12 DC-DC Converter, Q DC-DC Converter, and others. However, none of these meet the specifications of our project: an input voltage of 600 volts, an output voltage of 12 volts, and a current output of 30 amps. We could not find a DC-DC converter that meets our specific requirements from our market research, which is the primary motivation behind this project. However, we found an open-source DC-DC converter design on GitHub that could be modified to meet our design requirements. This is the design that we have based our converter on and have been closely communicating with the previous designers. We based our decision to pursue the DC-DC converter based on the following:

Pros:

- Converts V_{in} of 400-600 volts of power into 12 VDC highly efficiently
- Cost-effective low-voltage source
- Compact and energy efficient

Cons:

- No reverse polarity protection
- Lower thermal safety for some components
- Time-consuming and costly development

4.1.3 Technical Complexity

An LLC converter is a state-of-the-art converter topology that features numerous subsystems and complex technical concepts, which collectively contribute to its functionality. Each subsystem requires a deep foundation of electrical engineering knowledge to both understand and design, including magnetics design, controls, semiconductors, analog circuits, and electromagnetic interference. There are eight subsystems within the LLC converter: the pre-charge, half-bridge, resonant tank, transformer, rectifier, control loop, power supply, and overtemperature protection, all of which are interdependent.

The pre-charge circuit requires knowledge of op-amps, precision programmable references, MOSFETS, and other fundamentals, which all serve to properly charge up the input node of the circuit and filter out transients. The H-bridge circuit, which contains a half-bridge switching network, converts the DC into an AC square wave. The resonant tank then filters the harmonic content of the square wave to refine it into a sinusoid, enabling more efficient power conversion and reduced switching losses across the MOSFETs. Since the resonant tank circuit also has a gain curve, altering the switching frequency alters the overall gain of the system. An LLC controller chip uses this frequency-dependent gain curve to regulate the output. Proportional feedback from the output is used to adjust the switching frequency, maintaining an output voltage of 12 volts.

Next, the transformer is designed to serve two separate functions simultaneously. Its primary function is to step down the input voltage of 600 volts to 12 volts through coupling. However, the transformer is also gapped and tuned so that its leakage inductance serves as an inductor for the

resonant circuit. Tuning the transformer to the required specifications to meet both functions is a feat of considerable complexity. Finally, the secondary side of the transformer contains a synchronous or active rectifier to convert the low-voltage AC back into low-voltage DC. This circuit is much more complex than a simple diode rectifier because it uses MOSFETs that switch synchronously based on a voltage sense to decrease conduction losses; rather than suffering from the .7 volt drop from a diode, the current only has to travel through a few milliohms of RDS on. Finally, an output pi filter is tuned to properly smooth the waveform from ripple.

The complexity of the LLC converter speaks for itself.

An LLC converter is complex on its own; however, we did not make one from scratch. We used an open-source design as a reference for our application of an LLC converter. The main change that we made is that the open-source project steps down 600-Volts to 24-Volts. However, ours stepped down to 12 volts, which comes with many complex ramifications. In addition, we needed to substantially increase our output current from 21 amps to 30 amps.

We needed tune and test different parts of the converter to ensure that everything is suitable for the 12-volt output. Many parts of the open-source system were specifically designed to work with 24-Volts.

4.2. Design Exploration

4.2.1 Design Decisions

The main three decisions that we have had to make regarding our LLC converter was whether we make a transformer from scratch or find and order one that fits our requirements, whether we should use active or passive rectification, and what kind of transformer we want to create. We decided that we were to create our own transformer and use active rectification, solving our first two design decisions. To solve whether we should use a premade transformer or make our own, we started by searching for what premade transformers exist. The purpose of doing this was to learn how realistic it would be to use a transformer that was premade. We then ran through the pros and cons of making our own or buying one during one of our weekly meetings. In the end, we decided that creating our own transformer would allow us to save money, give us the flexibility to modify and tune it, and be more technically complex. This had a large effect on the project because creating the transformer was a challenging portion of our project. Furthermore, the transformer is what is actually stepping down the voltage in our converter, meaning that it is crucial to the functionality of our converter and how efficient the converter is.

The second design decision that we had to make was whether we should use passive or active rectification. For this particular decision, we decided it was important to get some more feedback on the decision, so we discussed it in a meeting with our faculty mentor. Furthermore, we reached out to other people that have created LLC converters before, and found that while passive rectification would likely work for our circuit, there was some risk involved that the diodes needed for passive rectification would burn out when dealing with the high currents involved in a LLC converter. Because of this, we chose to proceed with the plan of using active rectification in our circuit. This decision increased the complexity of our project but ensured that the converter does not break quickly after being created due to diodes shorting out. Therefore, this decision ultimately

allowed the project to be more successful, as using active rectification created a significantly more robust LLC converter.

The third design decision that was necessary for our project was determining what kind of transformer we should use in our circuit. This decision came up after we decided that we would build the transformer ourselves. This decision was difficult, as there are a large number of different transformers that can be used to create an LLC converter, meaning there were a large number of different options that we could go with. In the end, we decided to use a center tapped transformer with an internal resonant inductor. We made this decision by discussing during a group meeting the pros and cons of each of the different options and coming to a group consensus on which option we decided to go with. We chose to use a center tapped transformer with an internal resonant inductor because it is a commonly used transformer design for LLC converters, meaning that there is a lot of information that we can utilize in the design of our own transformer. Furthermore, it allows us to keep the general size and functioning of our transformer like that of the transformer that is used in the LLC converter we are referencing when designing our own LLC converter. This choice allowed the project to work very efficiently and our project technically complex without making it too complex to finish in one school year, thus ensuring the success of our project.

4.2.2 Ideation

When we were deciding what type of transformer that we would create, we started by having each group member do some individual research and come up with a couple different options each. From this stage, we came up with a couple of different choices. The first choice was whether we would use an internal or external resonant inductor. The second choice was whether we would center tap the windings for the inductor or not. What this means is whether we would use a wire coming out of the middle of the windings so that we could have more windings while still getting the exact ratio we need to step our signal down from 600 V to 12 V. The third decision to make was the core shape. Some of the options that we came up with for the core shape included an ETD form factor, and a PQ form factor. All in all, there were three major design decisions with two options each, totaling six total choices, and eight different possible combinations from the options available. During our following weekly meeting we had each member research pros and cons of each option for a couple of minutes. While doing this part, we made sure everyone was researching it individually to try and minimize the possibility of us missing any pros or cons. After this, we discussed as a group to try and come up with what we all thought was the best idea to go with for the project. We ended up deciding on using an internal resonant inductor, with a center tapped winding, and a ETD form factor. The reason that we chose the internal resonant inductor was largely because it simplifies the design of the circuit outside of the transformer. This allows us to get the same functionality out of a less complex circuit, at the cost of making the transformer a bit harder to make. There is plenty of information out there regarding creating a transformer with an internal resonant inductor, so we decided that this would be a safe decision to go with. We also decided to use a center tapped winding, as the majority of LLC converters that we could find were used for large voltage drops, like the one we are creating, use center tapped transformers in order to make a transformer that is more typical. This will make it easier to find information regarding how to actually design the transformer and will not come up with any drawbacks in terms of efficiency. Finally, we decided to use an ETD form factor because it makes it easier to use windings

with a large cross section. Being able to use windings with a larger cross section will help with the leakage induction and the isolation, meaning that the transformer will be more efficient and will also be better galvanically isolated. Galvanic isolation was a big reason as to why we chose to use an LLC converter, so choose an option that helps support that allows our project to better fit the design parameters of the client.

4.2.3 Decision-Making and Trade-Offs

The process that we used to identify the pros and cons of our ideated options was to have every member do individual research, before collating our research to create a large pros and cons list. From there, we discussed the benefits and drawbacks of each option to try and come up with the solution that made the most sense. While discussing which option to go with, we made sure to prioritize benefits and drawbacks that related to the design parameters we have received from the client, as well as focus on options that better fit some of our focus points in our LLC converter. Some of these focus points include making a converter that is highly efficient, small, lightweight, and has galvanic isolation. While discussing options, we did discuss the benefits and drawbacks that were not related to these focus points, such as the availability of resources to learn about the options, but we made sure to put more weight into the pros and cons that help us deliver these focus points that our client desires.

The reason that we chose the method that we did was twofold. First, we had every group member do individual research so that if any group member missed something, at least one member of the group would likely have seen it in their own research. This let us be more thorough when creating our list of pros and cons. Furthermore, it made it such that every group member was well versed in the different options when it came to discussing the options and deciding which to go with. The reason that we discussed each option as a group was because at the start of the semester, we all agreed that we would discuss major design decisions as a group before choosing one, and we want to stick to this promise that we made. The reason that we decided to make this agreement was because we believe that it allows us to come up with the best solution to fit the needs of our project, due to the fact that the discussion allows group members to each advocate for what they believe is the right solution. This helps to ensure that everybody sees all the good and bad for each option. This also helps to make it such that by the end of debating what option works best, we will be able to come to a complete and well-informed decision on what option works best for our project. It's worth noting that when we have discussions on design decisions such as this, we typically will listen more to whatever group member is the specialist for the part of the circuit being worked on, but every group member still gets a say on what the final decision is.

4.3. Final Design

4.3.1 Overview

The high-voltage DC/DC LLC converter takes power in the form of voltage and current from the high-voltage car battery as an input. The 600VDC input must then be converted to AC through the use of switches (MOSFETs) and turned into a typical sinusoidal wave via the “resonant tank”. Once the 600VDC input changes into a typical 600VAC wave, it may then go through a transformer that

decreases the voltage by a ratio to 12VAC while increasing the current by the inverse of that voltage ratio. Lastly, the 12VAC wave must be changed back into a DC signal by use of rectification, as seen in the figure below (Figure 3).

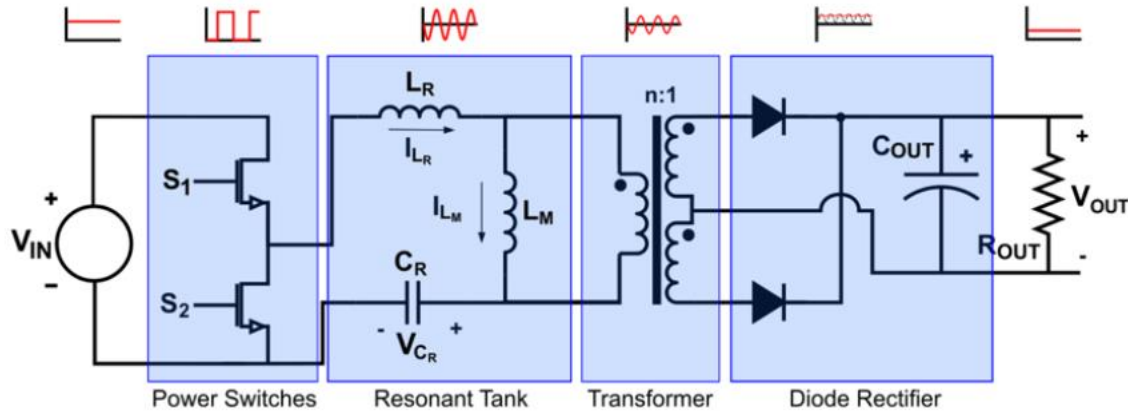


Figure: [How a Transformer works](#)

4.3.2 Detailed Design and Visual(s)

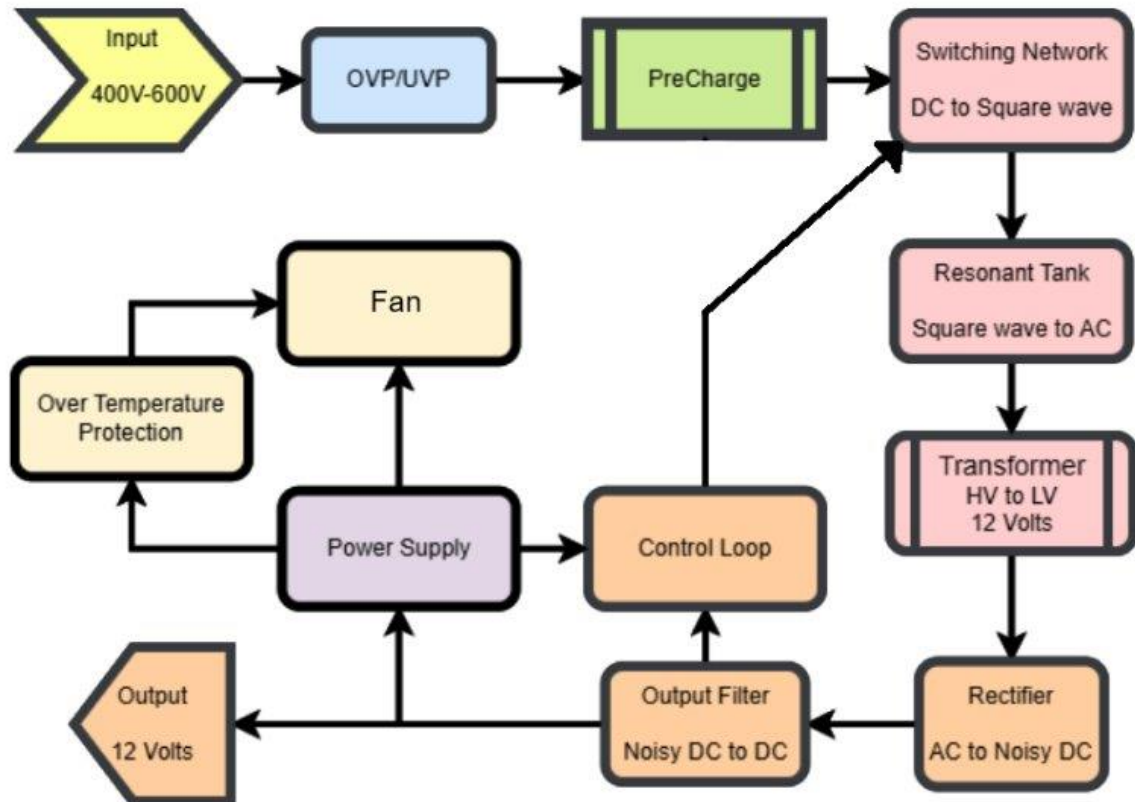


Figure: Flowchart of Systems

- The subsystems are split up into the following:
- Pre-charge/Discharge
 - OVP/UVP
 - Pre-charge
- Oscillator & Control Loop
 - DC/DC Control
- Primary stage
 - FET Half Bridge
 - Gate-Driver
- Transformer
- Secondary Side
 - Rectifier
 - 12V Output
- Power Supply
 - Buck Converter 12 to 3.7V
 - Supercap 3.7V
- Overtemperature Protection
 - Temperature Surveillance
 - Fan 3.7/12V

The Pre-charge and discharge circuits help pre-charge and discharge the DC link capacitor which smooths out the voltage from the high voltage battery. The DC/DC control loop regulates how much current is demanded from the high-voltage battery. From there, the primary stage converts the 600VDC to 600VAC through the use of the gate driver, which controls the FET Half Bridge. Next down the line, the Transformer converts 600VAC to 12VAC along with increasing the current. The secondary side converts the 12V AC to 12V DC, resulting in a 12V output. This voltage is too much for the Supercap, so it is stepped down with the Buck Converter from 12 to 3.7V. The supercap is utilized to act as a battery of sorts, so the board can be powered for long periods of time while off. From there, the temperature surveillance system monitors the transformer and the surrounding system and controls a fan.

Precharge/Discharge Circuit:

The primary function of the Pre-charge & Discharge circuit is to charge the capacitor prior to supplying 600VDC current to the DC link capacitor and discharging the 600VDC of the DC link capacitor once the circuit is turned off, respectively. The Pre-charge circuit uses a constant current to reduce the amount of power going through the resistor and therefore makes the resistor much smaller physically.

Going through the entire process of the circuit, we first need a 12VDC supply denoted by V_3 to supply the comparators and NMOS'. From there, we must look at the "Constant Current Pre-charge", V_3 is used as a reference in the U_{10A} , while the current going through Q_4 & R_6 determines the comparing voltage. As the comparing voltage drops or rises, U_{10A} increases or decreases its voltage output, respectively. By throttling the voltage output from U_{10A} , the gate of Q_4 allows current to R_6 and thus creates a constant current to the DC link capacitor. It's worth mentioning that the purpose of R_{51} , R_{49} , & D_{13} is to make the circuit act ideally; R_{51} is there to increase the

stability of the U10A, R49 is to allow current drive to the gate of Q4, and lastly D13 is to prevent a false positive of Q4 during startup that could cause the circuit to turn on without input.

The “Pre-charge Close” has the function of shorting the DC-link capacitor to the 600VDC source when the capacitor is close enough to 600VDC so as not to cause a large inrush of current. Once more, starting from the V3 source, the 12VDC is used as a reference in conjunction with R47 and R50. However, the 12VDC also flows through R53 to control the input of the U10B. As the DC-link capacitor charges, the voltage above D12 & R52 goes towards 0VDC. Once the voltage above D12 & R52 gets below 11.3VDC, the V3 source connects to ground through Q4 and activates R52 & R53 as a voltage divider and causes the output of U10B to go high. Once U10B goes high, it activates Q3 and shorts the DC-link capacitor and permanently keeps U10B high. Once again, R7 & R8 are used for stability and serve the same purpose as R51 & R49, but C38 is used to prevent false positives in the event of keeping Q3 closed.

Lastly, the passive discharge is comprised of R47 & R2 by shorting the DC-link capacitor and slowly discharging it in the event neither Q3 nor Q4 is conducting. Due to the size of R47 & R2, the DC-link capacitor discharges slowly, but this is done to reduce the amount of current going through the resistors, therefore making them smaller.

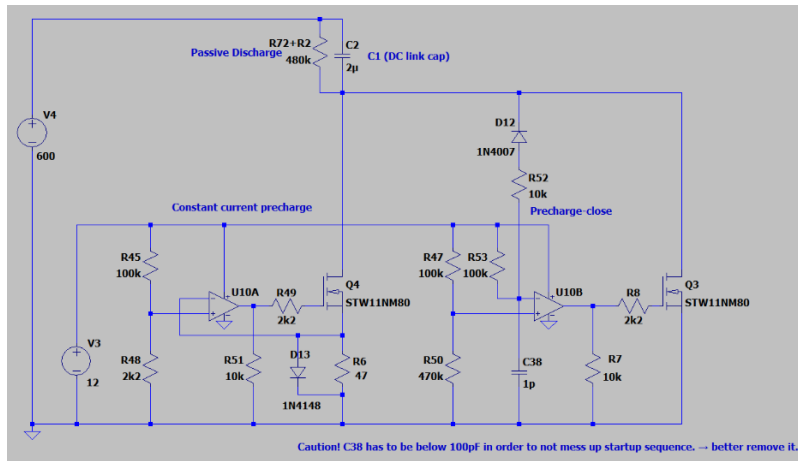


Figure 4: Pre-charge Circuit

Control Loop:

The purpose of the control loop is to regulate the output of the converter at 12V DC. To do this, it measures the output voltage, and if the output voltage is above 12V, it increases the switching frequency of the switching network which causes the converter's gain to decrease. It continues to increase the switching frequency until the converter's output is 12V. If the measured output voltage is 12V or less in the first place, the control loop doesn't do anything. In effect, this delivers instantaneous voltage regulation holding the output of the converter at the required 12V.

The purpose of the control loop in the LLC converter circuit is to maintain zero voltage switching and keep the output voltage of the circuit constant at the required voltage level. This makes it so that if voltage spikes on the primary side of the circuit, the converter's output voltage will remain constant. This is incredibly important, as the converter powers several different sensitive electronics that could get destroyed if the converter outputs too high of a voltage for a sustained period.

The main component of the control loop is a TL431 comparator. This component compares voltage to an internal reference voltage. The internal reference voltage of this comparator is 2.5V. The output voltage is sent through a voltage divider with the output connected to the reference pin. The resistors for the voltage divider are set up such that the voltage coming out of the voltage divider is just below 2.5 V. This makes it so that if the voltage at the converter's output spikes, the voltage going into this reference pin will exceed the comparator's internal reference voltage. If this happens, a current runs through the comparator, increasing the current at an optocoupler. The optocoupler then gets the control chip (the control chip is covered more intensely in the primary side section of this document) to increase the switching frequency, which in turn decreases the converter's output voltage.

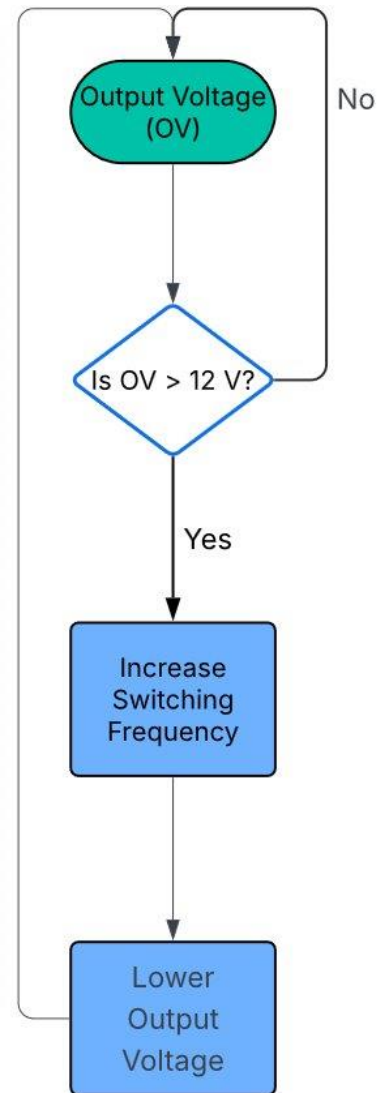


Figure 5: Block Diagram of Control Loop

Components Used:

Primary Side:

At a high level, the primary side consists of a switching network and a rectifier. The switching network is used to turn the input DC voltage into a square wave signal. The rectifier is used to turn the square wave signal into a sinusoidal AC voltage. The purpose of turning the DC input into an AC voltage is to allow the transformer to step the voltage down with a high degree of efficiency. This prevents losses due to things like waste heat that would occur if we tried to directly step down the DC voltage. A block diagram detailing the primary side from a high level is included below.

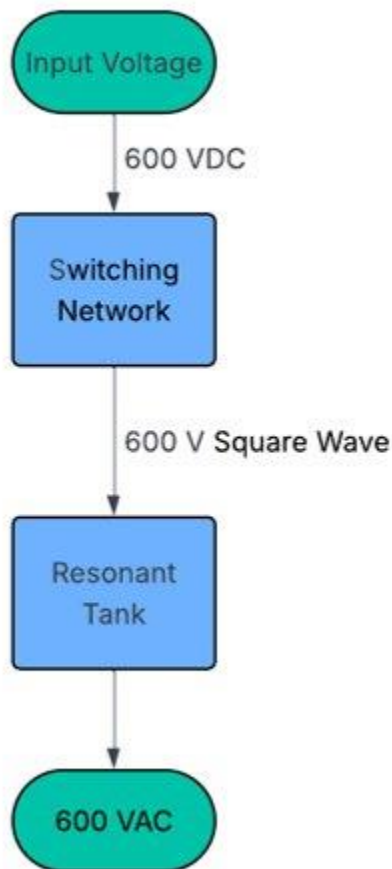


Figure 7: Block Diagram of Primary Side

The primary side uses two MOSFETs (IMBG65Ro50M2HXTMA1) to convert the DC 600V input into an AC signal, which can then be stepped down and rectified to 12VDC. The MOSFETs are driven by the UCC25600DR to open and close the MOSFETs and are boosted by the UCC21520DWR gate driver, which supports the higher current required to drive the MOSFETs.

Programmable dead time is necessary to optimize the converter's efficiency. Using a higher resistance value connected to the DT pin on the UCC25600DR to shorten the dead time. The UCC25600DR also has a programmable frequency, which is needed to match the frequency of the

resonate tank and transformer. The OC pin is used for overcurrent protection. When the OC pin exceeds 1V, the power stage shuts down, utilizing the sensing circuit, which uses the bottom-resonate capacitor voltage, proportional to the load current. If the load current is too high, the OC protection pin voltage will increase, causing the U3 chip to limit the current and potentially restart the circuit.

Going through the Circuit, U3 creates the clock signal using a 12V DC input; the dead time is set with the 10k resistor to minimize switching loss. The RT pin is connected to the two series-connected resistors and the output of the optocoupler on the secondary side of the converter, which sets the minimum and maximum switching frequencies. The SS pin is tied to ground through a capacitor to set the soft-start timing.

The soft start works by limiting the inrush of current when the capacitors in the circuit are completely discharged. If the current were not limited by the soft start, the system would attempt to charge the capacitors instantly, potentially causing damage to the circuit and excessive switching of the MOSFETs on the primary side. Using a 470nF capacitor connected to the SS pin and tied to ground sets the soft start time to 256ms.

The resonance of the switching network is crucial in creating a functional LLC converter; if the resonance is off, the MOSFETs may end up turning on simultaneously, shorting the circuit, which would be catastrophic. The resonate tank is used to set the resonance of the LLC converter and is implemented in this design using split capacitors to achieve this. Using two smaller voltage-rated capacitors connected in series, less current is sent through each capacitor, as only half the RMS voltage is applied across them.

The resonance of the converter is affected by the load; the figure below shows the resonant curve of the reference converter under various loads. The higher the load, the higher the gain is for the converter, with the resonate frequency shifting slightly to the left. At lower loads, the gain decreases, and the resonate frequency shifts to the right. It is essential that the resonance remains on the left side, as shifting it too far to the right could cause the switching network to short, as previously mentioned.

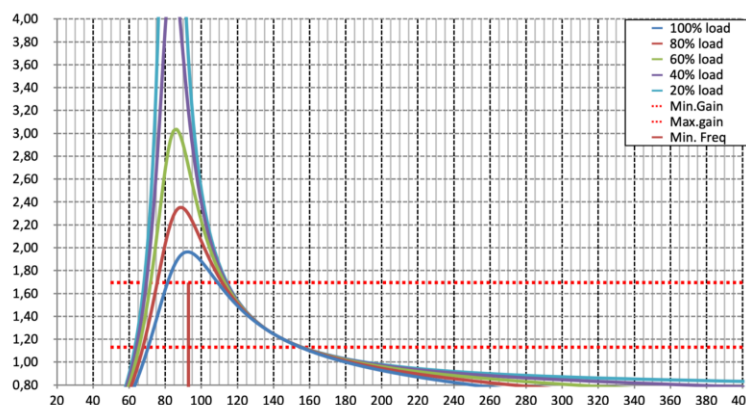


Figure 8: Resonate curve in frequency domain

GD1 and GD2 are connected to the INA high side and INB low side of the U1 gate driver, respectively, and grounded through GND. Additionally, disable and other unused pins on U1 are

grounded. VCCI, VDDA, and VDDB are connected to the same 12V input as U₃ to supply power to both the primary and secondary sides of the gate driver. OUTA and OUTB connect to the gates of M₁ and M₂. The MOSFETS M₁ and M₂ are connected in series and will alternate when they switch on and off based on the input clock signal. The node that connects M₁ and M₂ is defined as HB_out, which is sent to the primary side of the transformer.

Components used:

- UCC21520DWR – Gate Driver
 - 33ns propagation delay
 - 8kVpk isolation
- IMBG65R050M2HXTMA₁ – MOSFET
 - Very low switching losses
 - V_{gs} of 4.5V
- UCC25600DR
 - Programmable dead time
 - Programmable minimum switching frequency
 - 50/50 duty cycle
 - Over current protection

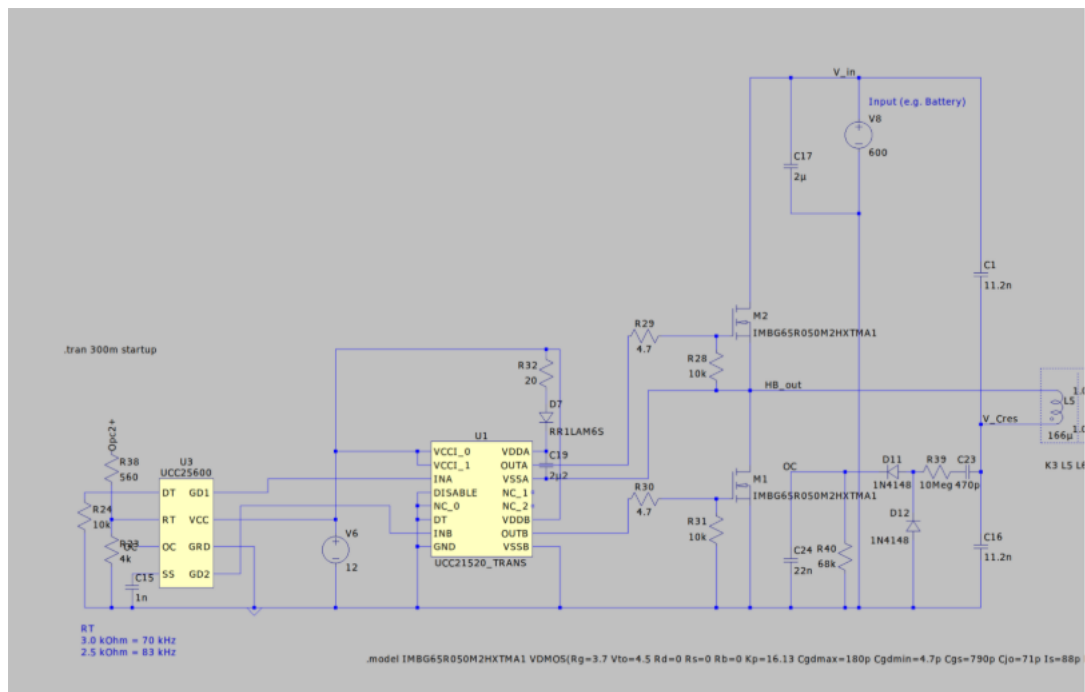


Figure 9: Primary Side

Secondary Side:

The secondary side is defined as the circuit that connects the transformer output, the secondary winding, to the load. The secondary side's primary function is to rectify the output, 12-volt AC signal, into a stable, smooth 12-volt DC signal. It does this by sending the 12V AC signal from the transformer to a rectifier, which turns it into a 12V noisy DC signal. That signal then goes through an output filter to provide a cleaned up 12V DC signal. Additionally, the secondary side contains a voltage-controlled feedback circuit to provide galvanically isolated, proportional feedback to the control circuit. To debrief further we will cover each circuit below in more detail:

Rectifier:

The first circuit on the secondary side is the center-tapped full wave synchronous rectifier. The purpose of this circuit is once again to convert 12-volt AC to a rough 12-volt DC signal. A standard center tapped full wave rectifier is shown below, where the center-tap serves as the output to the load (high-side) and the ends of the transformer are tied to ground via diodes (termed low side). These diodes serve to rectify the AC signal. However, the main issue with using diodes or passive rectification in our application is that we have a relatively low-voltage, large current output, so the voltage drops, and power dissipation of the diodes becomes a significant problem and the bottleneck of the DC-DC converter.

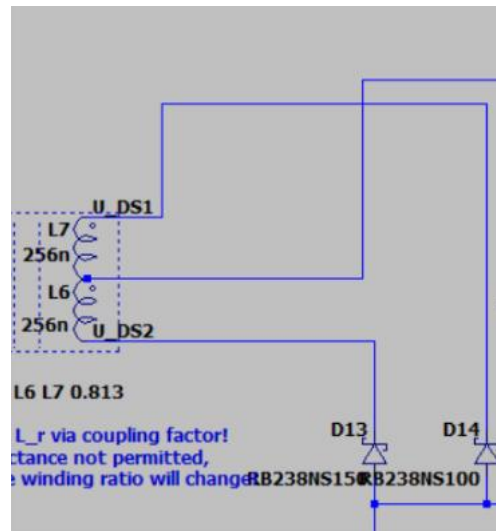


Figure 11: Center-Tapped Full Wave Passive Rectifier

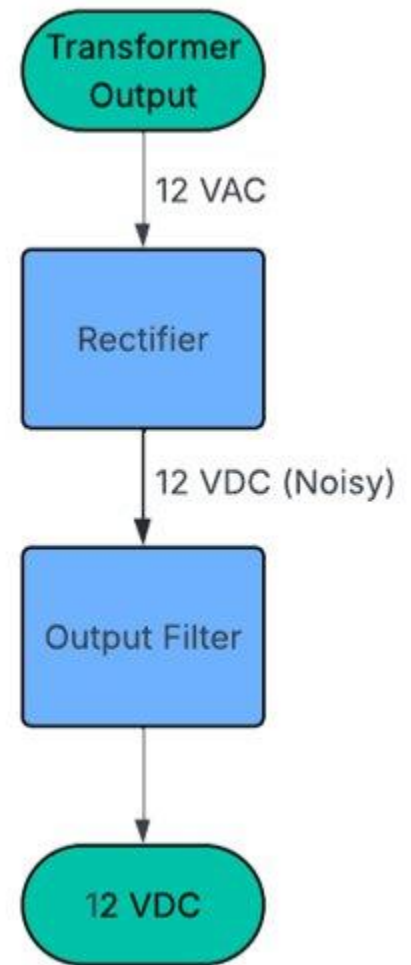


Figure 10: Block Diagram of Secondary Side

diode of the MOSFET is conducting (forward bias mode) and therefore, the MOSFET is switched on to provide a less resistive path and reduce conduction losses. When a voltage of nearly 0V is sensed at the drain, this means that the diode is now reverse-biased because the current is travelling in the opposite direction through the MOSFET's channel; therefore, the MOSFET is switched off to mock the reverse-bias behavior of a rectifier diode. Essentially, the MOSFET mimics the function of a diode with the added benefit of lower conduction loss.

Components:

- SR Controller: NCP4305DMTTWG
- MOSFET: BSC027No6LS5ATMA1
 - $R_{DS-ON} = 4 \text{ m}\Omega$
- C31 decoupling capacitor specified from datasheet
- R13 and C6 serve to properly trigger the CS pin (validated through testing)
- R42 sets minimum off-time period
- R38 sets the minimum on-time period

MOSFET Calculated Conduction Losses:

$$P_{Cond} = I_D^2 \times R_{ds-on} = 3.6 \text{ W}$$

Equation 1: MOSFET Conduction Losses

Output Filter:

The purpose of the output filter is to smooth the rectified, pulsating output waveform into a constant DC voltage/current, eliminating current and voltage ripple. This is done through a low-pass pi filter topology, with two shunt capacitors and a series of inductor. A pi filter is useful because it provides multiple stages of filtering. The first stage, which consists of five 15 μF capacitors in parallel (75 μF), provides a fast frequency response and filters out high-frequency noise/ringing. The next stage, the LC low-pass filter, contains a 0.8 μH inductor and two 2200 μF capacitors in parallel (4400 μF) to further smooth out the waveform to a constant DC voltage.

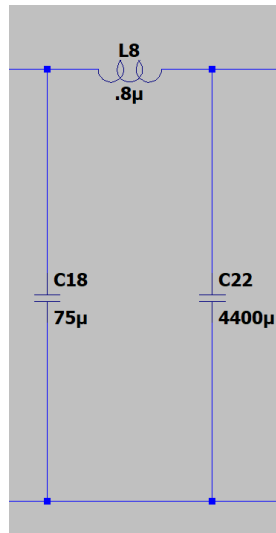


Figure 14: Output Filter Simulation

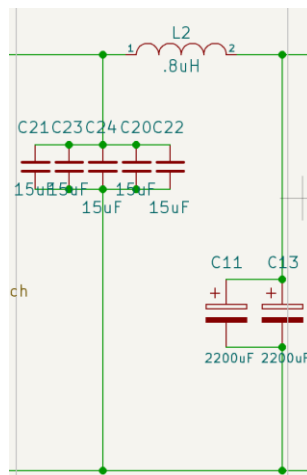


Figure 15: Output Filter Physical Implementation

When implemented into the system, we had a voltage ripple of around 7 mV and a current ripple of around 7 mA. The simulation results for the ripple current are shown below.

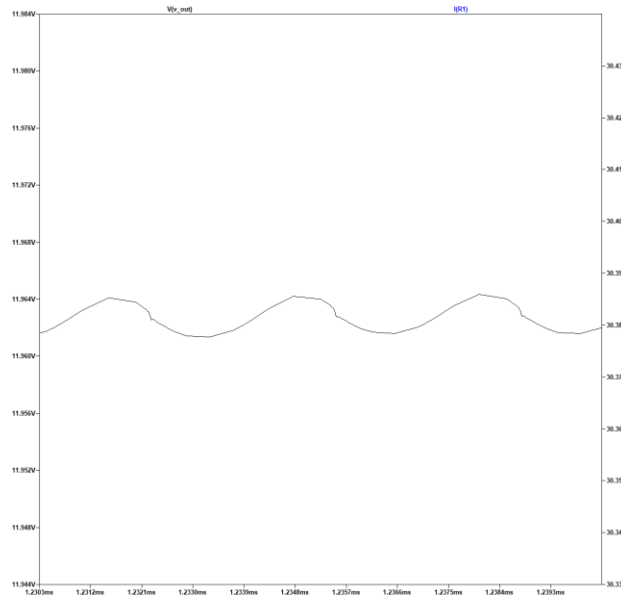


Figure 16: Voltage Ripple from SPICE Simulation

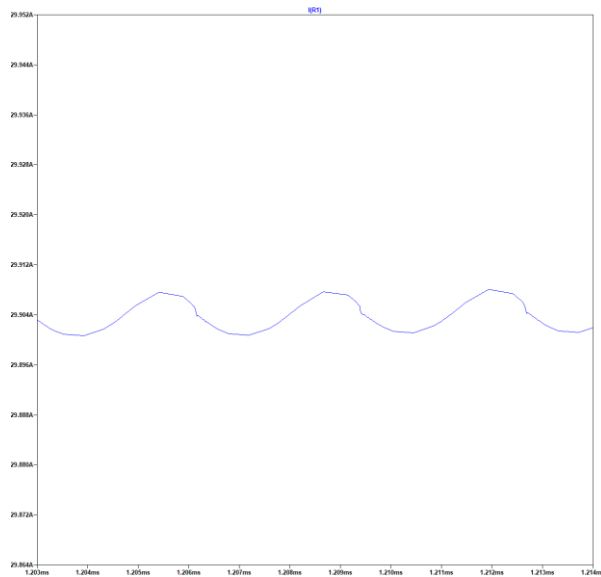


Figure 17: Current Ripple from SPICE Simulation

Transformer:

The purpose of the transformer is to step down the voltage from 600VAC to 12VAC and convert the current in the inverse ratio. This is done by using two ETD cores, with one gapped and the other not, and two separate wires wrapped around the ETD cores. In our case we used an ETD39 with a N87 material to reduce losses and maintain a small form factor. From here we used Remington Industries HF Litz wire 14awg (5X3/29/40) to make the coils; the size was chosen to allow proper amounts of current to flow without major cooling & for reduction of losses in the transformer windings. Lastly, an SLA printed bobbin will be used to encase the transformer to prevent windings and ETD cores from moving apart.

$$s = \frac{A_L^{\frac{1}{K_2}}}{K_1}$$

Equation 2: Transformer Air Gap Size Formula

$$A_L = \frac{L_p}{N^2} = \frac{166000 \text{ nH}}{28^2} = 211.73 \text{ nH}$$

Equation 3: Transformer Inductance Coefficient

$$s = \frac{211.73^{\frac{1}{-0.734}}}{196} = 0.90 \text{ mm}$$

Equation 4: Transformer Air Gap Calculation

The figure above shows the calculations used to determine the gap between UI cores to achieve a high enough magnetizing inductance, thereby avoiding the need for a series of inductors. The given values of L_p (magnetizing inductance) are from the Onsemi half bridge design spreadsheet, while K_1 and K_2 are from the datasheet of the UI cores. All this combined is used to calculate the center gap between the two UI cores.

The bobbin, as previously mentioned, is printed from temperature resistant SLA resin. The original bobbin design was an issue in construction because it had no guides for the LITZ wire resulting in massive deviations from our desired inductances. Due to these massive deviations, we added guides to a new revision of the bobbin to prevent the windings from moving during construction as shown in the figure below.

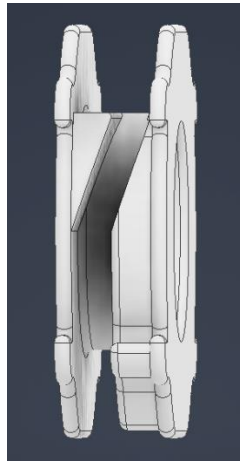


Figure 18: Final bobbin

The values that are required to have the transformer work efficiently with the rest of the circuit were given in the ONSEMI calculation excel sheet, the primary winding is meant to have a 166uH inductance value while the secondary windings (both) are meant to have a 56uH inductance value. By reaching these inductance values, the efficiency of the transformer stays high, the gain is minimized, and the leakage inductance is large enough to prevent the need for a power inductor. The windings, because of the transformer's tuning, are then glued to the bobbin to hold their desired inductance values. Likewise, the bobbin, once inserted onto the cores, is glued to prevent vibration and further variation in the inductance values. The transformer is then able to be placed on the board; however, the insulation used on the LITZ wire must be removed to conduct properly.

Power Supply:

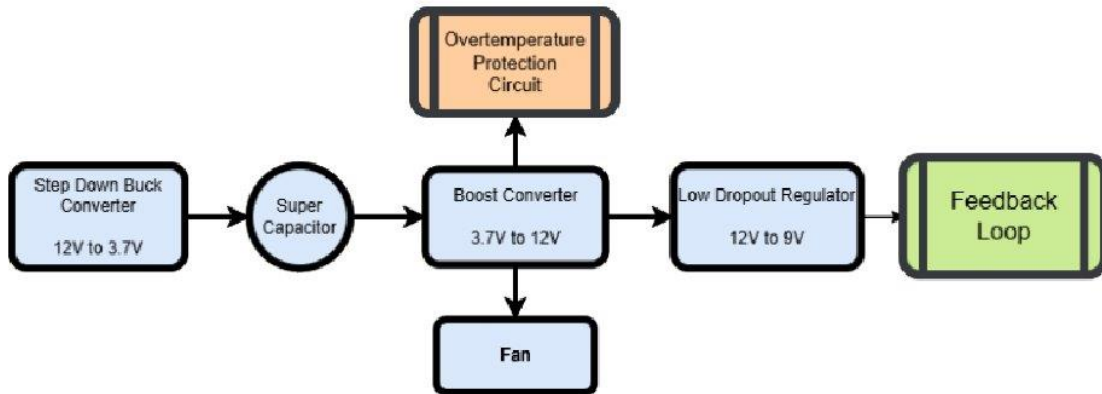


Figure 19: Block Diagram of Power Supply

The purpose of the power supply is to power the overtemperature system, help with the fan voltage select, and power the feedback loop. The power supply starts with taking in the stepped down rectified voltage from the transformer. This voltage then gets stepped down via U₃, a high efficiency step down converter to 3.7 Volts to charge the super capacitor C_{start1}. This is achieved by adjusting the voltage from the feedback pin via a voltage divider. The super capacitor can discharge the system startup for 100 seconds. It also acts as a battery as it takes the voltage from the feedback loop to keep the system running. U₃ is an MP4541_SOP8-EP, and we are using its common configuration provided by the datasheet.

FB Voltage: 1.0 V

$$V_{out} = 1.0 \left(1 + \frac{27000}{10000} \right) = 3.7V$$

Equation 5: U₃ Common Configuration Equation

Q₇ and D₈ are used for reversing current blocking. When Q₇ is off, the inductor releases a kicked-up voltage then is rectified by D₉. C₆₄ C₈ Stabilize voltage and provide instant current during switching. U₄ is a boost converter that uses the super capacitor voltage to boost the voltage back up to 12V. This 12V is used for port 1 of the fan select and it is used to power the overtemperature system. C₁₇ C₁₈ provides smoother 12V. U₂ is a low dropout voltage regulator; it takes the boosted 12V and converts it to a fixed 9V. This 9V is used to power the feedback loop.

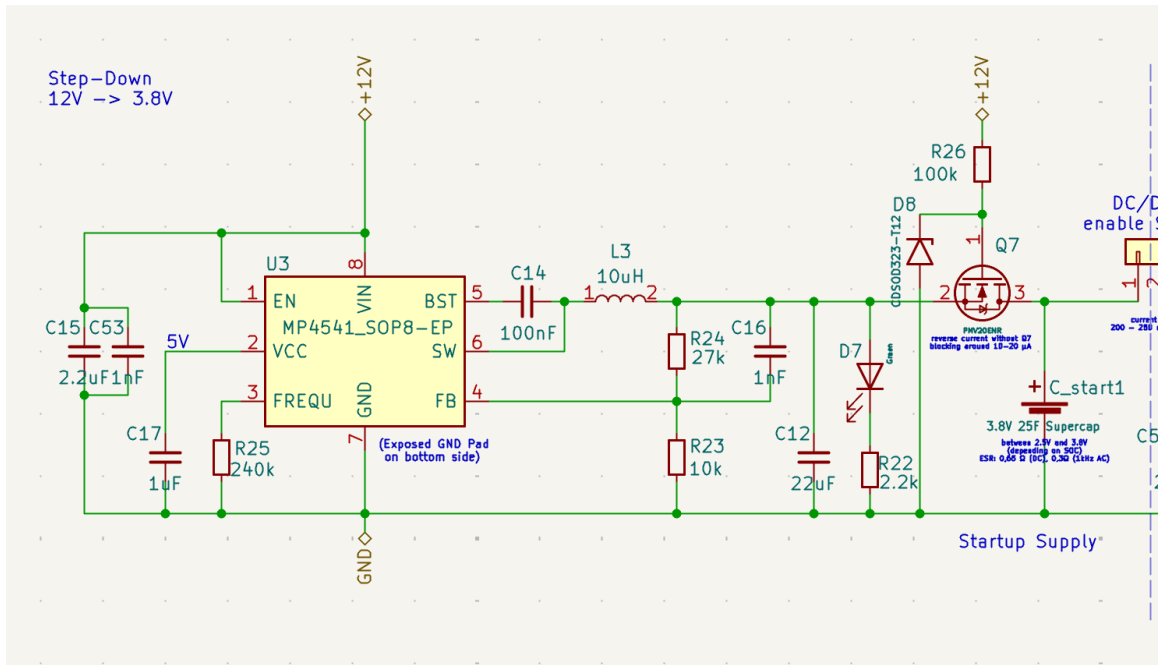


Figure 20: First Section of Power Supply

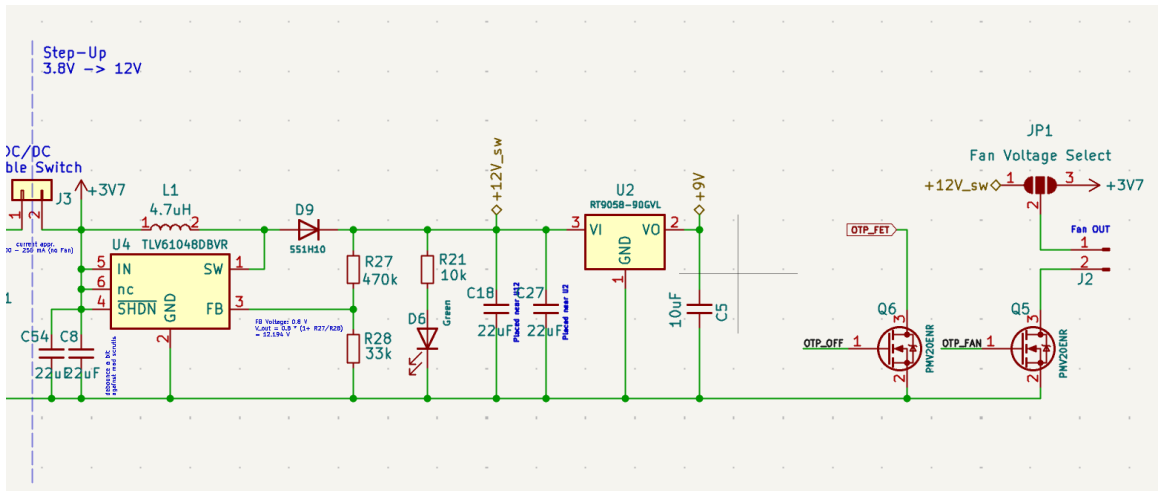


Figure 21: Second Section of Power Supply

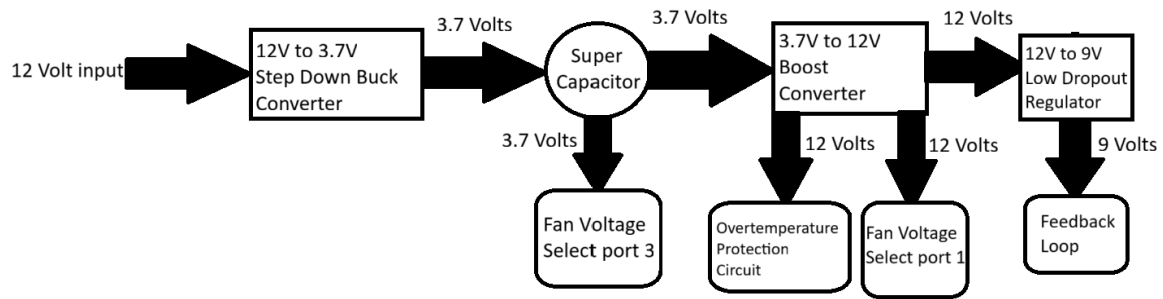


Figure 22: Power Supply Block Diagram

Overtemperature Protection:

The Overtemperature Protection system shuts down the device when the temperature surpasses the safe operating limit to prevent overheating. The Overtemperature Protection and Fan control are comprised of 2 op-amps. The Overtemperature control op-amp V_{in+} is connected to a D10 diode that is connected to the op-amp output for latching. The V_{in-} is connected to a TH2, which uses the diode to measure the maximum temperature based on the parallel combination of the $R_{36} = 4.7K$ resistor and parallel resistor. The TH2 impedance changes based on the combined parallel resistance and is converted to a temperature value measured in Celsius. TH2 is connected to the 2 parallel resistors. Right now, for R_{35} (330K) in parallel to the R_{36} , the TH2 has a resistance of 1404 ohms, which is a temperature of 113.5 Celsius, which is measured as 106.9 Celsius due to noise on the op-amp signals.

The fan control circuit regulates the fan speed by adjusting the current or voltage following it. When the fan op-amp is on, the fan receives 5.9466 V which converts to 4784 Ohms which converts to 77 degrees Celsius. When the fan op-amp is off, the fan receives 6.0533 V which converts to 4617 Ohms which converts to 68 degrees Celsius.

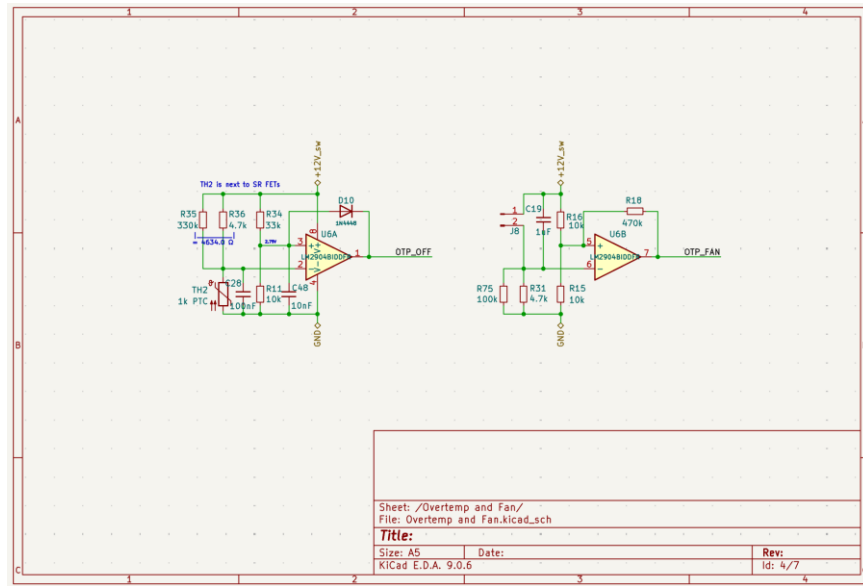


Figure 23: Overtemperature Protection Circuit

4.3.3 Functionality

The 600V LLC converter is to be used in conjunction with low-load electric vehicles. In a low-load EV scenario, it is expected that the LLC converter could be plugged into an existing system (seen in figure 4) and convert the inverter-supply voltage down to a usable 12 volts for low-voltage electronics. The LLC converter would be able to power all electronics in a low-load scenario, and the user would simply need to activate an enabled button to turn it on.

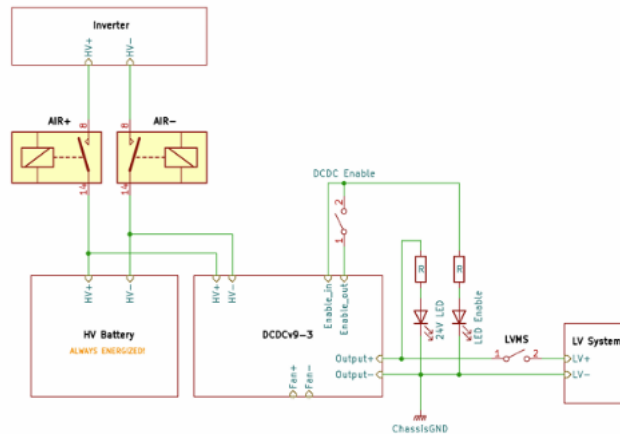


Figure 24: 600VDC LLC Converter Connection to Car

4.3.4 Areas of Challenge

All of our challenges in this project were in regard to technical progress and satisfying requirements. We were very successful in meeting user needs. The largest challenge that we ran into when designing our converter was concerning issues regarding the PCB print. Our first issue was that there was an extra pad included for a resistor that we had removed from our design. This was creating a break in our circuit which was hurting the results. To rectify this issue, we simply made a bridge between the pads with solder. Upon testing it after solving the issue, we found that the circuit was working as expected. Our next major issue was regarding a fuse that we had forgotten to add to the PCB footprint. To solve this problem, we ended up soldering it between two separate components that it should be connected to. We tested this solution to ensure it worked and found that the circuit functioned as expected. The last major issue that we ran into was regarding our transformer's inductance. We were having a very hard time tuning it to get the correct leakage inductance. To solve this problem, we reached out to our faculty mentor for advice and ended up finding that by shortening the leads we were able to solve the issue.

4.4. Technology Considerations

There are two distinct technologies used in the LLC converter. The first is the type of MOSFETs used in the switching network that converts the HV DC to HV AC. The types of materials used in the MOSFETs can alter the efficiency, heat output, activation voltage, switching time, among other factors. The MOSFETs used in the design are standard Silicon based MOSFETs that use dopants such as Boron or Phosphorus. However other MOSFETs can be used that utilize more cutting-edge materials such as Gallium Nitride (GaN) or Silicon Carbide (SiC) which offer benefits in these key areas, however they can be more challenging and expensive to implement compared to standard Si MOSFETs. For the design's first implementation, standard Si MOSFETs will be used to verify functionality with later revisions possibly adding GaN MOSFETs if time and budget allow it.

The other technology is in the rectification of LV AC to LV DC. There are two possible solutions, either using passive rectification utilizing diodes or using active rectification which utilizes MOSFETs that operate as a diode. Passive rectification has some advantages such as its simplicity to implement. It can have some drawbacks such as it can be less efficient and can break down and no longer function if the specifications don't meet the requirements. Active rectification on the other hand is more complex to implement requiring additional circuitry to control the MOSFETs but tend to handle more current and have overall better efficiency, but thermals are still a concern especially with the expected output current of 30A so cooling will be an additional consideration. With our design, active rectification is planned to be used due to its overall better efficiency and the lack of diodes that are available within our needed specifications. However, with active rectification, additional consideration will be on how to cool the MOSFETs as keeping the FETs cool will allow for more current to pass through them. If the thermal management is inadequate, then the converter will not operate as it is designed to.

4.5. Assembly

4.5.1 PCB

After designing the converter, we ordered the board through JLC PCB with assembly of most of the SMD components. However, there were still numerous components that we had to manually solder ourselves. Note that we found several issues with the quality of JLC assembly as a few solder joints were faulty and even some components had incorrect footprints. Figure 22 below shows our disassembled board while Figure 23 showcases our assembled board without the transformer.



Figure 25: Unassembled Board



Figure 26: Assembled Board

4.5.2 Transformer

Concurrent with the assembly of the PCB, the transformer was being assembled and tested. The transformer was incredibly finicky and highly sensitive – simply touching the windings with bare hands caused an 11% difference in inductance values. Due to the transformer being so sensitive and having a litany of parameters attached to proper assembly, it took significantly longer than expected.

Twenty eight turns were applied in clockwise order to the primary side of the bobbin, where there was two wires wrapped once clockwise to the secondary side of the bobbin. The windings were then glued to the bobbin and the transformer was then glued together with clamping force applied.

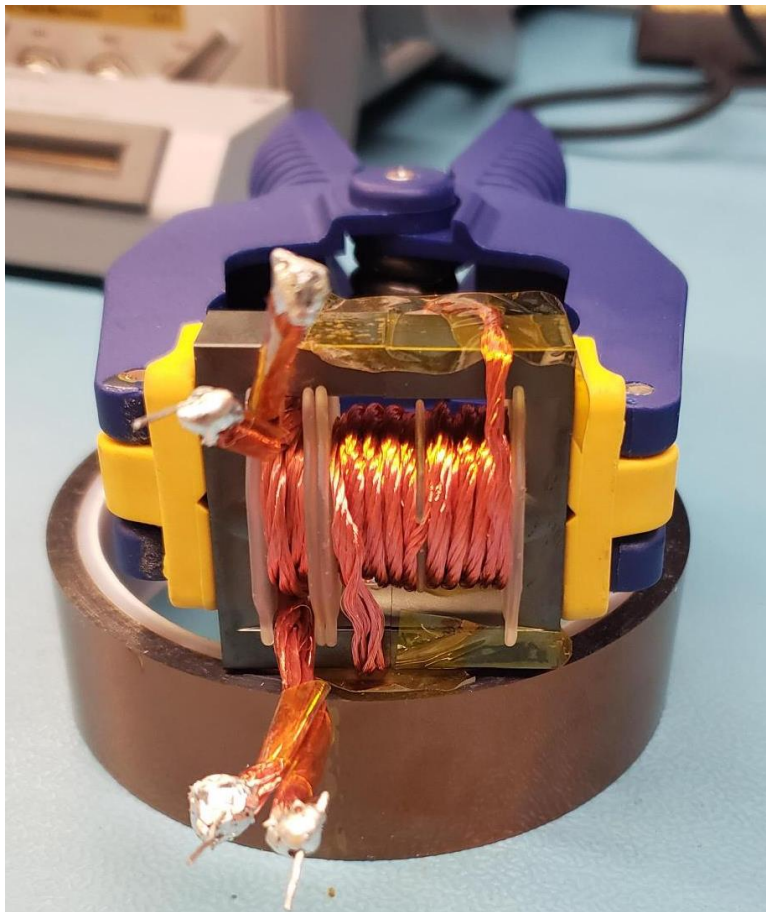


Figure 27: Transformer before being glued

4.5.3 Completed Design

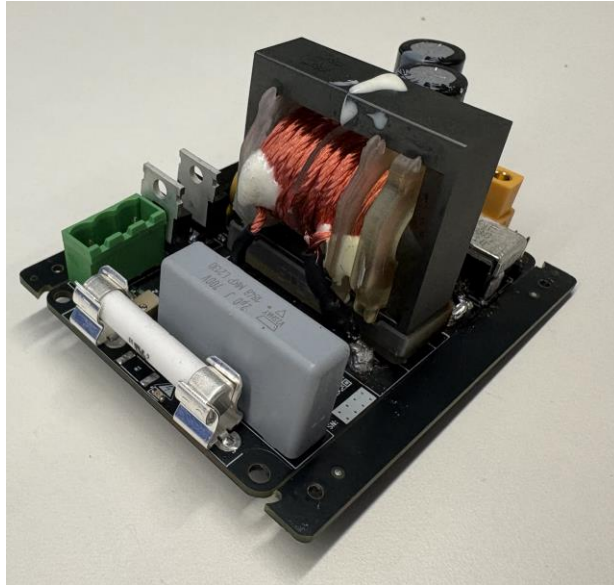


Figure 28: Primary side Isometric View of Completed LLC Converter

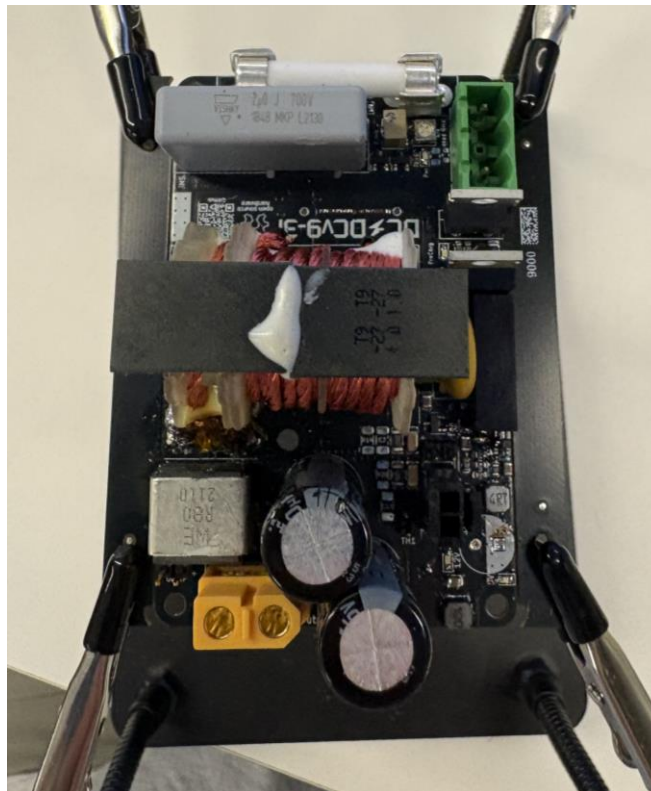


Figure 29: Top View of Completed LLC Converter

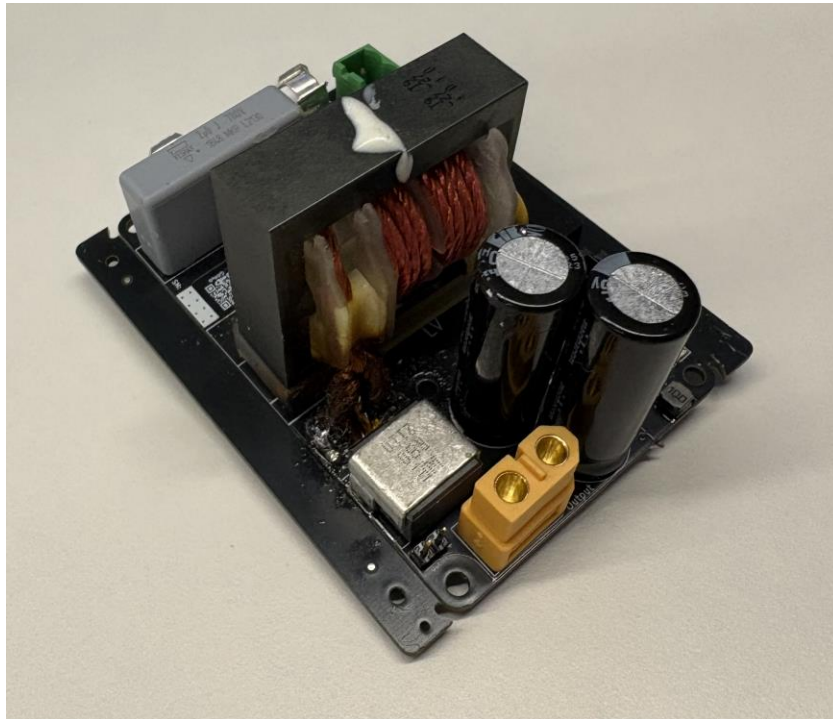


Figure 30: Secondary Side Isometric View of Completed LLC Converter

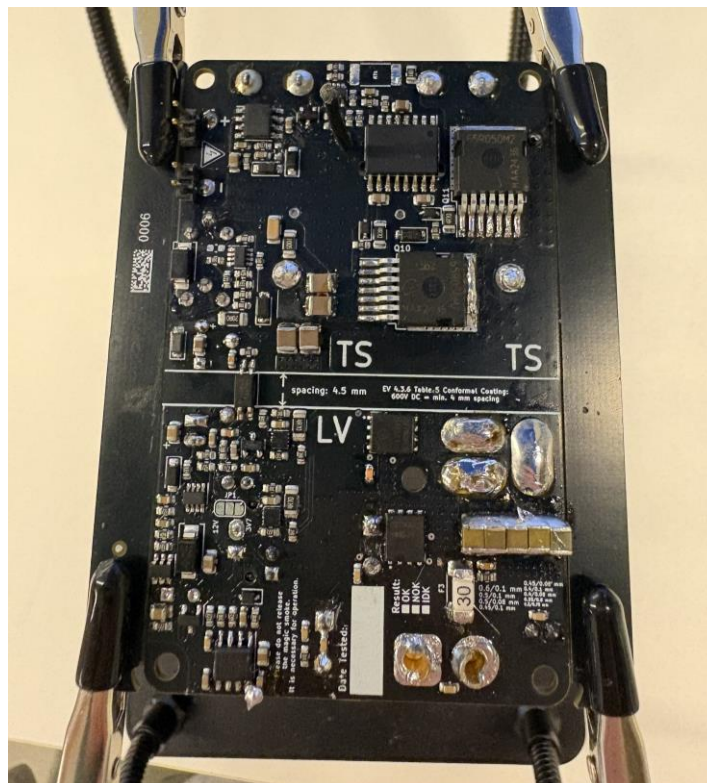


Figure 31: Bottom View of Completed LLC Converter

5. Simulation

5.1. Basic Passive and Active Load Tests

Before we assembled the board we created an LTSpice simulation to verify the design and functionality of the converter the full SPICE simulation Model can be seen below.

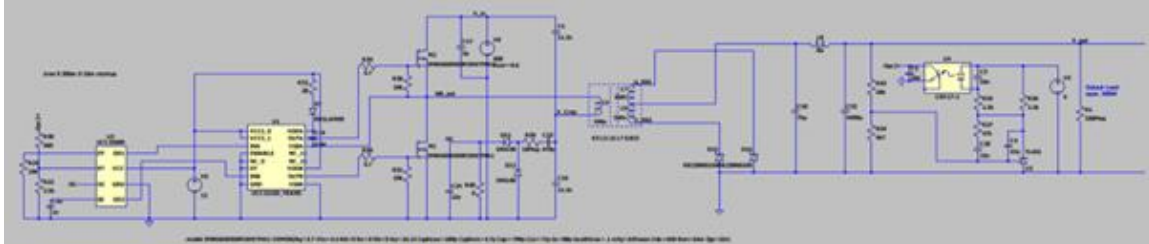


Figure 32: SPICE Simulation Model

The results displayed below showcase the converter's response to several passive loads. We first simulated the converter with a 144 Watt passive load and then a 360 Watt passive load (the maximum load that we have rated the converter for). Notice that simulated responses to the passive loads appear almost identical for both, stabilizing after about .5 milliseconds with marginal overshoot and settling time. The 144-watt passive load settles at about 11.98 volts, while the 360-watt passive load settles at about 11.95 volts.

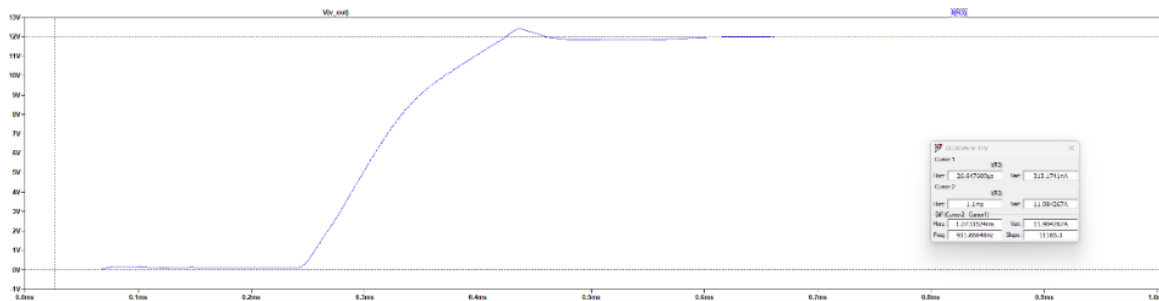


Figure 33: Results Using 600 Vin and 144 Watt Passive Load (1 ohm resistor)

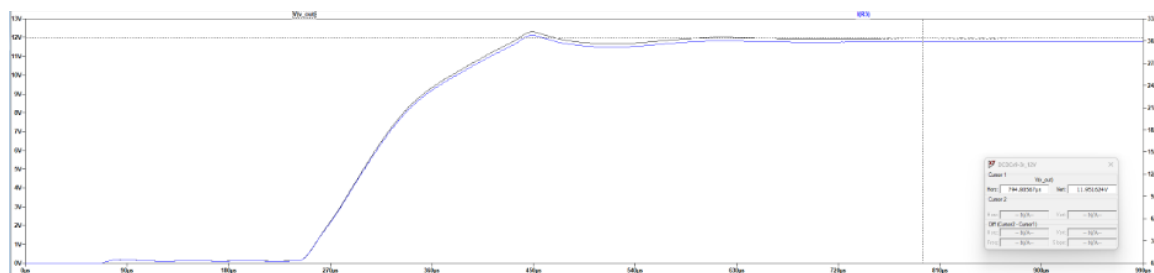


Figure 34: Results Using 600 Vin and 360 Watt Passive Load (.4 Ohms Resistor)

Next, we then tested the output line regulation. We did this by using a Vin of 430 volts, the minimum possible voltage of the tractive system battery at its lowest state of charge. As shown in the results, the output voltage ranges from 11.86 to 11.95 volts, within the required tolerance.

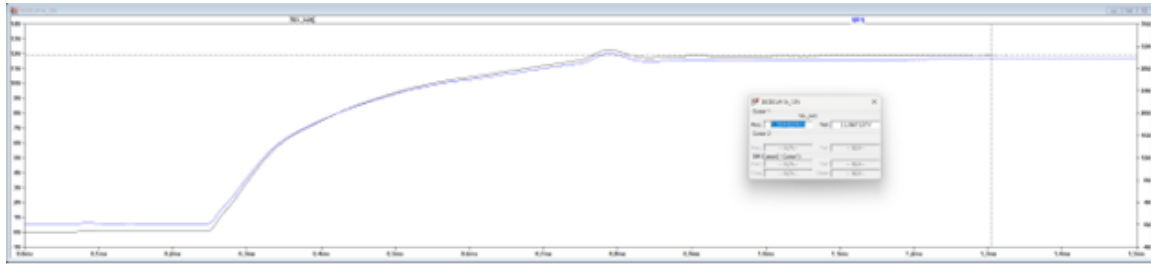


Figure 35: Results Using 430 Vin and 360 Watt Passive Load

The next set of results shows the converters' step response to an active load that pulses from 11 to 11.8 volts. As shown in the graph below, the converter responds well within a millisecond. However, it should be noted that Vin has a strong impact on the system's response, as it responds more effectively at lower input voltages, as you can see from the contrast between the two figures below. Note that there is no standard that our client has required us to hit for response times to loads, but it is good practice to ensure the response time is reasonable (within a few milliseconds at most). This is also a metric that we can improve in future iterations as our client continues to develop their EV.

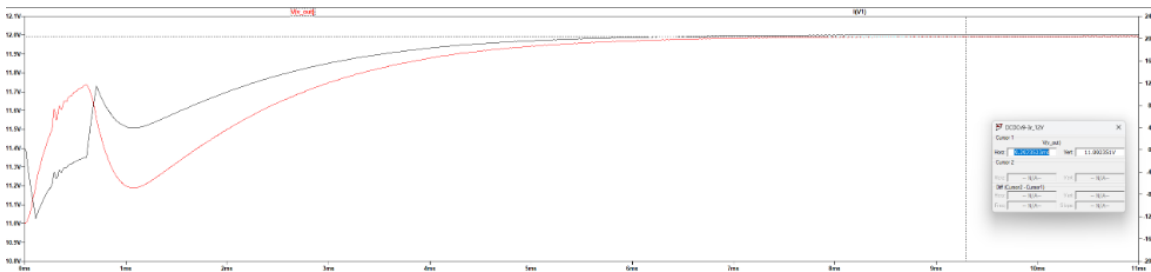


Figure 36: Results Using 240 Watt Active Load Step (Voltage Source with 48 mohm series resistance) Vin = 600 volts

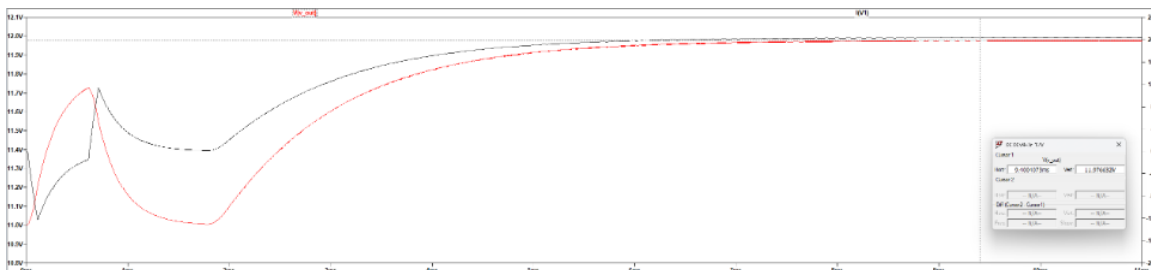


Figure 37: Results Using 240 Watt Active Load Step (Voltage Source with 48 mohm series resistance) Vin = 400 volts

5.2. Transient Response

After confirming that the overall converter simulation was functioning correctly, we decided to conduct more rigorous testing. To obtain a more accurate estimate of the system's response time, we simulated its transient response using a more elaborate test bench. As shown in the figure below, we created a dynamic load switching network using an array of voltage-controlled switches. We were then able to create a waveform to simulate load stepping, where the largest load is 576 watts, the intermediate load is 384 watts, and the smallest load is 192 watts. This allowed us to test the transient response of the converter just above its maximum load at 360 watts. Note that the input voltage was set to 600 volts, the maximum voltage, for each of these simulations.

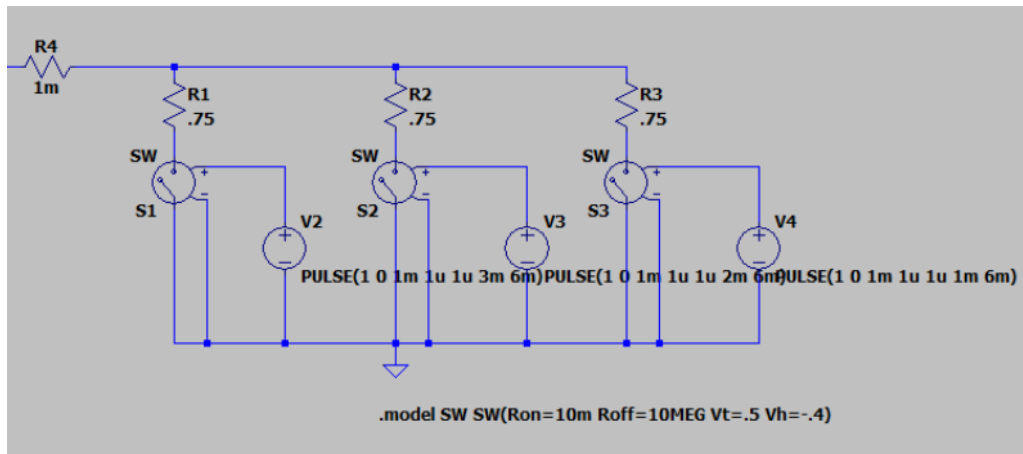


Figure 38: Dynamic Load Transient Response Testbench

We then measured the output voltage and current to obtain transient response performance metrics of the converter. The first figure below is the load step-down transient response of the converter. Note that the current waveforms in the figures for step-down are negative due to the way LTSPICE interprets the direction, but this was corrected for the step-up figures.

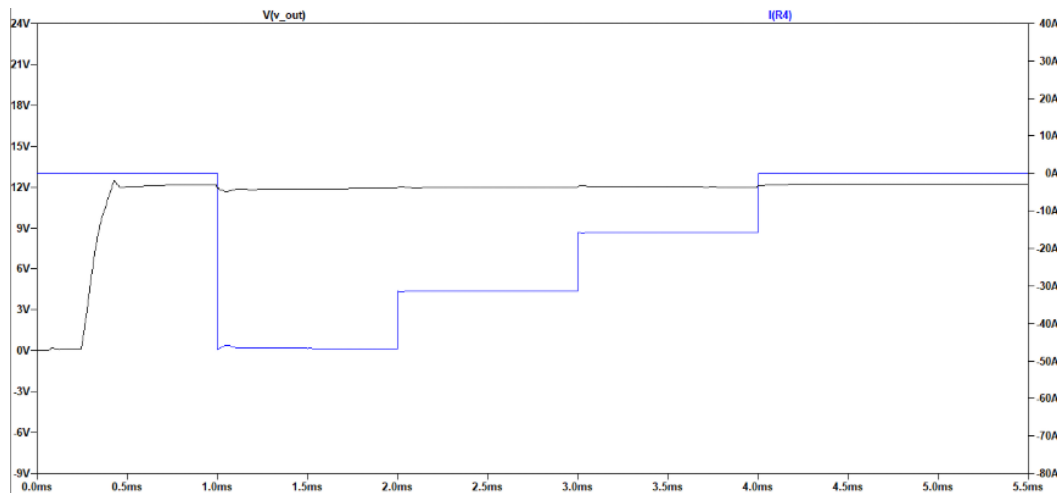


Figure 39: Step-Down Transient Response

Looking closer at the current waveform below, the second load step, where it reaches 384 watts, is close to the maximum output current of the converter. At this point, we can observe a small amount of oscillation in the current. However, this is practically negligible as this is only an overshoot of about .2%, well within the 2% settling criterion standard.

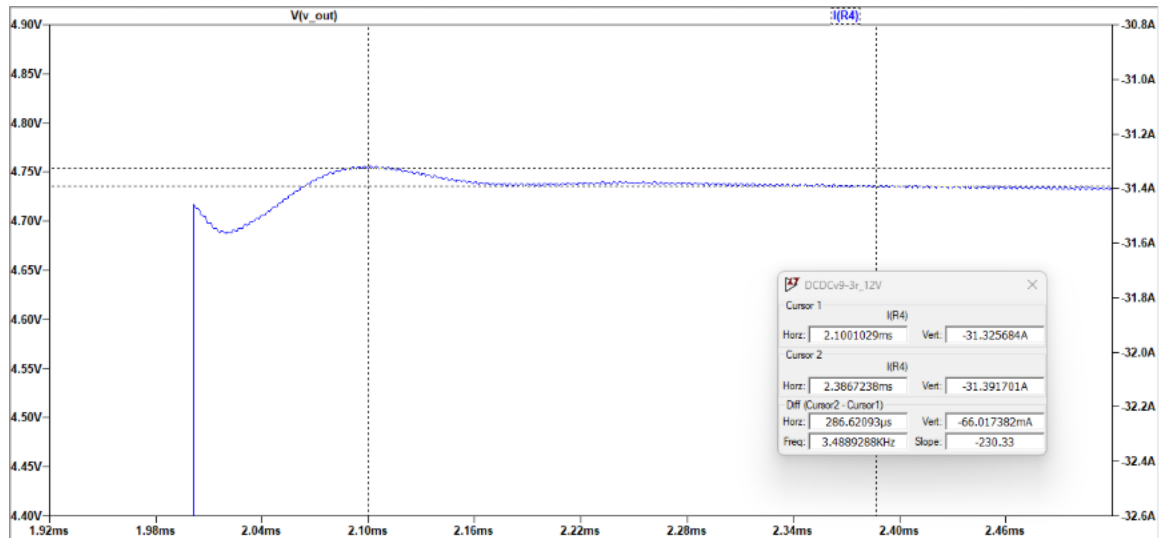


Figure 40: Current Transient Response, 384 Watts, Load Step Down

Looking closer at the voltage waveform, we once again see a very strong transient response to the change in load current. The max overshoot of the voltage is fractions of a percent, as shown below.

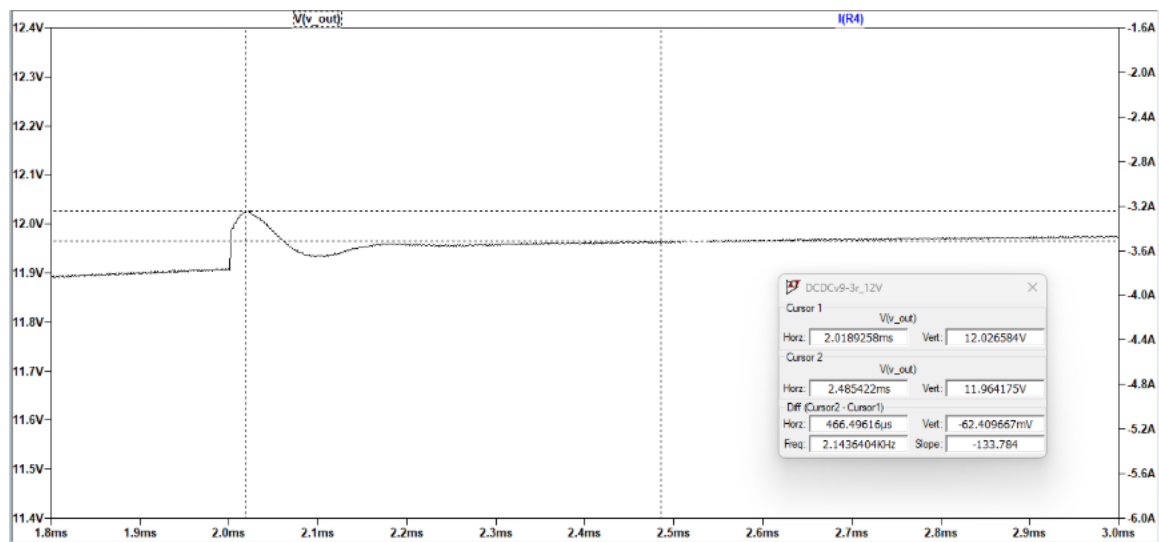


Figure 41: Voltage Transient Response, 384 Watts, Step Down

We then performed the same tests, but this time stepping up the load rather than stepping down. We received about the same metrics overall in the responses near the 384 watts transition. However, we also calculated the voltage droop of the converter. At the second step, the voltage noticeably dropped to 11.83 volts, a droop of around 170 millivolts. In addition, we calculated a slew rate of about 47 A/us. In conclusion, the converter has an appropriate overall transient response.

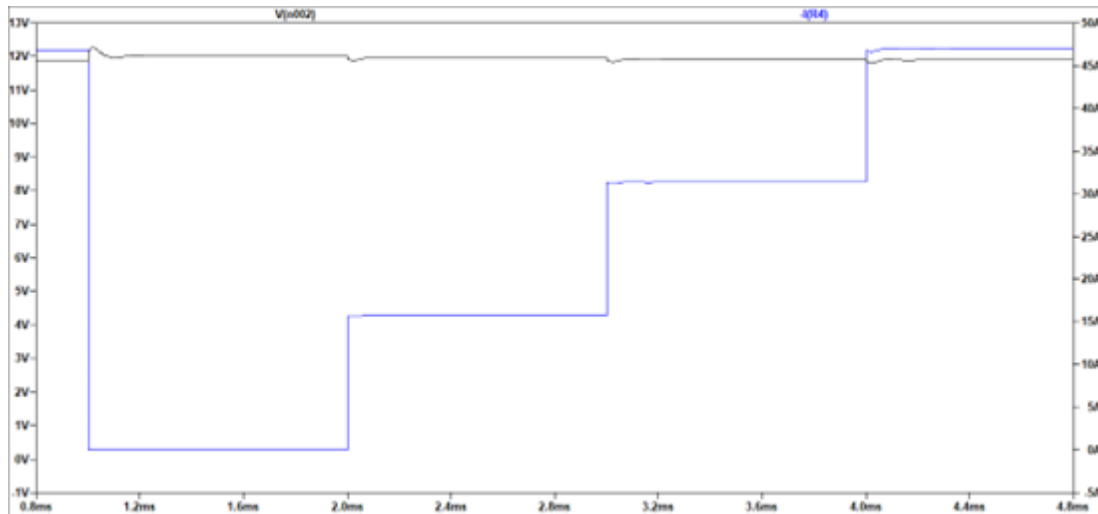


Figure 42: Step-up Transient Response

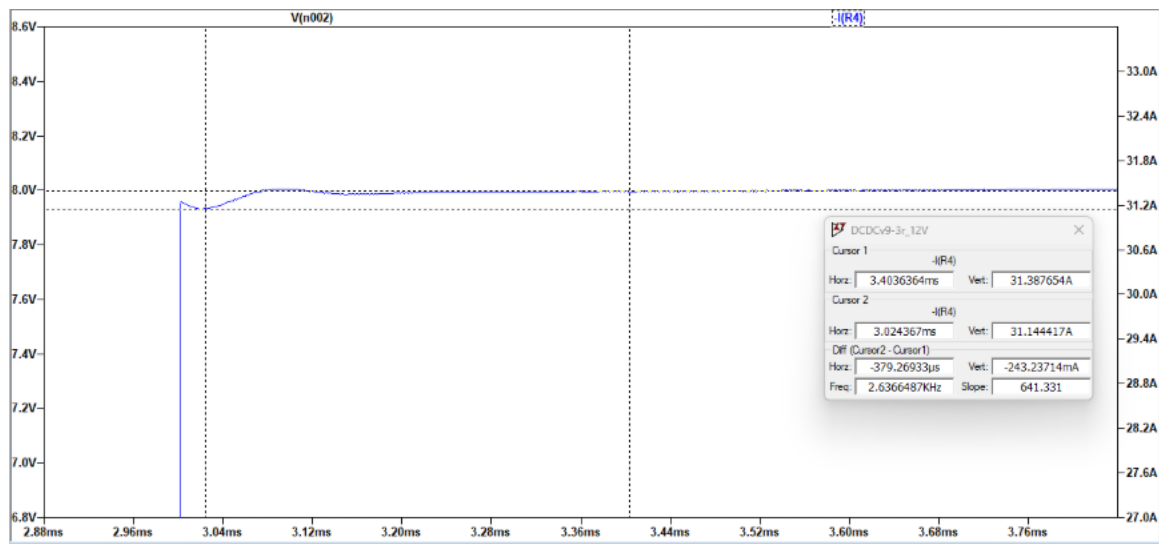


Figure 43: Current Transient Response, 384 Watts, Step-up

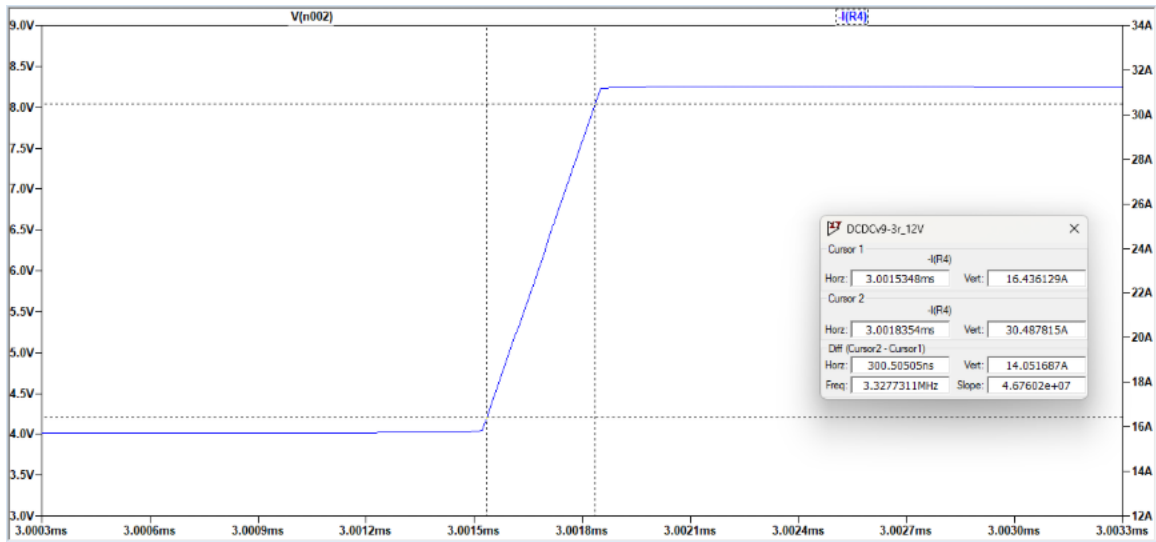


Figure 44: Current Slew Rate (47 A/us)

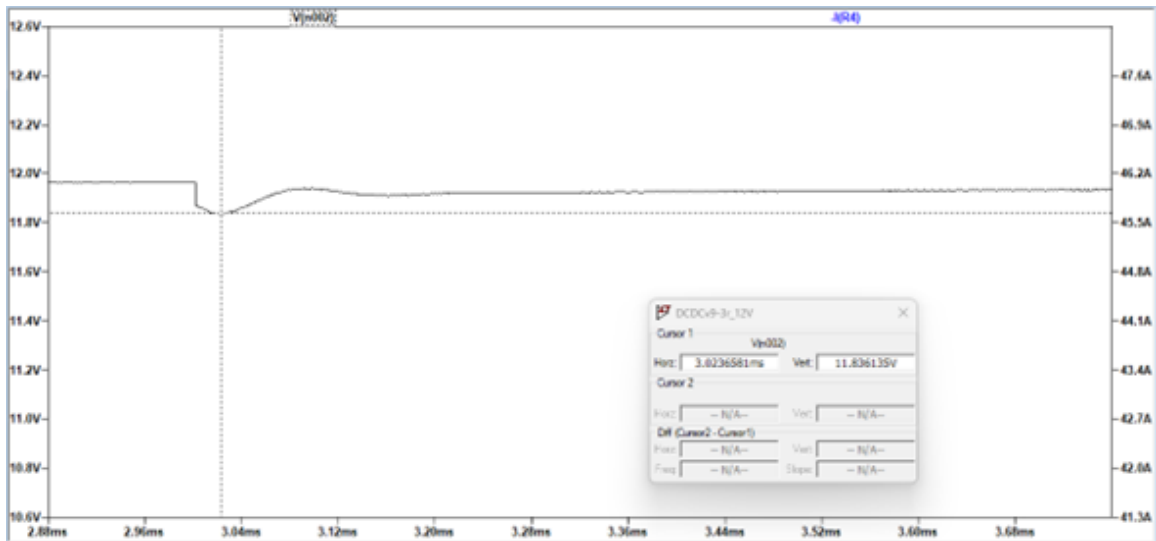


Figure 45: Voltage Droop at Transient, 384 watts

5.3. Power Dissipation/Losses

In the next test, we wanted to obtain the efficiency of the system. However, in order to get a better estimate of the rectifier losses, we created a diode model to simulate the conduction losses of the synchronous rectifier MOSFET. This was done by creating an ideal diode with an RDS-on of approximately 4 milliohms, as shown in the figure below.

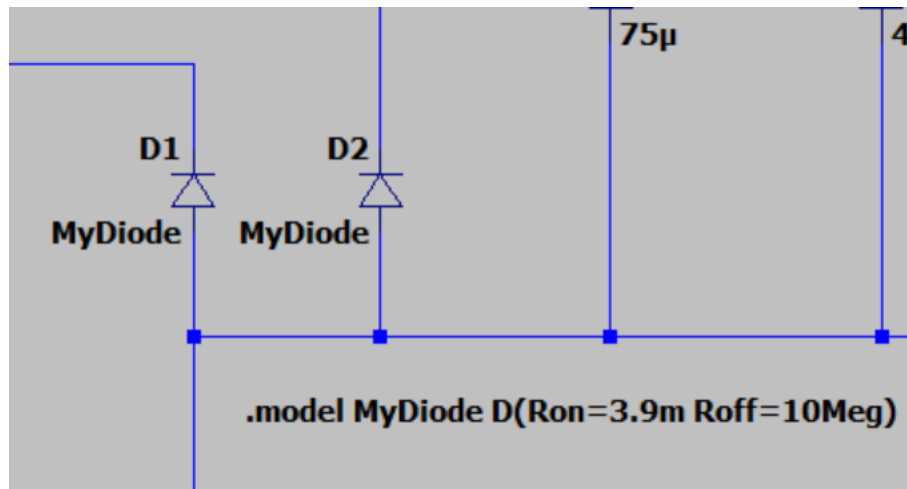


Figure 46: Synchronous Rectifier Conduction Loss Diode Model

This was done by measuring the average input power of the system and comparing it to the output power. The figure below showcases the instantaneous input power (blue) and output power (black). We then took the average of each waveform as shown below. The average input power was found to be 368 watts, while the average output power was 359.6 watts. Therefore, the total power dissipated throughout the system was 8.4 watts, and the converter's efficiency is 97.7%. However, this is once again an ideal model, so we expect to see much less efficiency during physical testing.

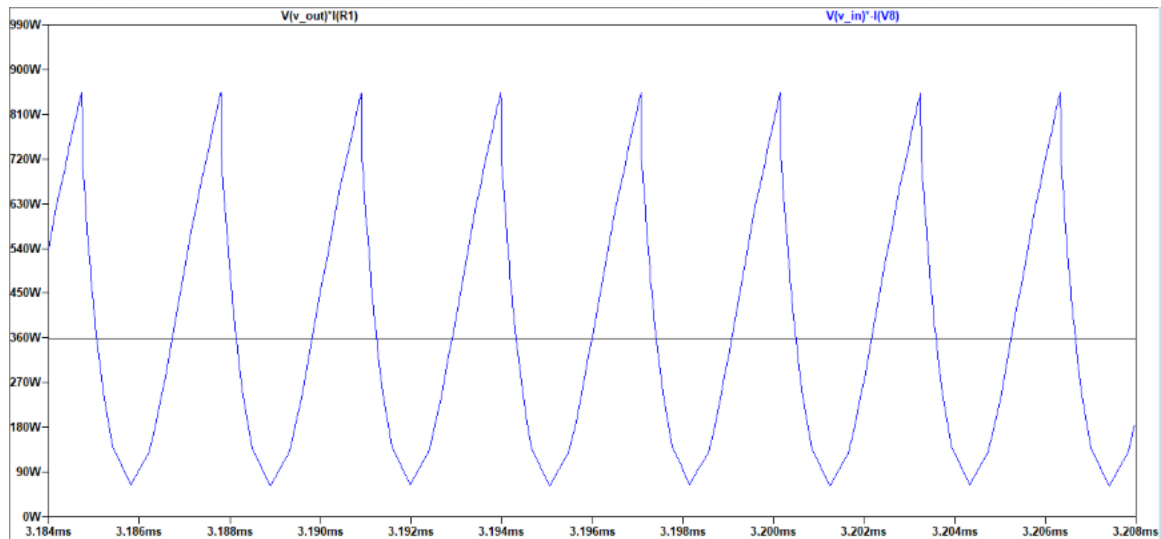


Figure 47: Instantaneous Input and Output Power

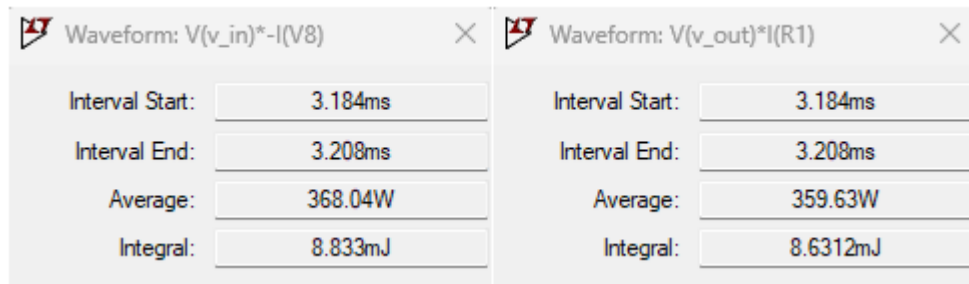


Figure 48: Average Input (Left) and Output Power (Right)

Finally, we decided to simulate the losses across individual components to investigate the major sources of power dissipation. This also allowed us to evaluate whether each component could tolerate the power dissipation. The following figures showcase the power dissipation across each individual component in the simulation: the half-bridge diodes, rectifier diodes, and output filter components. Note that since the transformer model did not account for core losses, we did not measure any power dissipation across the transformer. Overall, each component saw a tolerable amount of power dissipation.

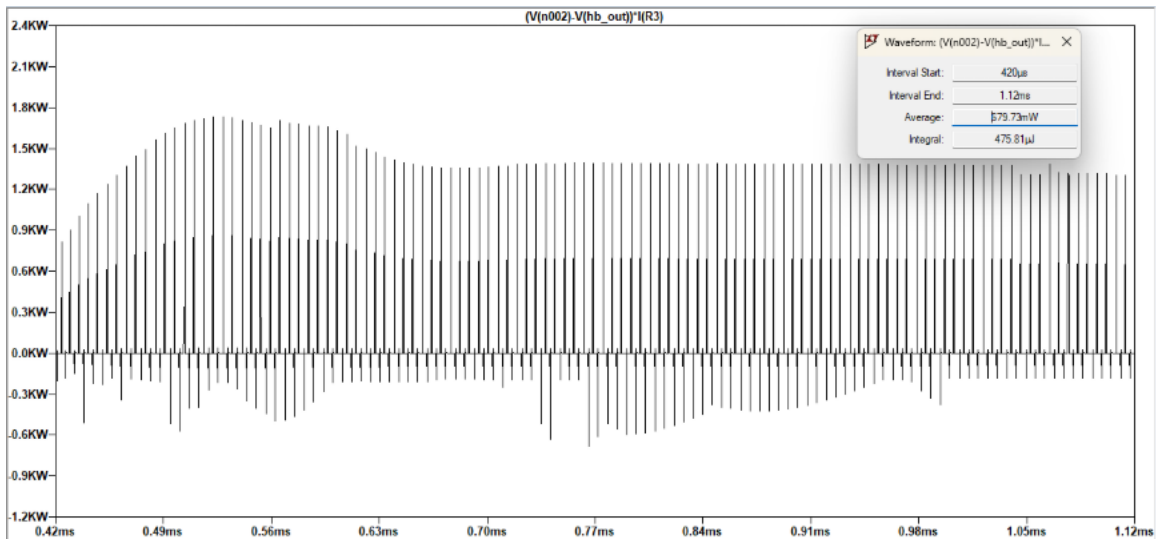


Figure 49: Conduction Losses Across M2, Pull-Up Half Bridge N-Channel MOSFET (679 mW)

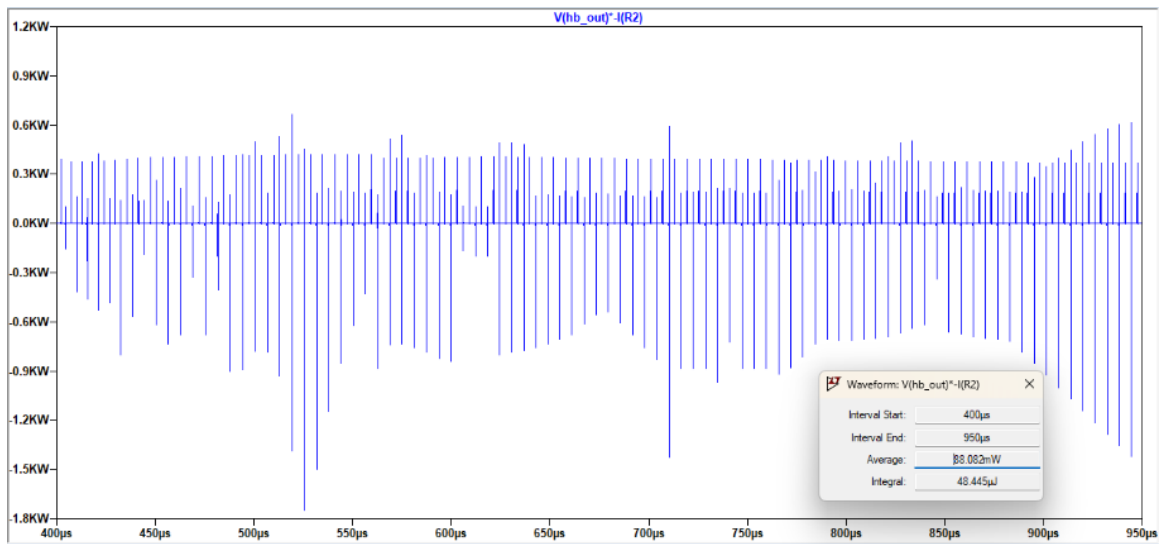


Figure 50: Conduction Losses Across M₁, Pull-Down Half Bridge N-Channel MOSFET (88 mW)

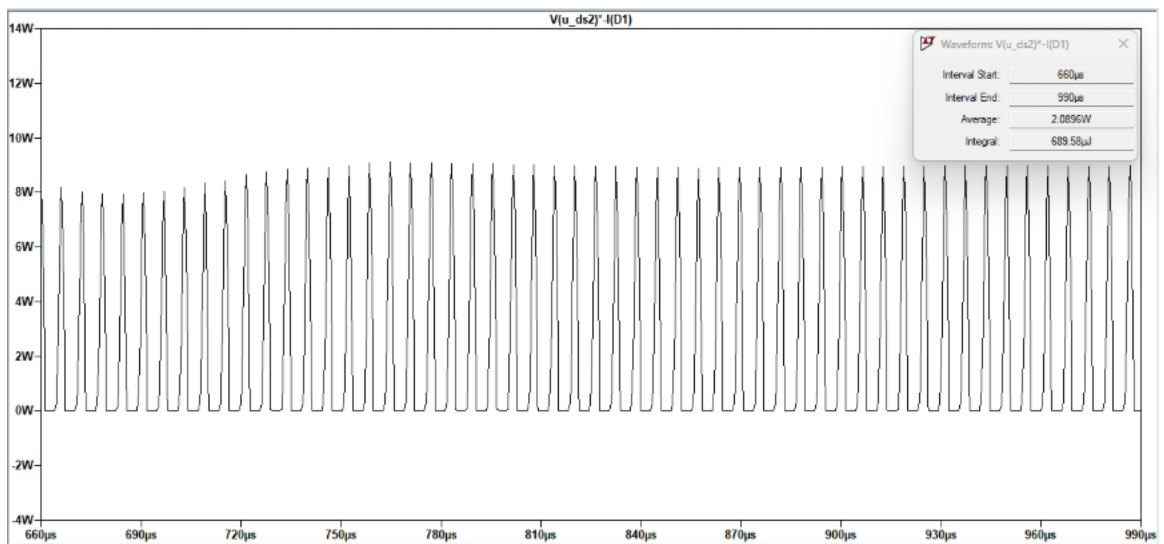


Figure 51: D₁ Synchronous Rectifier Diode Power Dissipation (2.08 W)

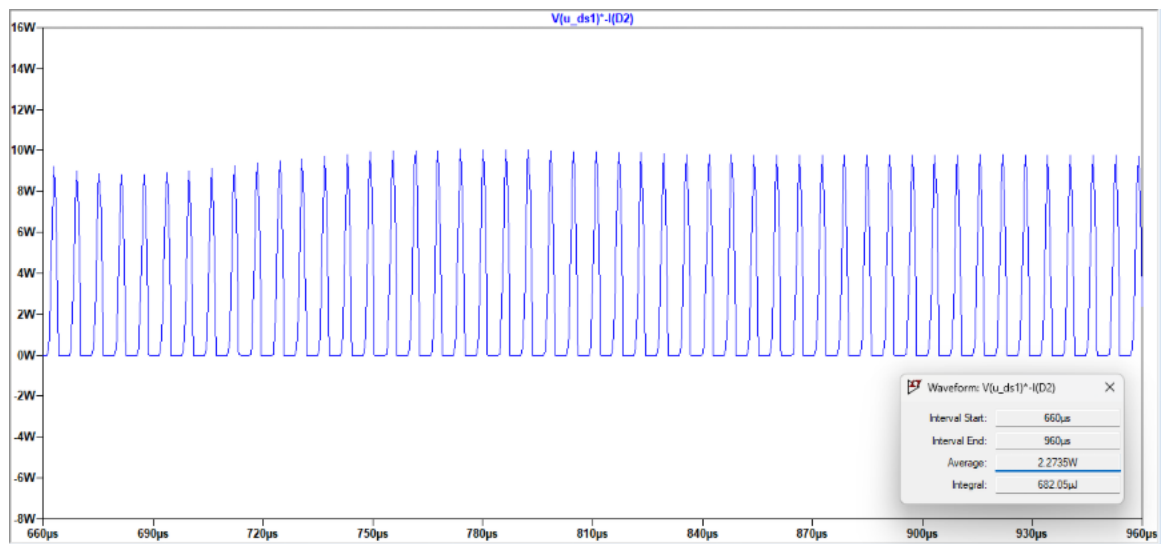


Figure 52: D2 Synchronous Rectifier Diode Power Dissipation (2.27 W)

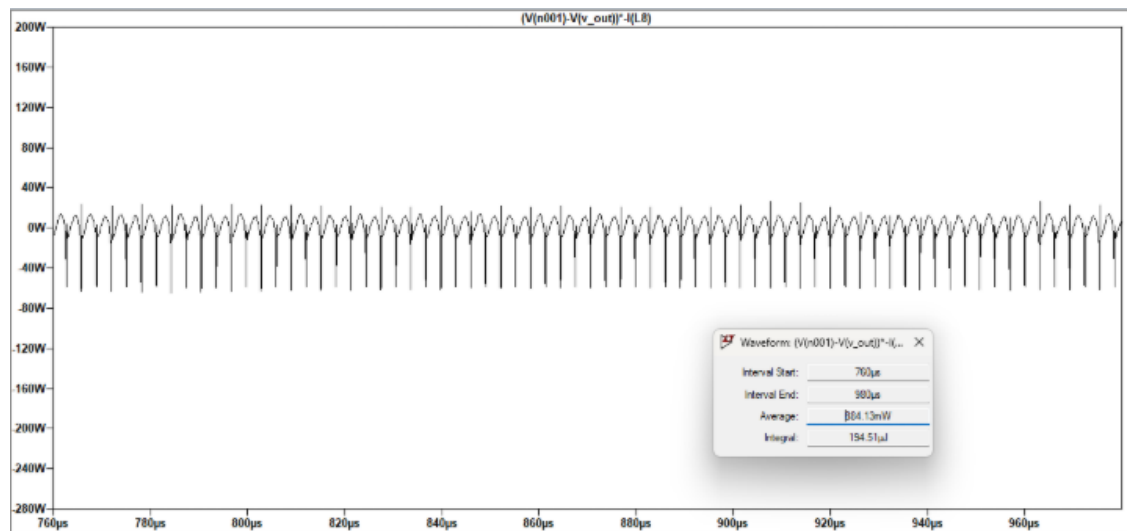


Figure 53: L8 Output Filter Inductor Power Dissipation (884mW)

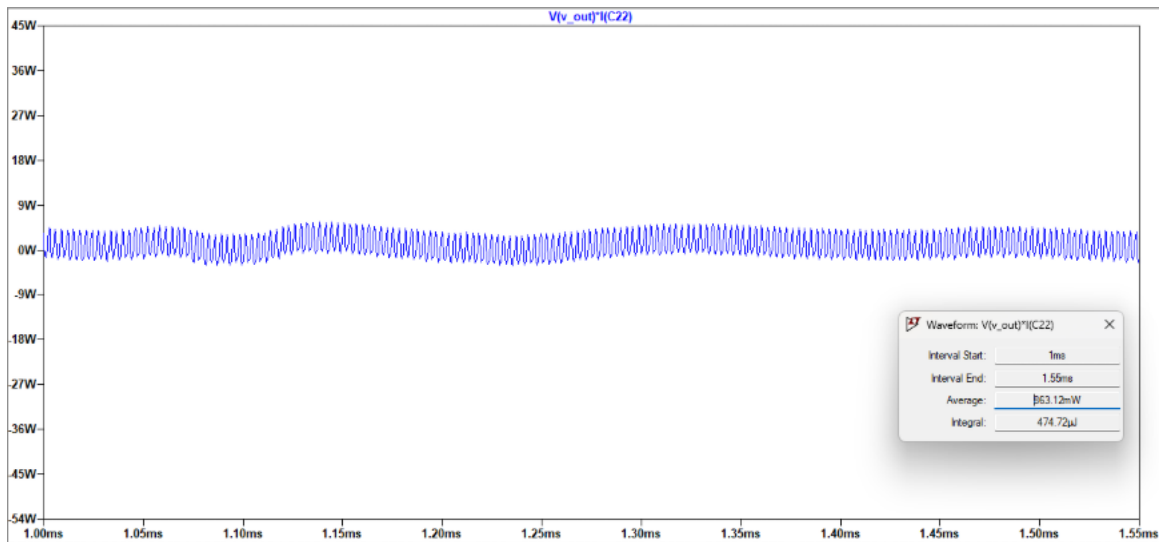


Figure 54: C22 Output Filter Capacitor Power Dissipation (863mW)

5.4. Conclusions regarding full-system simulation

Although these simulations provide a good baseline for overall function, this simulation does not incorporate a few essential elements of the system. Most importantly, it does not include our synchronous rectifier, which is a crucial component and a common failure point in this design. We attempted to build a simulation for the synchronous rectifier, but we were unable to find the proper SPICE models for the components used. Additionally, ringing and EMI issues often plague switching converters, and these issues will be uncovered as we begin physical testing, as outlined in the test plan in the unit test section.

5.5. Precharge Simulation

Below is a simulation of the Precharge voltage curve, where the voltage shown in the DC-link capacitor. The objective is to get the capacitor to charge as fast as possible with as small of a resistor as possible.

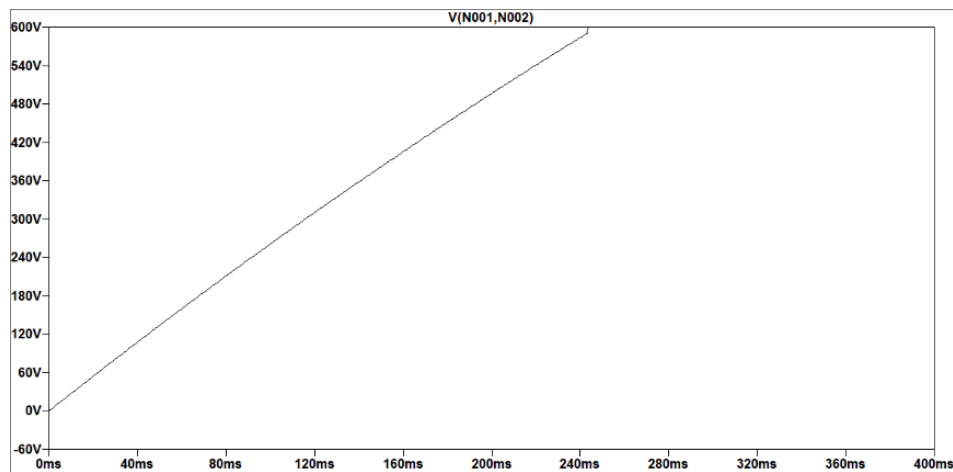


Figure 55: Precharge results for a 600VDC source

6. Testing

6.1. Unit Testing

For unit testing, each sub-circuit of the design was tested separately. With the transformer off the board, isolating the low-voltage and high-voltage sides, individual testing of the two sections can be performed. Some other components on the board were also left off or later removed to disable certain circuitry.

Testing the low-voltage side of the converter, the C_start capacitor is left off the board, and the DC lab power supply is set to a current limit of 100 mA. The DC power supply is connected to the LV-side output and slowly ramps up to 12V. The 5V LED should be illuminated. Using a multimeter set to VDC, to measure the voltage of the + pad of the C_start cap. The voltage should be between 3.6 and 3.8V, which is the voltage of the startup cap.

Using a wire to short the enable pin, the pwr and 12V LEDs will illuminate, and measuring the voltage between pins 1 and 2 of U₁ should be 12 to 12.5V. Once these values are verified, the lithium supercap C_start can be soldered onto the board. The current through the enable pin can also be measured with a multimeter, with an expected current of around 200 mA.

Removing R₆₁ on the board is then needed to disable undervoltage and overvoltage protection (UVP/OVP). With the enable pin still shorted, the square waves on pins 5 and 8 on U₁₂ and pin 1 on Q₁₀ and Q₁₁ can be measured using the oscilloscope by applying 12V with an increased 500mA current limit to the LV output and 10V 100mA current limit to the HV input. The transformer is still off the board at this stage.

Once the LV tests are complete, HV testing can be performed with the transformer finally installed on the board. Several 50V DC lab power supplies will be connected in series to achieve the required 600V output to power the LLC converter. A variable load will also be connected to the converter's output to ensure it can handle 30A of current. Two multimeters will also be used to verify and measure the input and output voltages, respectively. A 35,000-RPM fan provides active cooling to the board and the transformer during high-voltage tests. After the converter passes this initial HV test, the Cars 600V battery will be used to test and confirm that the converter can operate within the battery's voltage range of 470V to 600V.

Board Testing:

For results for unit testing on the LV side of the converter, the voltage of the C_start cap was measured at 3.7V, well within the desired range, and the 5V LED lit up. With the enable pin shorted and C_start soldered on the board, we measured 12V between pads 1 and 2 of U₁, and the 12V and pwr LEDs were also illuminated. When R₆₁ was removed, and 10V was applied to the HV input, the following square waves were recorded on the oscilloscope. The figures below show the square waves of the oscillator's high and low sides as well as the square waves for the switching network and the active rectifier. The current through the enable pin was also measured to be at 240mA, which is higher than the expected 200mA. The testing setup for the board is also shown in the figure below.



Figure 56: Square Wave of High Side of Oscillator

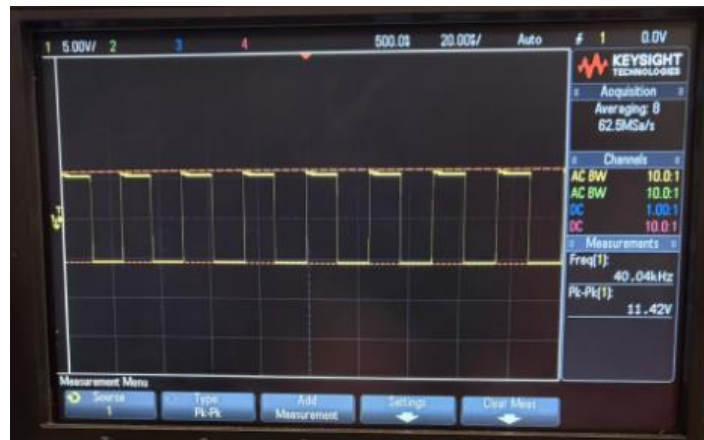


Figure 57: Square Wave of Low Side of Oscillator



Figure 58: Square Wave of Switching Network



Figure 59: Square Wave at Gate of the Low Side of the Half Bridge



Figure 60: Square Wave at Gate of the High Side of the Half Bridge

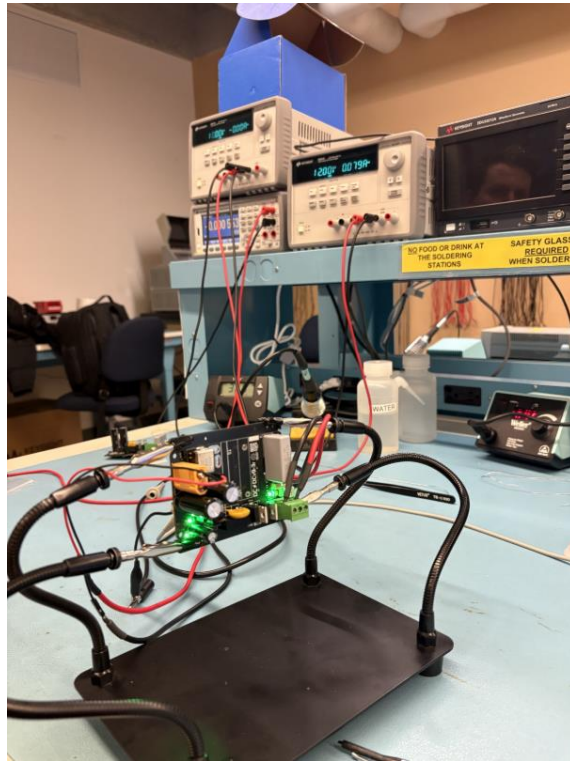


Figure 61: Board Testing Setup

Transformer Testing:

Testing the transformer proved to be finicky and riddled with misconceptions of the fundamentals themselves. We tested the leakage inductance of the primary (W_1) and secondary (W_2 & W_3), these values were compared to the given values of the previous calculation ($W_1 = 166\mu\text{H}$). The figures below show the found inductance values of the primary and secondary coils.

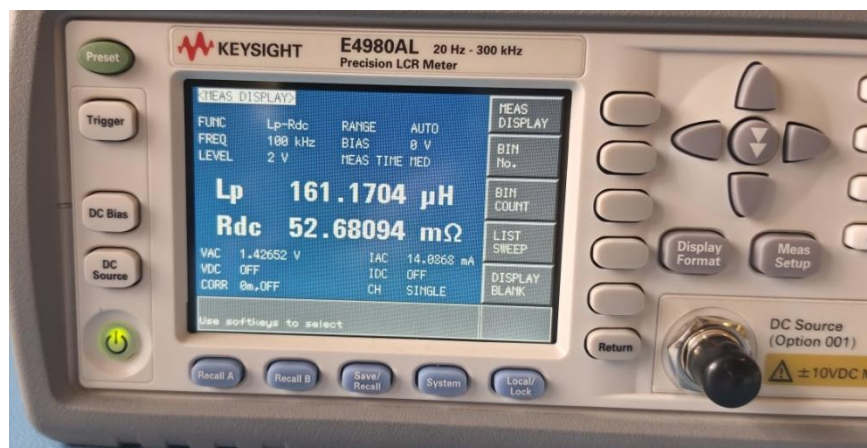


Figure 62: Primary Coil Inductance Value



Figure 63: Secondary Coil Inductance value

As previously mentioned, once the inductance values were reached by changing gapping, winding length, orientation, tension, etc., the windings were glued to the bobbin to prevent them from moving.

Isolation testing was also conducted by using a Instek GPT-9803 Hi-Pot Tester, with the help of a graduate student. The test was conducted by connecting the transformer in a typical fashion to the tester – positive to primary, and negative to all secondary wires. The test was conducted at a ramping voltage increase to 1.8kV in order to simulate the voltage that can be observed within the coils during normal operating instances.



Figure 64: Transformer Connected to Isolation Tester



Figure 65: Transformer Passing Isolation Test



Figure 66: Transformer Passing another Isolation Test

6.2. Interface Testing

The interfaces that we needed to test in our design were a 5V LED, a power LED, and a 12V LED. The 5V LED is meant to show that we are getting 5V on the 5V rail of the converter. This will allow the FSAE team to know that voltage is passing from the high to low side of the converter. During

testing, we applied a voltage to the converter's output to ensure the LED lit. The 12V LED is similar, but shows that there is 12V on the output. This allows the team to know whether the output voltage meets the requirements for the systems powered by the converter to operate. To test this interface, we applied 12V to the converter's output and confirmed that the LED turned on. The purpose of the power LED is to ensure that current flows on the converter's output side. This shows the user that there isn't a break between the high and low sides of the converter. To test the power LED, we applied a low voltage to the converter's output to ensure it turned on.

6.3. Integration Testing

The major integration path with the converter's design is with the FSAE teams' formula car. We tested our converter with the team's battery; however, the battery's voltage would sag from 580 volts to about 312 volts under load. This, in turn, meant that our converter would regulate to about 11.23 volts since the input range was slightly lower. We identified that the most likely cause was an issue on the client's end.

6.4. System Testing

Approach:

During the system testing of our converter, we began by testing each stage from left to right. Testing in this order would help isolate issues to a particular circuit and prevent them from affecting other components down the line. For each test, we began at low voltage and gradually increased it to 580 volts.

Testing Equipment and Set Up:

To ensure power was delivered to the board during this process, we used a separate 12-volt source. The power supply circuit on the board was previously validated through unit testing. Initially, we used a 60-volt power supply. We then used the isolation tester (used for our transformer testing), working up to about 100 volts before it reached its current limit of 10 mA. Since we needed something that could drive more current, we decided to place eleven 50-volt power supplies in series with our 60-volt supply to generate 580 volts. For load testing, we used two DC-electronic loads in parallel to draw the required 32 amps of output current. To measure high-voltage waveforms, we used high-voltage differential probes. We also used multimeters to track the input and output voltage.

Primary Testing:

We began by testing the converter's power switches and resonant tank at low voltage and ramping up. We connected only the primary side of the transformer as shown below in Figure. Kapton Tape was used for isolation at the transformer's output.

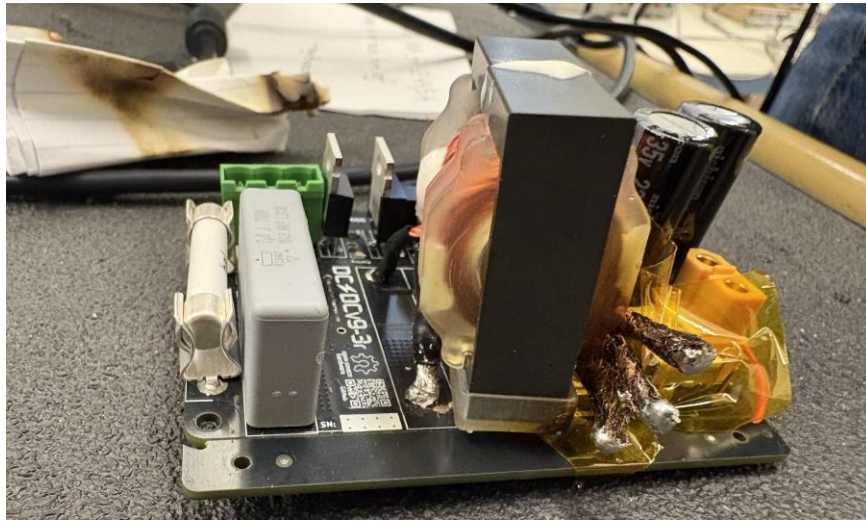


Figure 67: Setup for Primary Testing

During testing, we discovered an issue at about 20 volts when using the 60-volt supply. The oscillator would turn off at about 20 volts, and on our low-voltage VCC node, we measured the high voltage we were applying to the input, indicating that our isolation was broken. This break in isolation could have been very dangerous at higher voltages. However, we quickly traced this back to a simple mistake in the undervoltage/overvoltage protection circuit. There was a discrepancy between our BOM and a part that JLC placed on the board. Rather than placing a 100 M Ω resistor in the resistor divider referencing high-voltage, JLC placed a 100 m Ω resistor. This meant that the high voltage fried the U14A IC, shorting it directly to the low-voltage supply. However, by simply removing the component, we were able to proceed. Although not absolutely necessary, we ordered 100 M Ω resistors to enable this protection circuitry in the future.

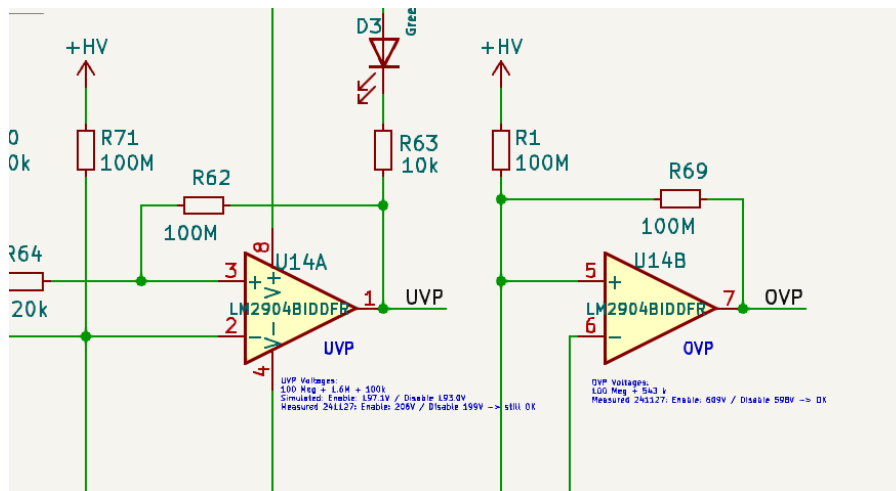


Figure 68: Schematic Location of Circuit Fault

After resolving this issue, we were able to apply up to 580 volts, the maximum input voltage that the converter will see, and observe the correct switching waveform at the node HB_out, pin 1 of the transformer. We then measured pin 2 of the transformer, which is the waveform that has been

filtered by the resonant tank and scaled by a factor dependent on the switching frequency. We measured a near-perfect sinusoidal waveform, indicating that our leakage inductance and capacitor values were correctly tuned. The figures below show both the square and sinusoidal waveforms at 460 volts. Unfortunately, the 580-volt waveform was not saved during this phase of testing, but 460 volts still shows the switching network operating at high voltage.



Figure 69: High Voltage Testing Setup

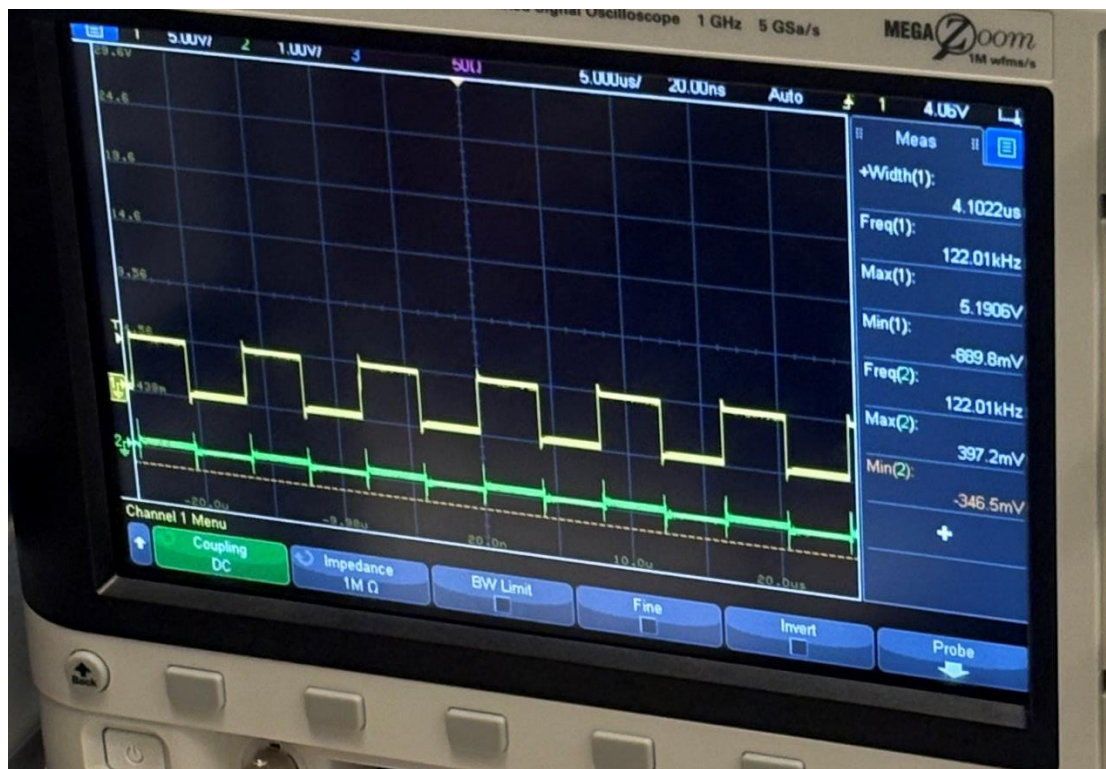


Figure 70: High Voltage Square Wave at Transformer

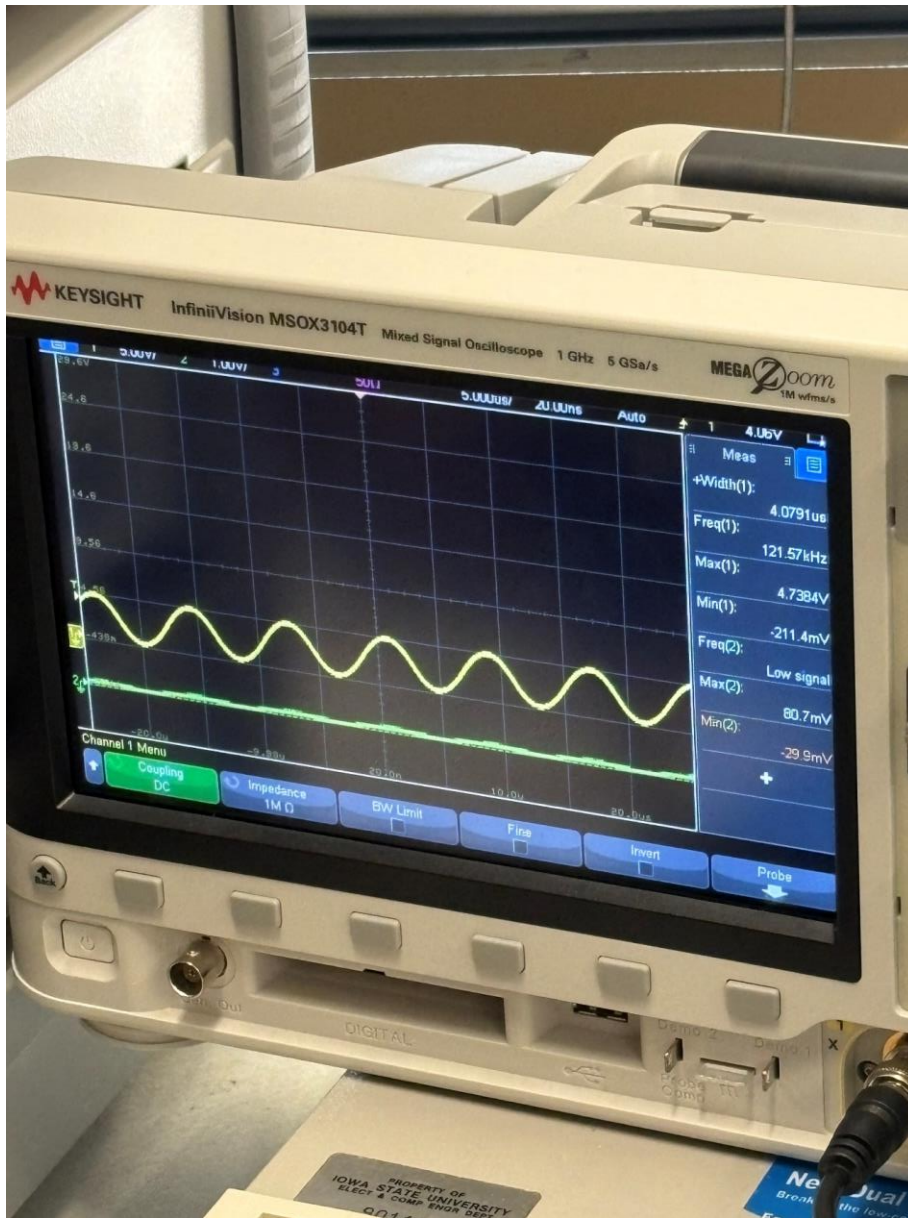


Figure 71: High Voltage Sinusoid at Transformer

Secondary Testing:

Finally, after verifying that the primary side of our converter was functioning properly, we soldered our transformer to the secondary side to begin testing the entire system, including the synchronous rectifier and output filter. At this phase of testing, we set up multimeters to monitor the converter's input and output voltages and applied two DC electronic loads to the output. In addition, we used an external fan for cooling (the on-board fan will be included later). As before, we started at no load and ramped up the voltage slowly to 580 volts, monitoring the square wave at the switching network. At this point, we then applied a load of about 500 mA to the output. Using a multimeter and differential probes, we measured 12 volts at the output. We then slowly increased the load to 32 amps, meeting the project requirement. Figures showcase our test setup and results. Although the

load tester does not show the output current rating, it can be inferred from the input current measured on the power supplies. Figure also shows the waveform seen at pins 3 and 5 of the transformer, the drains of the synchronous rectifier.

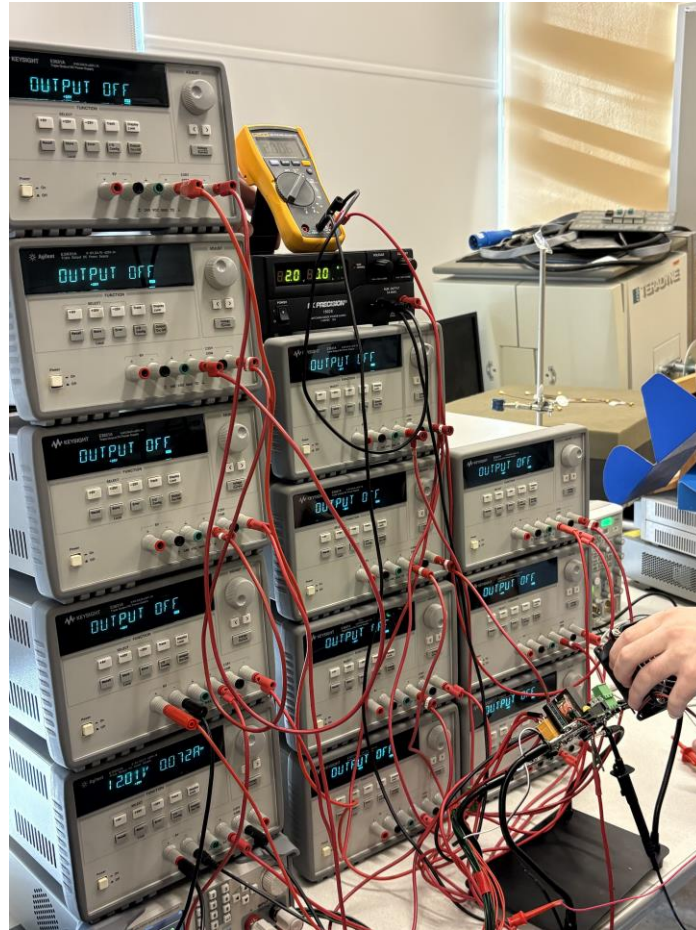


Figure 72: High Voltage Test Setup



Figure 73: High Voltage Test Setup ($V_{in}=570$ V, $V_{out}=12$ V, $I_{out}= 32$ A)



Figure 74: High Voltage Test Setup

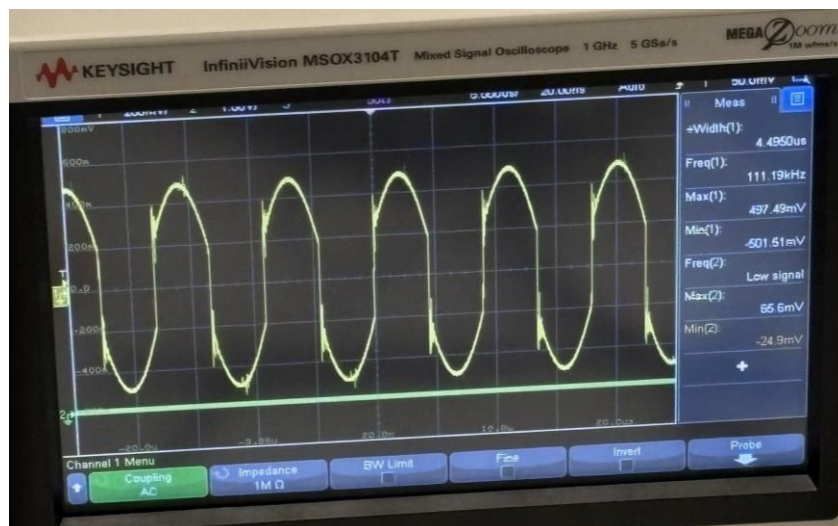


Figure 75: Voltage at Pins 3 and 5 of the Transformer

Notes on Tuning:

In addition, we found that reducing the operating frequency from 120 kHz to 111 kHz resulted in a slight increase in efficiency and a wider input voltage range. At 111 kHz, we achieved a V_{in} range of 350-600 volts and an efficiency curve shown in the results section below. The switching frequency can be tuned by simply adjusting RV2.

6.5. Regression Testing

Although there weren't any new additions to the design, we did have an issue with ordering the wrong resistors. Instead of 100 Megaohm resistors, JLC placed 100 Milliohm resistors. This caused issues with our first board. We removed the resistors from the second board to fix this issue. In addition, we also ended up not using the supercapacitor for the power supply circuit since our client, FSAE, had changed plans and needed to also have a small 12-Volt battery on board. We decided that, rather than depending on the supercapacitor for power, the 12 Volt battery would be used to power the board when it was off while using a 500uF capacitor to ripple voltage. We simulated this change in our testing as well.

6.6. Acceptance Testing

To demonstrate that the design requirements were functional, we had to test our converter with high voltage to see if it was functioning properly and meeting our clients' desired specifications. Testing with the FSAE team proved unfruitful because they were ill-prepared when we attempted it the first time. Once they were prepared, they encountered a system issue that prevented the converter from functioning properly. Overall, the acceptance testing is inconclusive due to client issues.

6.7. Security Testing

Since our project is a purely hardware project, we don't have any security testing that would be applicable to the project since there is no software angle to be attacked from.

6.8. User Testing

For user testing, our group wanted to test the LLC converter in the FSAE team's car to ensure that the converter's high voltage input range is suitable with the FSAE team's high voltage battery output range. By the end of the semester, we managed to conduct one test with the FSAE team's high voltage battery. Unfortunately, the test did not go as well as we hoped. We found that when we connected the converter with the high voltage battery and used the car's fans as a test load (this was a 20 A load), the output voltage of the high voltage battery fell from 580 V to 312 V. This meant that the input voltage for our converter, which is the same as the output of the high voltage battery, fell enough such that the output of the converter was only able to regulate at 11.23 V. This is below our design requirement of 12 V and would not be sufficient to fully run all of the low voltage systems on the car as well as charge the 12 V battery. We discussed the issue separately with both our client, the FSAE team, and our faculty advisor and both us and our client agreed that the issue is most likely to be a problem with the pre-charge system on the client's high voltage battery. Regardless of this issue, thanks to our separate high voltage testing, we know that our converter functions properly for the design requirements. Unfortunately, there was not time to test the converter with the high voltage battery again before the end of the semester. We have handed off

the converter to the client, and hope that once they update their battery the system works as expected.

6.9. Testing Results

The results of our first high-voltage test exceeded our expectations. Following the same testing setup and method outlined in the unit testing section, the input voltage is increased by turning on each 50V DC power supply, reaching a max input voltage of 580V; at 400V, the converter was able to achieve a steady 12V DC at the output and maintain 12V DC at the output as the input voltage reached its peak. The load tester was then turned on and set to constant current, and the current was slowly increased from 1A to 30A. The converter continued to regulate 12V at the output with efficiency increasing as the load increased. Figure 22 shows the converters' efficiency reaching 95% at a switching frequency of 115 kHz. We then planned to test with the high-voltage battery from the FSAE team, but we did not have enough time to gather data with it due to delays on the client's end. However, our team achieved and, in some cases, exceeded our clients' requirements for input voltage range, output current, and efficiency.

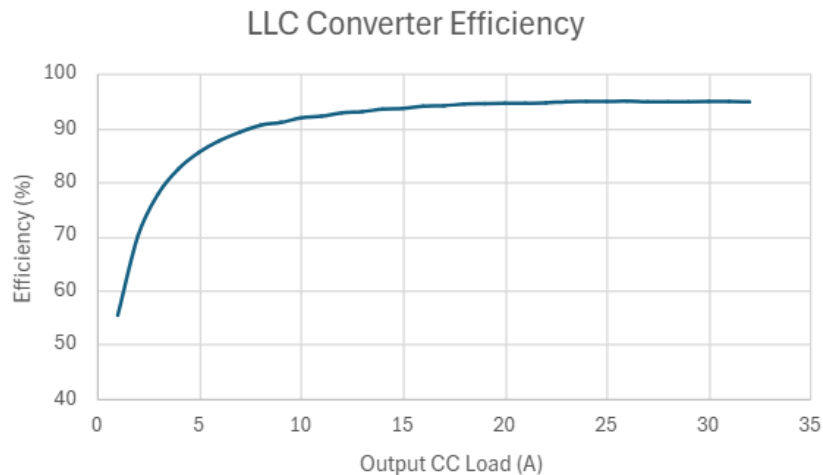


Figure 76: LLC Converter Efficiency Curve

7. Implementation

We have successfully designed, constructed, and implemented all the subsystems in our final design, with a few changes noted below to the overall implementation into the client's system.

Adaptations:

The client made changes to their design since last semester. Rather than having our DC-Converter connected directly to the high voltage battery, they wanted it after the isolation relays (IRs). The client also wanted to include the battery in parallel with the 12-volt battery, so that when the high-voltage tractive battery was not enabled, meaning no high voltage was present at the converter's input, the low voltage systems would still receive adequate power. Figure 75 shows the schematic implementation of the converter.

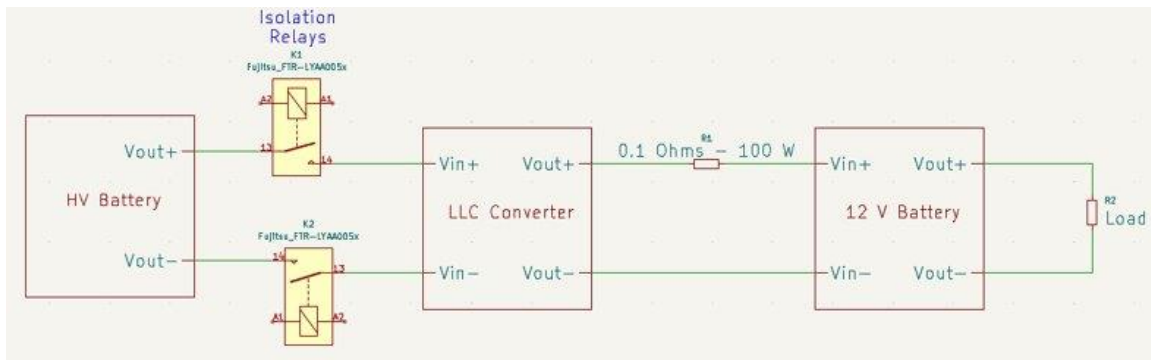


Figure 77: Implementation Schematic

Due to the client's change in plans, we were able to eliminate one of the most complicated, problematic elements of the design, the start-up supercapacitor. During board testing, we found that the supercapacitor discharged considerably faster than expected, so if the board was left powered without any high-voltage present, it would likely fully discharge and require manual charging before the board could be used again. However, since the client now also wanted to include a 12-volt battery in the car, we could simply draw constant power from the 12-volt battery instead to power the converter. However, having the option to include the start-up supercapacitor in the future provides added flexibility, as the client may potentially change their overall EV architecture.

Furthermore, having the low-voltage battery constantly powering the converter with a low quiescent current also allowed us to eliminate the DC-enable switch. Furthermore, because the initial load at startup was relatively low, we were also able to eliminate the load switch. Since both of these switches were eliminated, the user will not have to interact with the DC-converter at all while it's in operation, eliminating any interfacing complications.

However, there was also another stipulation to the change in design. For added functionality, when high voltage is enabled, the converter would recharge the low-voltage battery. Incorporating the 12-volt battery in parallel eliminates the need for a supercapacitor, but a new complication arises. Since there are two non-ideal voltage sources in parallel, a slight difference in their voltages will create a short circuit. Therefore, we decided to use a droop control resistor to limit the potential inrush current between the battery and converter. For a start, we decided to limit the possible inrush current to 30 amps for a 3-volt difference in voltage (worst case). This is the worst-case scenario, as the 12-volt scenario will deplete to about 9 volts at its lowest SOC. We decided to use a .1 ohm, 100-watt resistor for this. The resistor can be seen in Figure 75.

I also proposed that rather than having these sources in parallel with a resistor burning power, they could use a SPDT relay to switch between the battery and converter for powering the low-voltage load. However, this approach has the drawback that 12 volts must always be present to power the low-voltage load. This is to ensure the proper safety circuitry is powered to monitor the high-voltage battery at all times. If power is cut, the high voltage battery will shut off due to a perceived fault; therefore, the relay would have to be fast enough to bypass this fault. The client was not confident in implementing this approach.

We adhered to FSAE and IEEE standards and guidelines such as transformer isolation, EMI, Low voltage and high voltage signal isolation, high voltage labeling, PCB trace widths and lengths, and PCB dimensions.

Pending Issues With Implementation:

Since the client was not available for testing until April 30th, we did not have adequate time to test and troubleshoot the implemented design. When, we tested our implemented design in the Formula Car, the converter did in fact power the load, but we encountered an issue. When using our converter, the voltage at the output of the isolation relays (IRs) sagged from 500 volts to about 312 volts. Since our minimum input voltage was set to approximately 350 volts, the output was only able to regulate to about 11.23 volts as shown in Figure 82 and 83. After consulting our advisor and other technical experts, we determined that the voltage drop before our converter is likely an issue on the client's end. The culprit is likely either their precharge/discharge circuitry or BMS system. We will assist them in troubleshooting this before competition in the coming weeks.

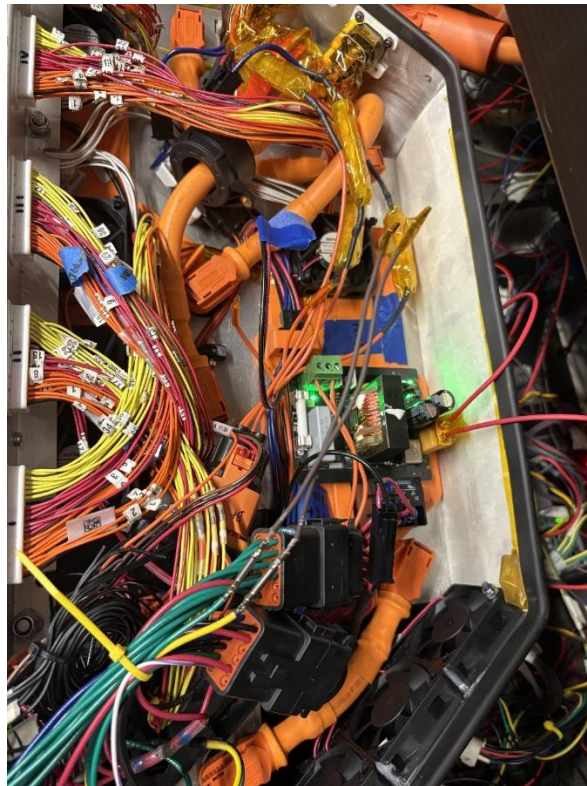


Figure: Converter Implemented and Powered In Car

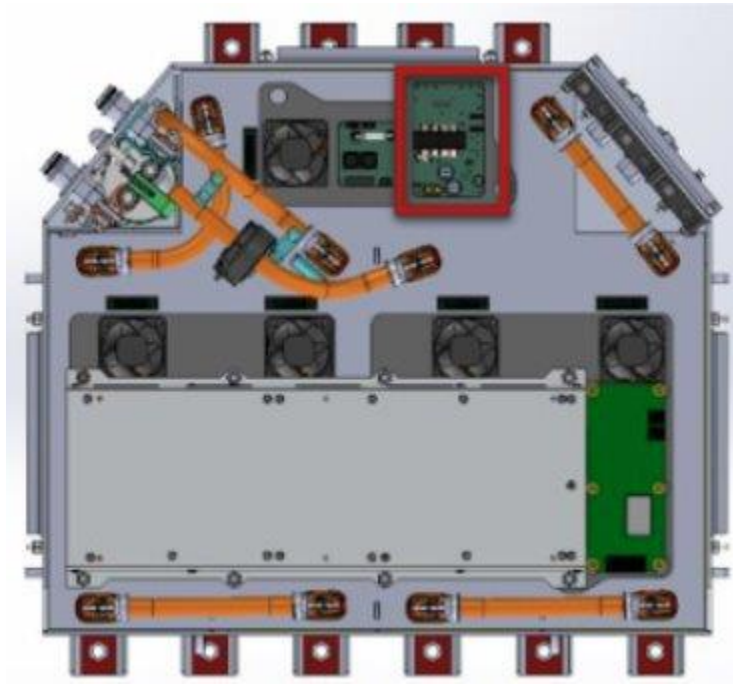


Figure: 3D Model Showing Location of Converter



Figure: Input Voltage



Figure: Output Regulation

7.1. Design Analysis

All in all, our implemented design works very well. We have conducted high-voltage testing of our LLC converter across the full range of input voltages and output loads it will experience and have found that it regulates to a stable 12 V output throughout each range, with a little extra contingency room in each. This means the converter is meeting the specifications required by our client, the FSAE team. Furthermore, we have found that the converters' efficiency is 95%. This exceeds our client's requirements. Finally, we have conducted isolation testing on our transformer to verify that our converter provides the galvanic isolation our client requested. When all this is put together, it means our converter either meets or exceeds all the requirements defined by our client. Our converter works well because it was designed to the specifications we were given, and we simulated it before building to ensure it meets the requirements. Furthermore, we were very thorough when ordering parts to ensure that each part's tolerance range wouldn't prevent our design from working. Once the converter was built, we conducted rigorous testing to ensure that every part functions correctly and that there will be no issues with the converter when it is installed in the car. This test has proven that the converter meets all the requirements set by our client. For further information on our testing processes and how the results align with our clients' specifications, refer to the testing section of the design document.

One change we would like to make to our converter is to add a proper 3D-printed mount for the fan that cools our transformer. Currently, the fan is zip-tied to the transformer, which is a fully functional solution. Furthermore, the converter is located within the car's battery system, so once it is in use, no one will see it. Having a proper mount for the fan would be a more professional way to set up the cooling system and would also make the converter a bit more robust.

Another thing about our design that could be improved is testing our converter with the actual battery; although we did try to use the battery, it still needs testing. Whenever a load was connected, their battery would drop in voltage; however, our converter still regulated, just not to 12 volts, since it was below the input range. Unfortunately, after further research, we found that this was an issue on the client's end, so we were not yet able to perform full integration testing by installing the converter in the car. While we have separately verified full functionality of our

converter, being able to do the final step of testing the converter inside of the car would be a great way of demonstrating that the converter works fully to the clients' requirements as well as being able to survive in the environment it would be in rather than a lab setting to ensure it is a finalized working product.

8. Ethics and Professional Responsibility

8.1. Areas of Professional Responsibility/Codes of Ethics

Area of responsibility	Definition	Relevant Item from IEEE
Work Competence	Complete tasks in a reasonable and timely manner with effort, quality, diligence, and integrity. Come to meetings and workdays consistently.	Maintain and improve our technical competence and undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations.
Financial Responsibility	Compile a cost-effective and reasonable bill of materials and items to buy and order necessary products in a cost-effective, efficient, and safe manner. Be responsible with PCBs and other materials and make sure they are taken care of.	Avoid unlawful conduct in professional activities and reject bribery in all its forms.
Communication Honesty	Deliver honest and full reports of progress and keep the team updated with transparent progress so far. Also contribute sufficiently during workdays.	Seek, accept, and offer honest criticism of technical work, acknowledge and correct errors, be honest and realistic in stating claims or estimates based on available data, and credit properly the contributions of others.
Health, Safety, Well-being	Consider the mental and physical wellness and health of group members and assign responsibilities in a way that does not infringe on their mental or physical well-being. Check on the mental and physical state of all group members and adjust accordingly.	Hold paramount the safety, health, and welfare of the public, strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and disclose promptly factors that might endanger the public or the environment.
Property Ownership	Respect the intellectual work and design of group members and sources. Cite all sources and don't steal work without permission. Also don't steal materials and only use materials owned or purchased.	Avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors or any other verbal or physical abuses.

Sustainability	Respect the environment and order items that are more environmentally friendly in sourcing and operation.	Hold paramount the safety, health, and welfare of the public, strive to comply with ethical design and sustainable development practices, protect the privacy of others, and disclose promptly factors that might endanger the public or the environment.
Social Responsibility	Create a project or design and work on it in a way that will benefit others or clients greatly. Be supportive and cooperative with teammates.	Support colleagues and co-workers in following this code of ethics, strive to ensure the code is upheld, and do not retaliate against individuals reporting a violation.

Table 5: Areas of responsibility, their ethical definitions, and relevant IEEE code

One area our team is strong in is property ownership. We cite our sources and try our best to alter designs we use so as not to directly copy someone else's work. We try to create a unique design as possible and try our best to come up with our own thinking or transformative information in our documentation. This shows that we are respecting the intellectual property of others and being intellectually honest in a way that does not infringe on others' online property. Another area that we have been doing well in is communication and honesty, as we have been transparent in our failures and successes throughout the semester.

We are also now strong with work competence, as we have been working diligently and productively in building our transformer, soldering our PCBs, testing them, and fixing any issues. Though not all our members are contributing as much or equally, we are making progress as a team, and some of our members have been putting in significant time and effort outside of meetings in related areas.

One area our team could improve is health, safety, and well-being. Some of our members were somewhat overworking sometimes, which led to losing sleep. Our team assigned tasks, however, did not have a fair division of work between members. We could discuss together ways to delegate tasks more evenly and remind our group members not to overwork on any day. We could also have group members bring up their workload over the past week and any overwork or disruption in sleep to help resolve the issue.

We could also improve financial responsibility, as we have spent a lot of money on our materials, and maybe we could have spent less. We could have spent more time planning our budget and considering our options before purchasing. We could have asked for advice or help from other advisors, teams, or salespeople when buying things.

8.2. Four Principles

	Beneficence	Nonmaleficence	Respect for Autonomy	Justice
Public health, safety, and welfare	Beneficence would mean that a group	Nonmaleficence helps public health, safety,	Respect for autonomy teaches people	Justice helps to give people the public health,

	should act in ways that are beneficial to the health and safety of the public as well as act in a way that is in the interest of public welfare.	and welfare by teaching individuals and groups to act in ways that are not harmful to public health, safety, and welfare.	to allow the public to make their own decisions in regard to their health, safety, and welfare.	safety, and welfare that they deserve to have.
Global, cultural, and social	Beneficence allows people to act in a way that is good for societies and cultures around the world.	Nonmaleficence helps people to not act in a way that is bad for global societies and cultures.	Respect for autonomy allows different societies and cultures around the world to keep their own identity and autonomy.	Justice provides global cultures and societies with fair treatment.
Environmental	Beneficence helps people to act in a way that is good for the environment.	Nonmaleficence helps you to act in a way that is not bad for the environment.	Respect for autonomy dictates that people should act in their own interests, allowing people to act in a way to protect the environment.	Justice provides environmental groups with the protection that they need to act in the interest of the environment.
Economic	Beneficence assists people in acting in a way that is good for their work, which is in turn good economically.	Nonmaleficence dictates that people should act in a way that is not bad for the economy.	Respect for autonomy allows people to act in their own interests. Often, the person's interest is to act in a way that is good for them economically.	Justice helps protect people's economic situation from unfair harm.

Table 6: Four principles and relation to categories of social ethics

Global, Cultural, and social beneficence allows people to act in a way that is good for societies and cultures around the world. We have tried our best to comply with the FSAE

standards and IEEE ethical standards, so our design will be the most effective and ethical for our clients. We have also been working diligently in building and testing our PCBs and fixing any issues, so our DC-DC converter will be completed in time and deliver the best results for the ISU formula racing club. We have consistently been discussing the FSAE expectations and reviewing them as well while soldering component parts during workdays.

Respect for autonomy would correlate to every individual team member since we are working on a project that is funded by Milwaukee tool for the FSAE team which means that the FSAE team will get to keep the part. Everybody on the team is fine with this, however we had to make sure that everybody on the team was on board. We have two members who used to be part of the FSAE team here at Iowa State, and the rest of us are not.

Environmentally, this project serves as a more power efficient solution to the FSAE car's low voltage systems. Since this project will replace the current solution of the team needing to use multiple batteries which will eventually die out and be discarded this part not only provides a better way to take power from the main battery that achieves the same goals of the low voltage batteries however it is more environmentally friendly. This is also a part created for the electric formula car which is better for the environment than a gas-powered car.

One area in which we are lacking is global, cultural, and social justice. Not all our team members contribute equally. Some are contributing very little, especially the substantive parts of PCB design and testing. Others are overcontributing, spending extra time outside workdays. Not everyone is given a fair chance to contribute to equal tasks. Of course, the lack of contribution from some members is compensated for by the extra contribution or overall contribution of the other team members. All team members are at least attending the workdays and helping when given the chance and contributing at the very least in smaller ways or in other areas. We have also at least tried to delegate specific tasks to each member. As for our design, it follows the FSAE rules and IEEE ethics, and we are creating a DC-DC converter that the entire ISU Formula SAE team can use to power its cars, which all members can be able to use.

8.3. Virtues

Virtues displayed by our team:

Responsibility – Being held accountable for one's actions and willing to act morally to fulfill their duty.

Honesty – Being transparent and authentic.

Integrity – Being truthful and sticking to one's words.

Individual Reflections:

Diego Rivera

The virtue that I have demonstrated thus far would be the virtue that is honesty. It is important to me since there have been multiple projects that I have been a part of where people would not be honest with the work they would say they would do. In turn, this meant that the rest of the team would suffer because of a team member not being honest. I believe that everybody so far on this team has been honest. I have communicated multiple times when I was able to make it to meetings, when I was able to work, and overall being completely transparent with the team. As for the rest of the team in my eyes, they have also demonstrated honesty with what work was done and letting everyone know if they were unable to make it to a meeting or not and communicating rather than not saying anything.

The virtue that I have not demonstrated in senior design so far would fall under responsibility. I feel like I do not yet have a full understanding of my part of the project which means that I should have been more responsible by putting more time into the project. This is important because I am the one that should have a detailed idea into what my part of the project does and be able to explain it clearly. I need to put more time into understanding how my part works and have the team ask questions about it to make sure that I understand it properly.

Wesley Mao

One virtue that I have demonstrated so far this semester during senior design is integrity. This is important as I want to be seen as an important member of the team to make up for the lack of contribution. I also believe it is important as it is the most important part of a team member, contributing to what I can contribute. I have attended all the meetings and workdays required. I have tried to complete the task delegated to me by Caleb and have at least understood and observed what my other team members are doing when building and testing PCB parts.

The virtue that I have not demonstrated in senior design so far would fall under responsibility. I feel like I do not yet have a full understanding of my part of the project, which means that I should have been more responsible by putting more time into the project. This is important because I am the one who should have a detailed idea of what my part of the project does and be able to explain it clearly. I need to put more time into understanding how my part works and have the team ask questions about it to make sure that I understand it properly. I also have not been doing much PCB building or testing, only observing others, and have not found any substantial tasks I could work on.

Caleb Rezmer

One virtue that I have exemplified during my senior design work is responsibility. I have always felt that responsibility is necessary because it means you care about what you are doing, and why do anything if you do not care about it? Especially as the team lead of the project, I believe I bear a great deal of responsibility. I try my hardest to keep the team on a rigorous pace by constantly reaching out members on their part of the project. In addition, I plan multiple meetings and

workdays a week, trying my best to keep a focused agenda and prioritize everyone's time. I've tried hard to attain a strong understanding of the entire system so that I can help team members. When tasks are urgent, I sacrifice hours of time to get things pushed through the finish line, whether it be through building SPICE simulation or pushing to order parts for prototyping. When we encountered the funding issue with the Department and the client, I took responsibility for it and sought sponsorship from Milwaukee Tool. I spent many hours assembling and testing the board and transformer in order to get our team back on schedule according to our Gantt chart. Whenever I or someone else makes a mistake on the team, I try to reflect on how I may have led them, which may have caused the issue.

One virtue that I could develop more in the future is patience. To counter my last point, I think I could be more understanding and patient with others' experiences, skill sets, and temperaments. Whether I've shown it or not, I have definitely been a bit impatient with some team members and could have been better at confronting them to resolve tasks. I may have come off as passive-aggressive by being overly urgent, which I do not intend to do but is a byproduct of my impatient and frustration. Being a calmer, more collective person is something I strive for, but it is hard as an electrical engineering student. This virtue will be especially useful during the build and testing process next semester.

Andrew Chandler

The virtue I have demonstrated in senior design would be responsibility. Throughout senior semester I have had to become more responsible and, in some situations, take the lead such as when Caleb isn't able to attend some of our meetings and he entrusted me to lead those meeting and ensure we stayed on schedule in his absence. I continue to uphold this virtue throughout senior design and later in my career by making sure my parts of the projects were completed on or ahead of time when I knew I would be busy. Ensuring that I wouldn't be dragging the team down by not completing my work.

The virtue I believe I could develop more is honesty as I feel I haven't been as transparent with the team when it came to managing my workload and agreeing to work on parts of the project on topics, I feel I wasn't fully knowledgeable and qualified in. I tend to not speak up for myself and put the project first by working on the tasks assigned to me even if that means working on parts I don't feel the best fit for.

Drew Bixler

One virtue that I have demonstrated in senior design is responsibility. Responsibility is a virtue that matters a lot to me, as it is incredibly hard to work with a group when you or others are not responsible for their fair portion of a project. I am always aiming to be a strong contributor to groups that I am a part of, and I believe the best way to do that is to be responsible for my own portions so that others don't have to worry about my portions being finished. Throughout the project so far, I have done my best to ensure that all portions of the project that fall under my responsibility have been completed in a timely manner and to the best of my ability. Some examples of this include working with the additive manufacturing lab on the printing of a

transformer bobbin, the updating of the control loop and oscillator portions of our team's converter and contributing to the development of the team's test plan.

A virtue that I believe I could do a better job of displaying is communication. Communicating is important to me because it is the most necessary virtue for a team to be able to work together. There were portions of this project in which I feel I should have done a better job communicating that I felt I needed extra time to handle a particular task. Furthermore, I should have done a better job communicating with the Additive Manufacturing Lab on completing our transformer bobbin. While I was timely giving the information needed, the lab was taking longer than expected to fulfill the order we placed, and I needed to go in person to speak with them about it. Once I went in person to ask for an update on the progress, they completed our part. If I had gone to them in person earlier, we could have had the part and potentially been able to complete some transformer testing by the end of the semester. I will make sure to keep this in mind when working in industry and work to make sure that I am better at communicating early and effectively.

Hunter Kanne

The biggest virtue I demonstrated during the senior design work has been patience. The project has been met with plenty of unexpected changes, downtime, and waiting for materials, all of which test patience to the utmost degree. While my teammates had been less affected by waiting due to their subsystem requirements, building a physical piece of equipment is heavily affected by waiting and ruined my personal timetable more than once. As such, being patient with what may come in the project proved useful and prevented me from losing my composure or making rash design decisions in the name of convenience. This value is important because it is the basis of working on a long project, and without patience, nothing worthwhile can be done.

The virtue that I did not demonstrate in the senior design project was diligence. Diligence is important because without it, nothing will be documented or done. To demonstrate this virtue more, I need to document every piece of information I come across that I use, and complete work when it becomes available, not when it is convenient.

9. Conclusions

9.1. Summary of Progress

During the first semester we set out to get our foundation established. Doing so requires us to define a concrete problem with limits, find a reference design that we could look to for help, get a support base of knowledge, designate roles to team members, and start design. We were able to meet all of these in the first semester right away and as a result we were able to start design early, make crucial tweaks and changes to our project to meet and excel our given requirements, and order parts to assemble the project for next semester. Second semester saw the greatest amount of movement within the team, with the boards being assembled and the transformer being manufactured there was much to test & tune. Once both the board(s) and transformer had been assembled testing began to validate our designs, this took only a month or so and was largely successful with the exception of a few incorrect parts that had to be replaced. Towards the end of

the second semester, we were able to finally test the project in its intended use case, for which the client was not ready. Overall, the project was a success, but we have yet to see it tested in its intended environment successfully.

9.2. Value Provided

The purpose of the project was to provide a compact, highly efficient, and reliable 600VDC-12VDC converter to be used in the FSAE car at Iowa State University. The FSAE team had planned to use a heavy and high-capacity low-voltage battery to power their low-voltage systems. Since the announcement of the LLC converter completion, the FSAE team has swapped out their low-voltage battery with a significantly smaller & lighter battery in anticipation, as such, the converter will be an invaluable resource to them in their quest to reduce the size and weight of their car to get more performance.

9.3. Next Steps

Further capacity could be added to the converter as we found that even without the fan for cooling the parts we could reach the theoretical maximum current and sustain it for long periods of time. It is unknown which part would fail first under these circumstances, but it would likely be the transformer or rectifier – both of which could easily be upgraded to meet further demands. In the event that the converter is needed to work simultaneously with the battery while providing power to said battery, the addition of a resistor between the battery and convert would be required, while changing the supercapacitor to a much lower 500uF capacitor would be recommended. The capacitor should be changed because there would no longer be a need for a start-up source on the board due to the addition of the battery; likewise, the resistor is to prevent a positive feedback loop in the converter therefor overloading itself onto the battery. Finally, to reduce manufacturing time it is highly suggested to create a new bobbin that would better fit a singular turn on the secondary winding. Likewise, to reduce the manufacturing time for the board applying the taller or larger parts last is highly recommended – even if Digikey is willing to put the larger parts on.

10. References

- [1] Rootthecause, “GitHub - Rootthecause/DCDC: An open source credit card sized DC/DC converter to convert up to 600 V to 24 V in a energy-, cost- and space efficient way.,” *GitHub*, Feb. 12, 2025. <https://github.com/Rootthecause/DCDC>
- [2] “PMP22083 reference design | TI.com,” *Ti.com*, 2020. <https://www.ti.com/tool/PMP22083#tech-docs> (accessed Dec. 01, 2025).
- [3] M. Ivankovic, *Application Note 26 V 1.0 2023-03-21 LLC design guide: 3300 W converter*. Infineon, 2023. Accessed: Dec. 01, 2025. [Online]. Available: <https://www.infineon.com/assets/row/public/documents/24/42/infineon-llc-design-guide-3300w-converter-applicationnotes-en.pdf?fileId=8ac78c8c86919021018713adb87d42e2>
- [4] *Two-Phase Interleaved LLC Resonant Converter Design With C2000™ Microcontrollers*. Texas Instruments, 2020. Accessed: Dec. 01, 2025. [Online]. Available: <https://www.ti.com/lit/ug/tiduct9a/tiduct9a.pdf?ts=1761712217376>

11. Appendices

11.1. Appendix 1 – Schematics and Layout

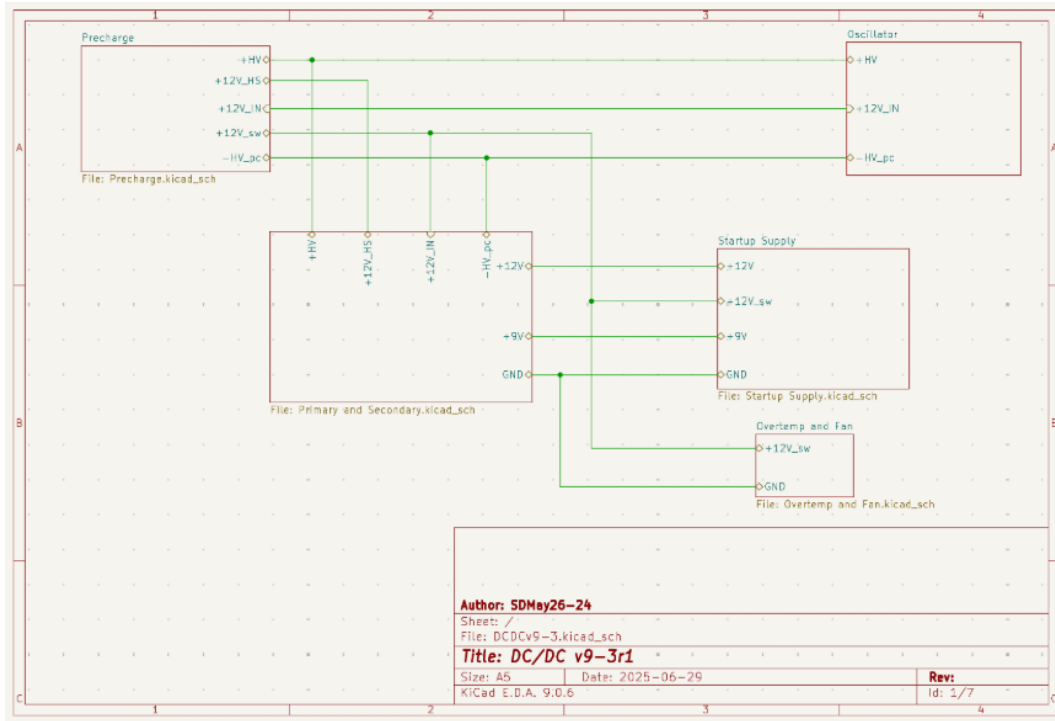


Figure 78: Circuit Block Diagram

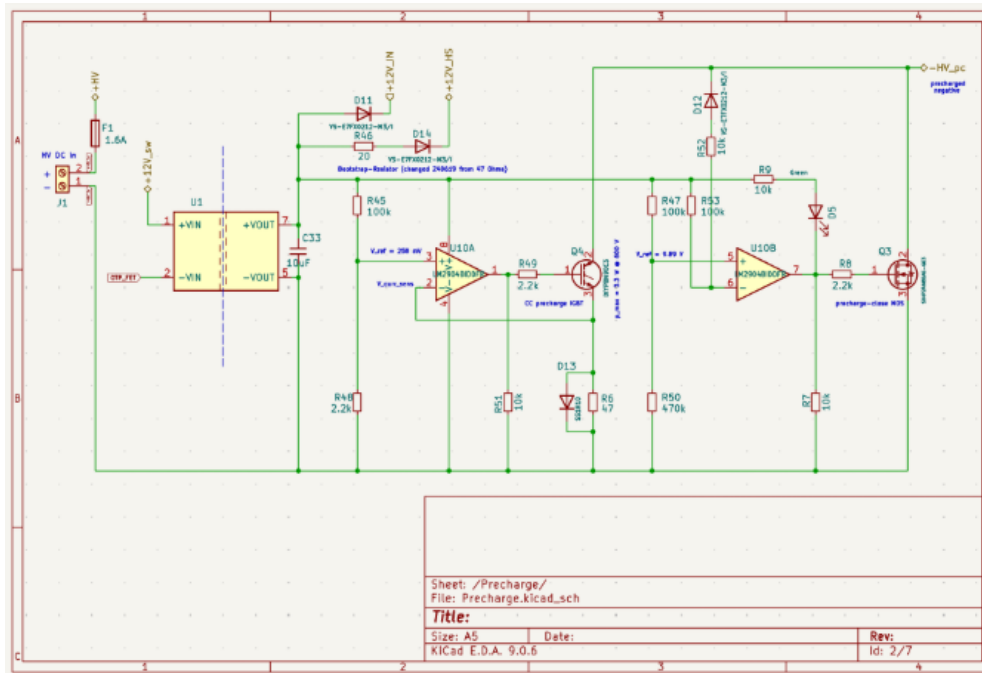


Figure 79: Precharge

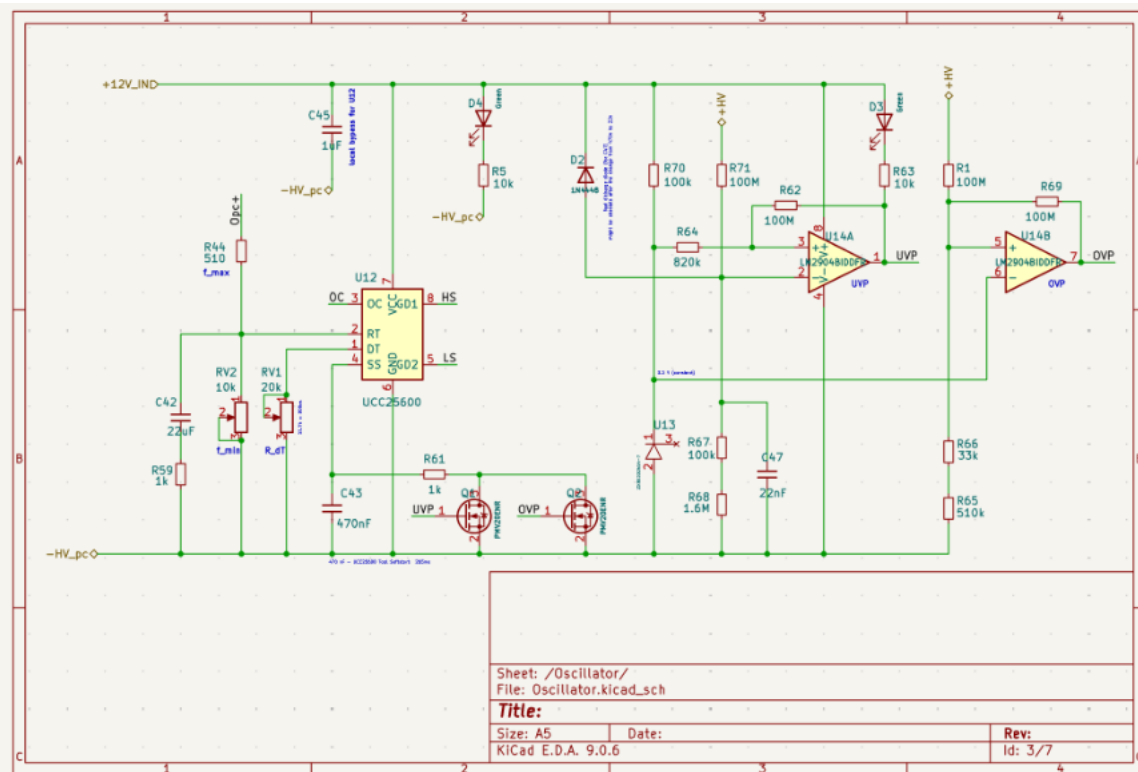


Figure 8o: Oscillator

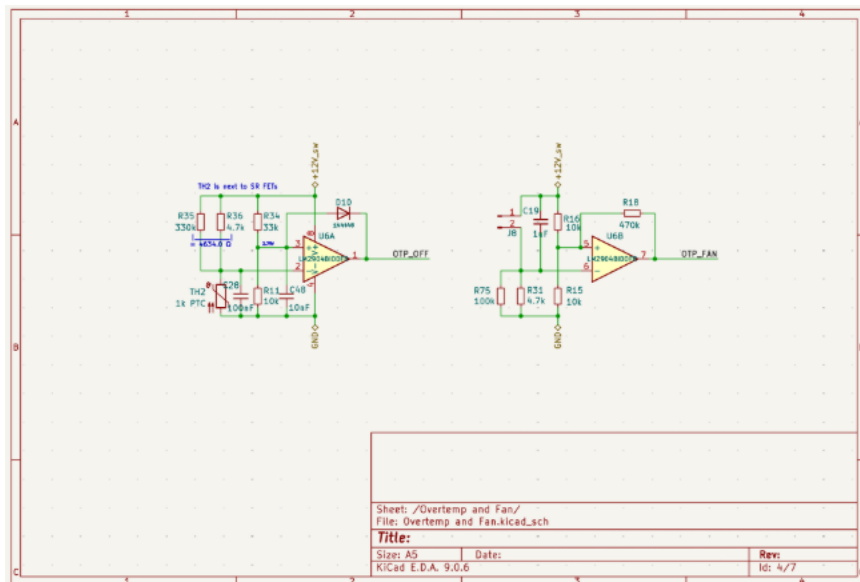


Figure 8i: Overtemperature protection and Fan control

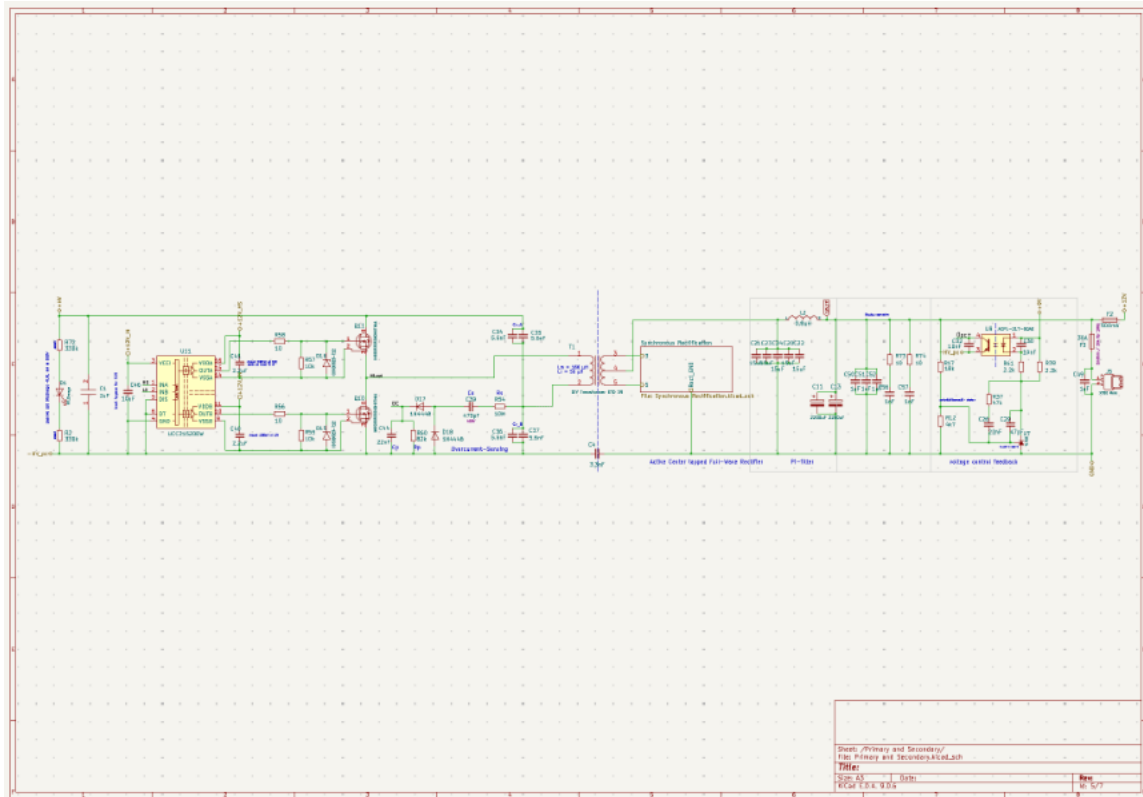


Figure 82: Primary and Secondary Side

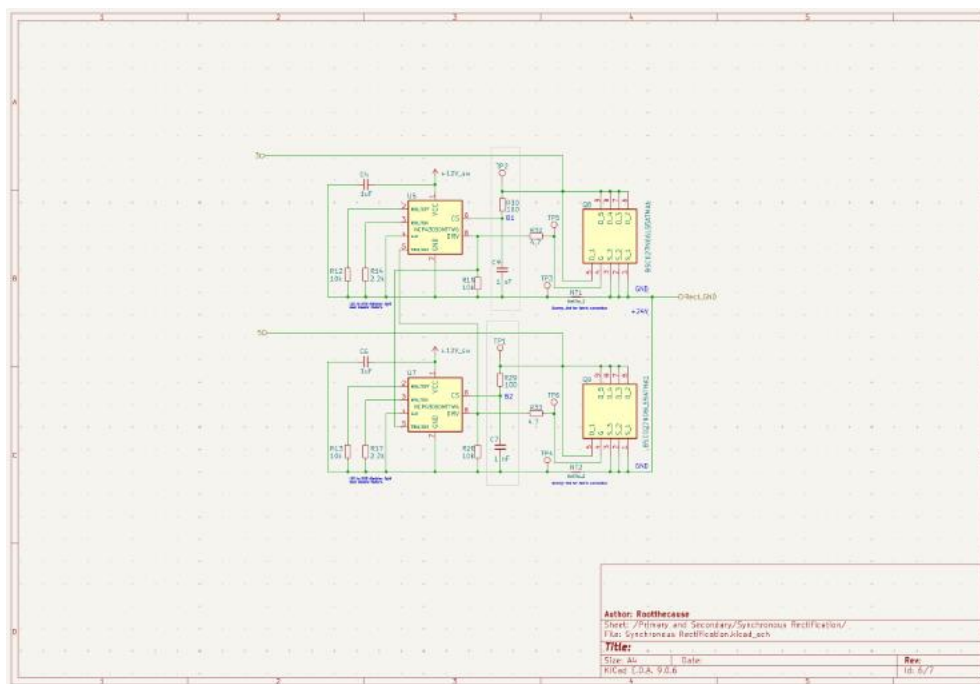


Figure 83: Active Rectification

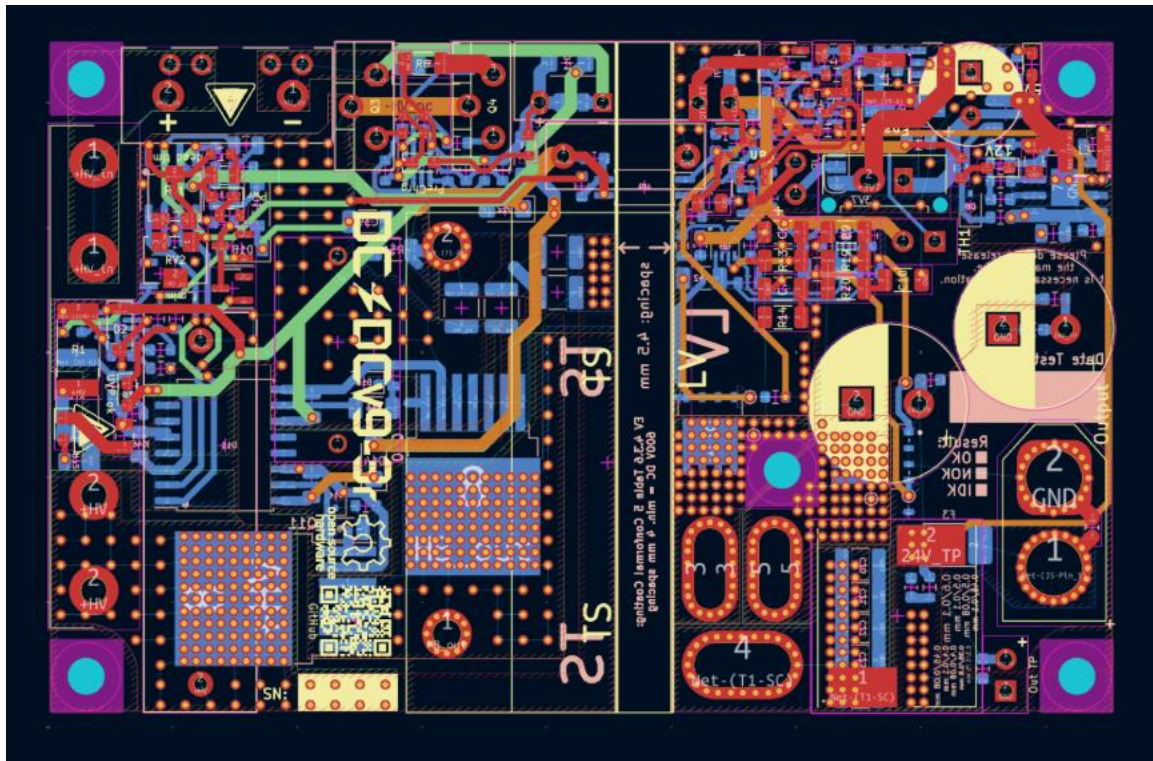


Figure 85: PCB Layout

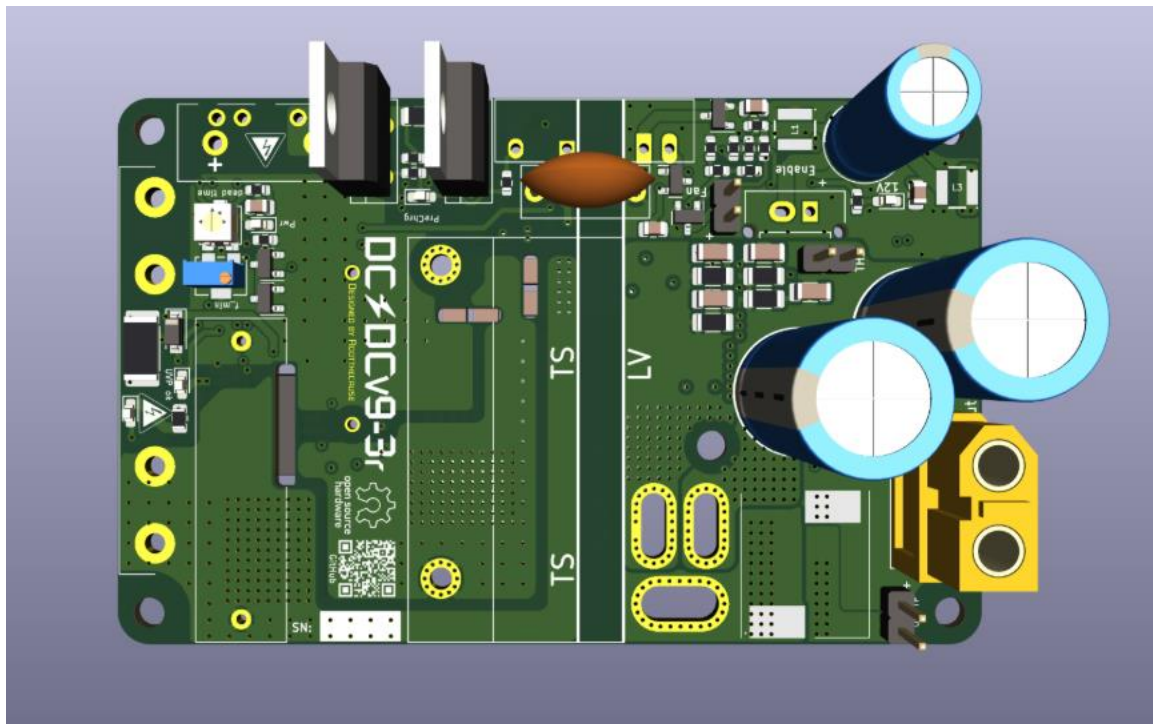


Figure 86: Front of PCB

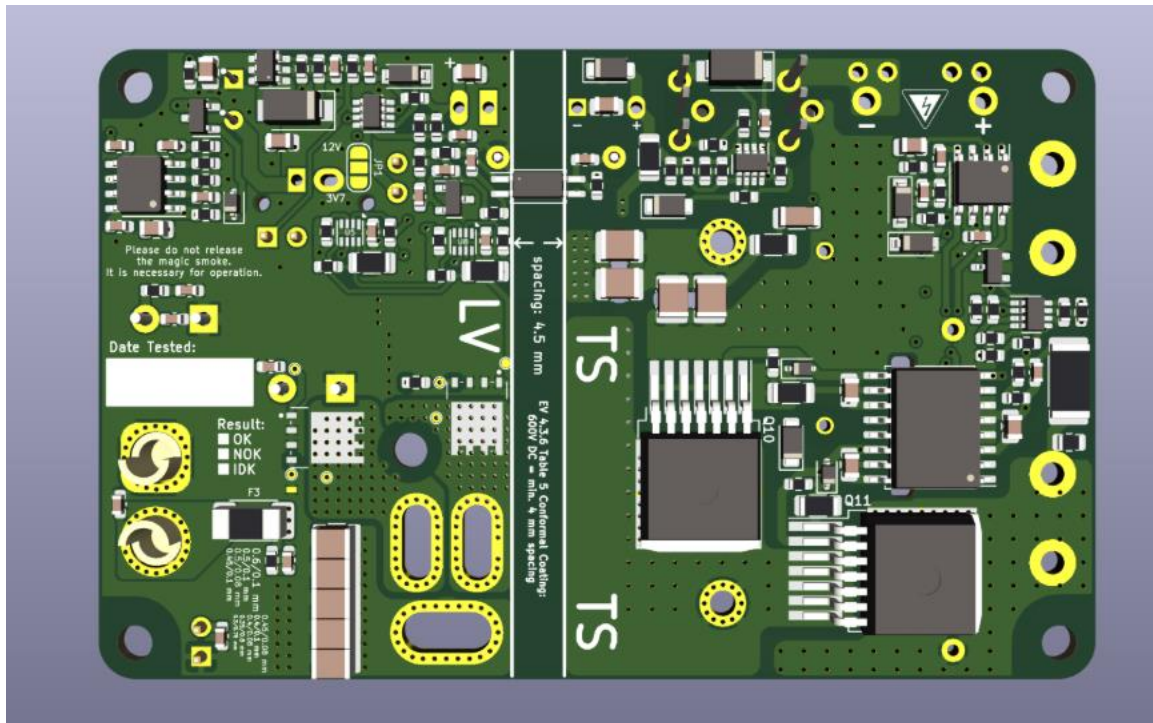


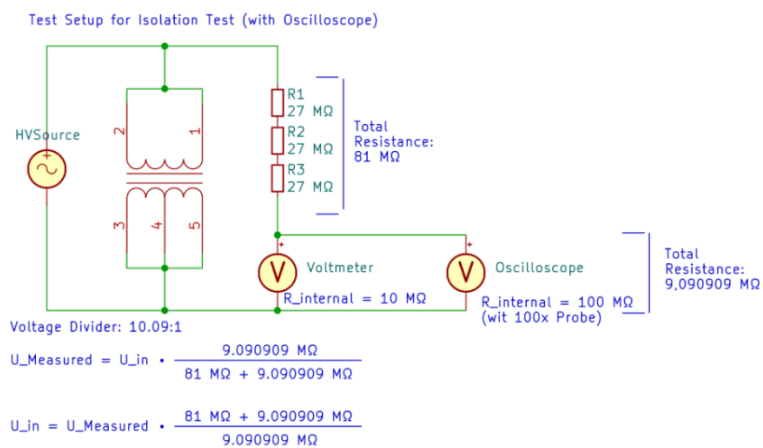
Figure 87: Back of PCB

11.2. Appendix 2 – Operation Manual

Some of the testing procedures were referenced from [1]Rootthecause's GitHub in the references

Transformer

Isolation Test:



Steps

Assembly

- Ensure transformers are fully de-energized and isolated from all external circuits.
- Connect transformer terminals according to the tested pair:
 - **High-Voltage Lead (HOT):** primary winding terminals tied together (H_1/H_2).
 - **Return/LOW Lead:** secondary winding terminals tied together (X_1/X_2), or transformer core/tank, depending on test.
- Ensure unused terminals are left floating unless the tester manual requires grounding.
- Attach the tester's safety ground to the test bench earth.
- Keep everyone at a safe distance from test area when energizing



Figure:88 Rev.3 of Transformer

Expected Values

- Applied Voltage: 1800 VAC @ 50 Hz.
- Duration: 60 seconds.
- Leakage Current Limit: 0.5mA
- Expected Behavior:
 - o No breakdown, arcing, or audible corona during test.
 - o Tester completes the test without triggering “FAIL,” “ARC,” or “HV OFF” fault modes.

Test Procedure

1. Setup and Safety
 - a. Verify terminals are clean and dry
 - b. Turn on & confirm GPT-9803 output is off
2. Tester Configuration
 - a. Select AC Withstand Mode.
 - b. Set Output Voltage to 1800 VAC.
 - c. Set Frequency to 50 Hz.
 - d. Set Ramp Time to default
 - e. Set Test Time to 60 seconds.
 - f. Set Leakage Current Trip Level: 0.5mA
 - g. Enable Arc Detection
3. Isolation Test
 - a. Test A: Primary to Secondary
 - i. Hot leads to the primary terminal.
 - ii. Return leads to the secondary terminal.
 - iii. Start the test.
 - b. Test B: Primary to Core
 - i. Hot leads to primary terminals.
 - ii. Return leads to core.
 - iii. Start the test.
 - c. Test C: Secondary to Core
 - i. Hot leads to secondary terminals.
 - ii. Return leads to core.
 - iii. Start the test.

4. Interpreting Results
 - a. Test Pass:
 - i. The tester completes the full duration.
 - ii. No visible discharge, change in smell, audible crackle, or insulation failure.
 - b. Test Failure:
 - i. GPT-9803 triggers leakage overcurrent, arc detection, breakdown, or shutdown.
 - ii. Any visible arc, smoke, or audible corona.
5. Post-Test
 - a. Turn HV output of GPT-9803 off.
 - b. Wait for GPT-9803 & Transformer windings to discharge

Referenced from rootthecause's GitHub

Load

Load Test:

Materials required

- High Voltage DC Source
- 120VAC wall socket
- 5-gallon bucket of cold water
- Current clamp
- Multimeter
- Transformer
- 4/0AWG wires
- Alligator clips or lugs

Steps

1. Connect the transformer to the wall socket with alligator clamps
2. Plug in immersion heater to wall socket with heater off
3. Set up the multi-meter to measure AC from the low side of the transformer
4. Connect the transformer to the single pole of the inverter with alligator clamps and 4/0awg wire, cap, or electrically isolate the remaining poles
5. Connect inverter to High Voltage DC source
6. Turn on the High Voltage DC source after making sure the inverter is pre-charged
7. Measure AC voltage from the low side of the transformer. If below 20Vac, proceed to the next step
8. Using a current clamp connected to the multi-meter, clamp around the immersion heater wires
9. Measure current & Voltage at the same time, assume power loss is negligible
10. Control the power to go into the transformer using the inverter.

PCB

LV Test

1. Keep DC power supply output OFF. Set DC power supply to 100 mA and 0V. Connect it to the output of the DC/DC with the same polarity.
2. Slowly turn the DC power supply output to ON. Set the voltage of the DC power supply to 12 V.
 - a. Make sure 3.6 to 3.8 V between pads of Cstart1
 - b. Check 5V LED lights up
3. Enable switch on DCDC
 - a. Check for 12 to 12.5 V between U1 pads 1 and 2
 - b. Make sure the 12 V and power LEDs light up
 - c. Be sure the input current at 23V is 20 - 30 mA.
4. Insert the super cap (Cstart1Cstart1).
 - a. Charge between 3.6 and 3.8 V.
 - b. Don't short the circuit
5. Remove R61 to deactivate UVP/OVP
6. Measure ground (U12 pins 5 and 8, Q10 and Q11 pin1(gate)) for square wave signals
7. Output should be below 23V for testing
8. Apply low voltage like 10V / 100mA to HV input

Referenced from Rootthecause's GitHub

HV Test

At the HV input:

- Voltage source of 0 – 600 V / 0 – 2 A
- Multimeters for voltage and current measurements

On the LV output:

- Power sink 24V / max. 0 – 31.25 A
- Multimeter for voltage measurement connected via meas. Points J4
- Multimeter for current measurement
- Laboratory power supply unit for charging the start cap (suitable for 5-24V)

On the DC-DC with oscilloscope:

- Voltage measurement with Cres with HV
- Insulated current measurement coil via transformer cable
- Drain source and gate source voltage measurement on an SR MOSFET (Q8 or Q9)
- C1 measurements (HV!) checking pre-charge
- HV input and LV output for ripple Volt meas.

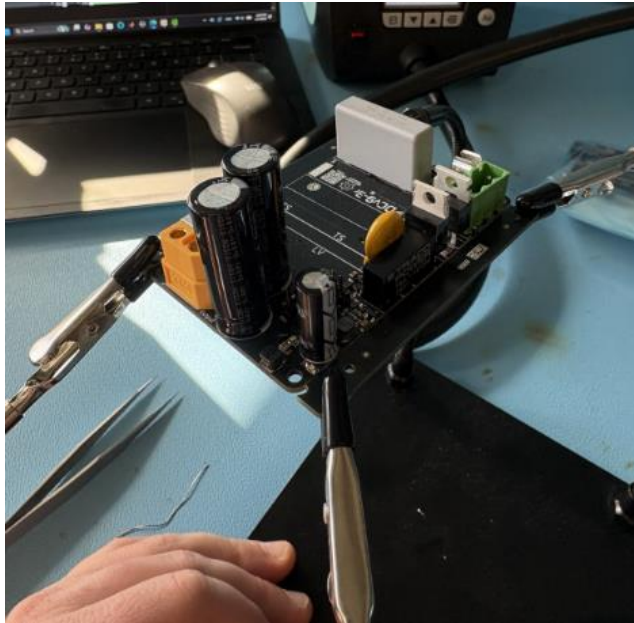


Figure 89: PCB Assembly of THT Components

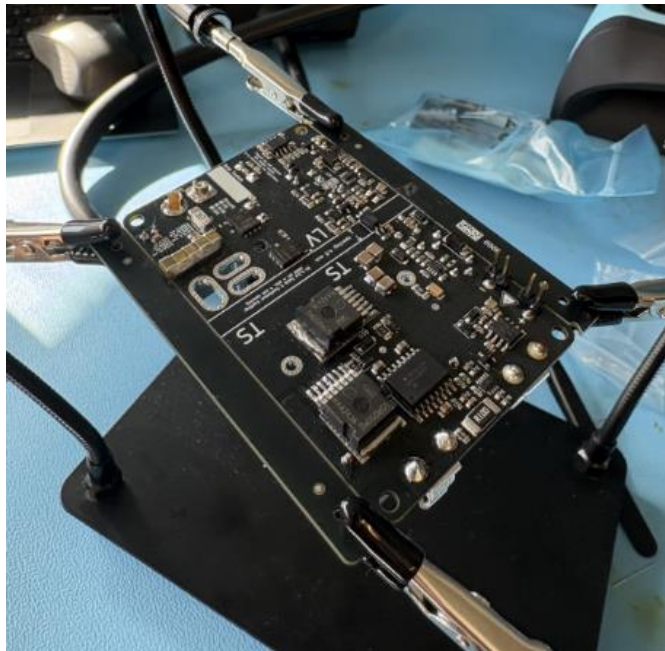


Figure:90 PCB Assembly of SMD Components

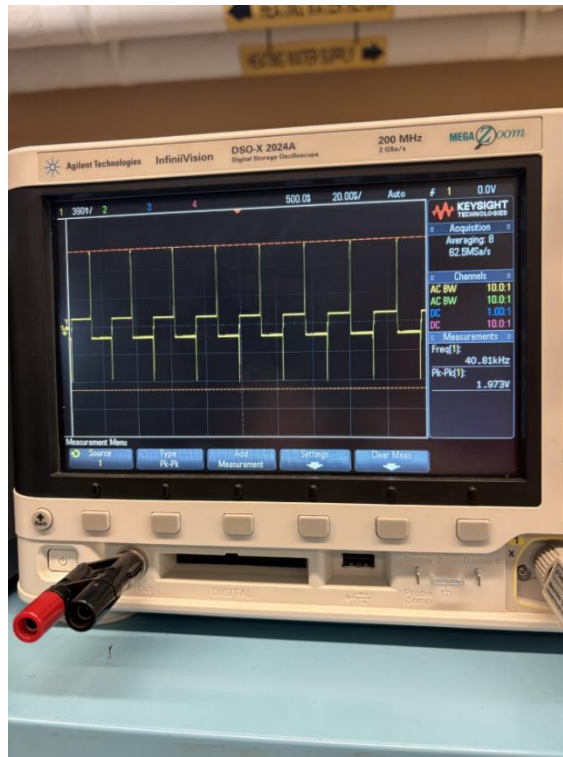


Figure 91: Unsatisfactory High Transient Square Wave Later Fixed

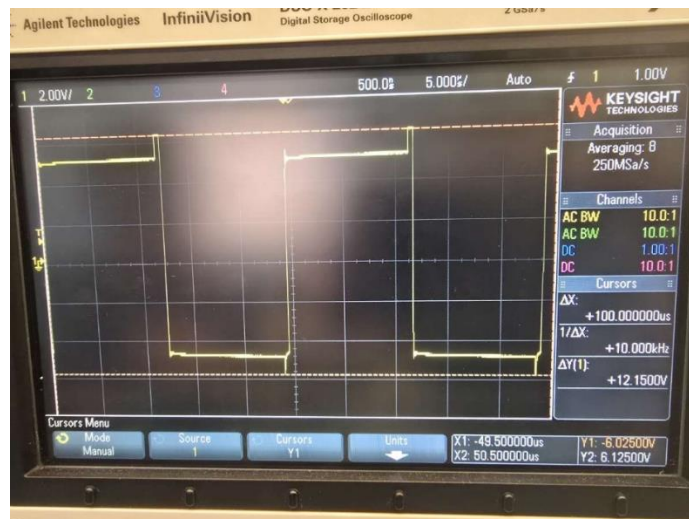


Figure 92: Deadtime Peak Measurement

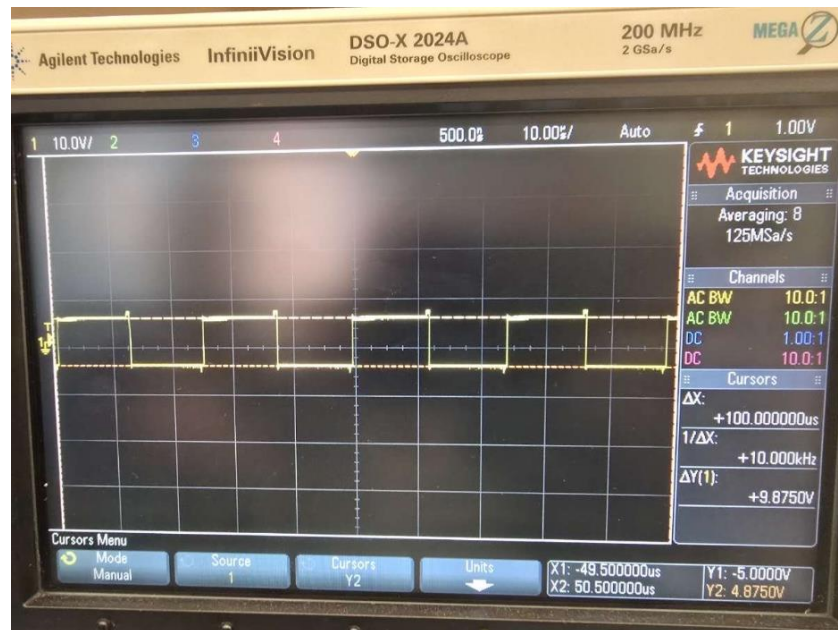


Figure 93: Half-bridge Output Square Wave

11.3. Appendix 3 – Alternative/initial version of design

Our final design did not change much compared to the initial design. The only alteration that we made is to remove the supercapacitor and the DC enable switch from our converter. The reason that we did this is because the client decided to add a 12V battery to their car, so that our converter would be driving the car's low voltage systems as well as charging their 12V battery. This greatly simplified our power supply as instead of using a supercapacitor to charge the components on our converter, we could use the 12V battery to charge the components on our board. Furthermore, we added a droop resistor at the converter's output to ensure it could properly charge the battery.

11.4. Appendix 4 – Other considerations

The major consideration with our converter is that it has not yet passed testing when connected to the 600V battery in our client's racecar. This should not be a major problem, as after discussing with the client and our faculty advisor, both the client and we believe it to be an issue with the 600V battery. Once their battery is updated to resolve this issue, the system should work as intended.

11.5. Appendix 5 – Code

Link to reference design: <https://github.com/Rootthecause/DCDC>

11.6. Appendix 6 – Team Contract

Team Members

TEAM MEMBER NAMES

Andrew Chandler
Caleb Rezmer
Diego Rivera
Drew Bixler
Hunter Kanne
Wesley Mao

Table 7: All team members' names

Required Skill Sets for Your Project

Magnetics/Transformer Design

Control Systems Design

Semiconductors

Circuit Analysis

LTSPICE Simulation

PCB Design

Skill Sets Covered by the Team

Skill Set	Team member covering it
Controls and PCB Design	Caleb Rezmer
Controls and Circuit Analysis	Drew Bixler
Power Systems and Transformers	Hunter Kanne
Circuit Analysis	Diego Rivera
Semiconductors and PCB Design	Andrew Chandler
Semiconductors and Circuits	Wesley Mao

Table : Skill sets and the team members covering it

Project Management Style Adopted by the team

During this project, we adopted a waterfall-style management approach. We created a Gantt chart to track progress on various tasks and phases of the project. We then updated the status and progress of the tasks each week, emphasizing certain tasks through color coding. We also set goals and milestones to further create a sense of urgency.

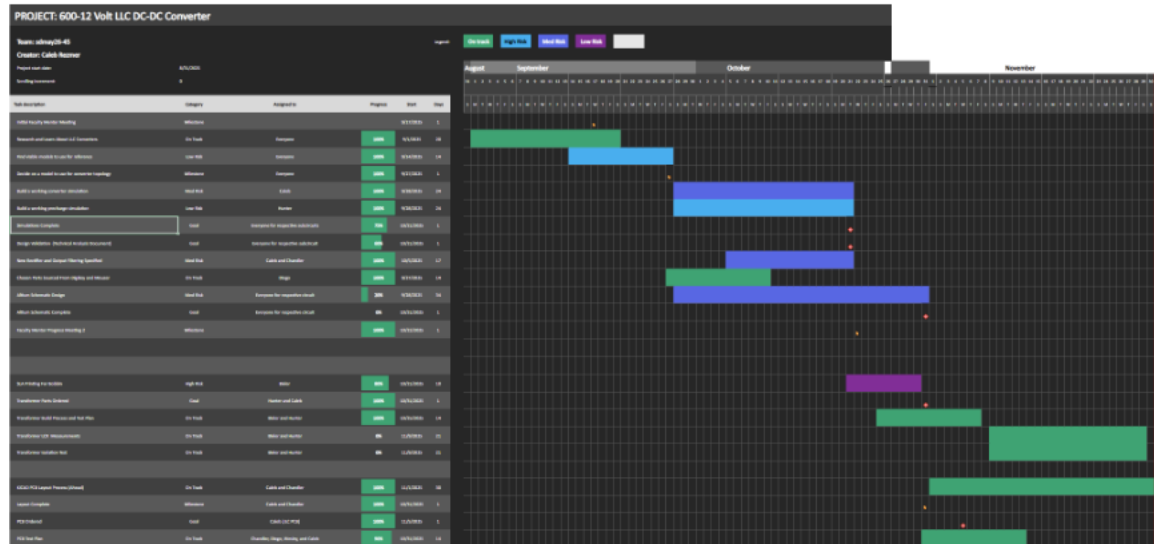


Figure 94: Gantt Chart

Individual Project Management Roles

Chandler - Primary Side

Caleb - Secondary Side

Bixler - Oscillator (aka Controls)

Diego - Power supply

Hunter – Pre-charge and discharge

Wesley - Overtemperature Protection

Team Contract

Team Members:

- 1) ___ Caleb Rezmer_____ 2) ___ Drew Bixler_____
- 3) _____ Andrew Chandler_____ 4) ___ Wesley Mao_____
- 5) _____ Hunter Kanne_____ 6) ___ Diego Rivera_____

Team Procedures

1. Weekly Meeting Time:
 - a. Wednesdays, 2pm - 4pm, face-to-face at a reserved room in Coover 2011
 - b. Saturdays, 10pm – 12pm, face-to-face at the signals and systems lab in Coover Hall
2. Preferred method of communication updates, reminders, issues, and scheduling (e.g., e-mail, phone, app, face-to-face):
 - a. Communication updates will be given during weekly meetings as well as in the announcements channel on the Microsoft Teams group.
3. Decision-making policy (e.g., consensus, majority vote):
 - a. The team will come to a consensus opinion during the weekly meetings. Each decision will be debated until a consensus opinion is reached.
4. Procedures for record keeping (i.e., who will keep meeting minutes, how will minutes be shared/archived):
 - a. Diego will keep meeting minutes
 - b. The minutes will be archived on the group's Teams channel

Participation Expectations

1. Expected individual attendance, punctuality, and participation at all team meetings:
 - a. Attend all team meetings (except when sick) in a timely manner and engage in meetings.
 - b. Members will inform the team via Microsoft Teams if they will not be able to attend a meeting.
2. Expected level of responsibility for fulfilling team assignments, timelines, and deadlines:
 - a. Complete individual assignments in a timely manner and have updates to give and work to present for the next meeting.
 - b. Deadlines, particularly design reviews, must be fulfilled unless extension is agreed upon.
3. Expected level of communication with other team members:
 - a. Communicate with the team at least once a week on progress during weekly meetings.
 - b. The group's Teams chats will be used to communicate as needed between weekly meetings.
4. Expected level of commitment to team decisions and tasks:
 - a. Spend at least 6 hours a week outside of team meetings working on assigned tasks.

Leadership

1. Team Member Roles:
 - a. Caleb Rezmer: Team Lead and Client Communications
 - b. Drew Bixler: Control Specialist
 - c. Andrew Chandler: Primary Stage Specialist
 - d. Hunter Kanne: Transformer Design Specialist and Advisor Liaison
 - e. Diego Rivera: Power Supply Specialist and Secretary
 - f. Wesley Mao: Secondary Stage Specialist
2. Strategies for supporting and guiding the work of all team members:

- a. During weekly meetings, members will present their work, teaching the rest of the team about the function and details of their portion of the project. Other members of the team will review their work and give their own feedback. This will allow all ideas to be reviewed by the full team and give an opportunity for members to receive feedback on their contributions.
3. Strategies for recognizing the contributions of all team members:
 - a. Team members will be recognized for their contributions during the weekly meetings as they present. At the end of the project, each member will be recognized for the work done on their subsection of the project.

Collaboration

1. Describe the skills, expertise, and unique perspectives each team member brings to the team.
 - a. Each team member will be a unique expert for smaller portions of the project. The portion that each team member is put in charge of has been chosen based on their prior knowledge. This will allow each member to be able to contribute in a meaningful way, while also taking advantage of each team member's unique expertise.
2. Strategies for encouraging and supporting contributions and ideas from all team members:
 - a. Each team member has their own specialization over smaller portions of the project. In weekly meetings, each member will present their ideas and contributions towards their portion of the project. This will encourage and support ideas from each member, while also ensuring that each team member understands what is going on regarding the project.
3. Procedures for identifying and resolving collaboration or inclusion issues (e.g., how will a team member inform the team that the team environment is obstructing their opportunity or ability to contribute?)
 - a. The team member is responsible for communicating their frustrations to the rest of the team, preferably informing the rest of the team in-person during a weekly meeting. The team will then discuss with the team member how they can contribute more, or change the collaborative environment, and find specific tasks to assign or change for the team member if necessary.

Goal setting, Planning, and Execution

- Team goals for this semester:
 - a. PCB and transformer assembled and tested this semester
 - i. All parts tested for errors
 - ii. PCB parts removed, added, or changed to work properly
 - iii. Transformer tested and made sure to function properly
 - iv. Transformer parts are also glued together or fixed
- Strategies for planning and assigning individual and team work:
 - a. We have delegated focus areas to each member as follows.

- Note: *Each team member should have a holistic understanding of the LLC Circuit, but these areas of expertise ensure a deeper level of understanding for specific components of the design.*
- - i. Precharge and Discharge: Hunter
 - ii. Primary Stage: Andrew and Caleb
 - iii. Control: Drew and Andrew
 - iv. Transformer: Hunter and Caleb
 - v. Secondary Stage and Auxiliaries: Wesley and Diego
- Strategies for keeping on task:
 - a. Specific goals and deadlines for the next meeting are defined and posted on the Teams channel after every meeting.

Consequences for Not Adhering to Team Contract

1. How will you handle infractions of any of the obligations of this team contract?
If any infractions occur, the team must talk to the team member committing said infractions about their lack of commitment to the aforementioned obligations.
2. What will your team do if the infractions continue?
If the infractions continue the team member will be reported to the course instructor(s)

- a. I participated in formulating the standards, roles, and procedures as stated in this contract.
- b. I understand that I am obligated to abide by these terms and conditions.
- c. I understand that if I do not abide by these terms and conditions, I will suffer the consequences as stated in this contract.

- 1) ___Hunter Kanne_____ DATE ___4/5/2026_____
- 2) ___Wesley Mao_____ DATE ___4/5/2026_____
- 3) ___Andrew Chandler_____ DATE ___4/5/2026_____
- 4) ___Drew Bixler_____ DATE ___4/5/2026_____
- 5) ___Caleb Rezmer_____ DATE ___4/5/2026_____
- 6) ___Diego Rivera_____ DATE ___4/5/2026_____